

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\109-C-Solis SS-Original file.pvdb

Date of Analysis : Sep 8, 2026 6:22pm

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 109-C-Solis SS-Original file Page 2 of 197
 Vessel Design Summary: Step: 31 6:22pm Sep 8,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 900.000 mm ID
 Vessel Design Length, Tangent to Tangent 6638.35 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 100 °C
 Channel Side Design Temperature 130 °C
 Shell Side Design Pressure 1300.000 KPa.
 Channel Side Design Pressure 520.000 KPa.
 Shop Shell Side Test Pressure 1690.000 KPa.
 Field Shell Side Test Pressure 4027.787 KPa.
 Shop Channel Side Test Pressure 676.000 KPa.
 Field Channel Side Test Pressure 1267.007 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-240 304	...	...	S30400	
Flange	SA-965 F304	...	...	S30400	
Flange	SA-965 F304	...	...	S30400	
Nozzle	SA-312 TP304	...	...	S30400	
Nozzle	SA-965 F304	...	...	S30400	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-182 F304	...	> 130	S30400	
Tubes	SA-213 TP304	...	...	S30400	
Tubesheet	SA-965 F304	...	...	S30400	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Blind Flange	520.000	103.42	975.000	0.0000	N/A	No
LHS CH Flang	528.822	103.42	2637.000	0.0000	N/A	No
Ch Shell	528.822	103.42	3487.000	0.0000	N/A	No
main shell	1308.822	103.42	3587.000	0.0000	N/A	No
RHS Ch shell	528.822	103.42	3487.000	0.0000	N/A	No
RHS Ch Flg	520.000	103.42	2646.000	0.0000	N/A	No
RHS Blind Flg	520.000	103.42	1149.000	0.0000	N/A	No

Liquid Level: 900.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:



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Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	59.2	56.0	56.0	56.0	42.9	...	1.00	1.00
Body Flg	194.2	135.0	105.0	105.0	92.8	...	1.00	1.00
Cylinder	1194.2	1000.0	12.0	12.0	1.8	...	1.00	1.00
Cylinder	5404.2	4170.0	12.0	12.0	4.3	...	1.00	1.00
Cylinder	6444.2	1000.0	12.0	12.0	1.8	...	1.00	1.00
Body Flg	6582.3	135.0	105.0	105.0	92.8	...	1.00	1.00
Body Flg	6638.3	56.0	56.0	56.0	42.9	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	56	42.8752	103.419	No Calc
20	30	105	92.8116	103.419	No Calc
30	Tsht	12	3.16387	103.419	1145.9
Tsht	40	12	No Calc	103.419	No Calc
40	Tsht	12	5.71422	103.419	641.968
Tsht	50	12	No Calc	103.419	No Calc
50	60	12	3.16394	103.419	1145.88
60	70	105	92.8116	103.419	No Calc
70	80	56	42.8752	103.419	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	999.97	27608.8	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	4170	94597.8	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	1000.03	27609.7	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.819	Shell	Passed
N3-Shell inlet	WRC-297	0.857	Shell	Passed
N4-Shell outle	WRC-297	0.857	Shell	Passed
N2-Tube Outlet	WRC-297	0.819	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

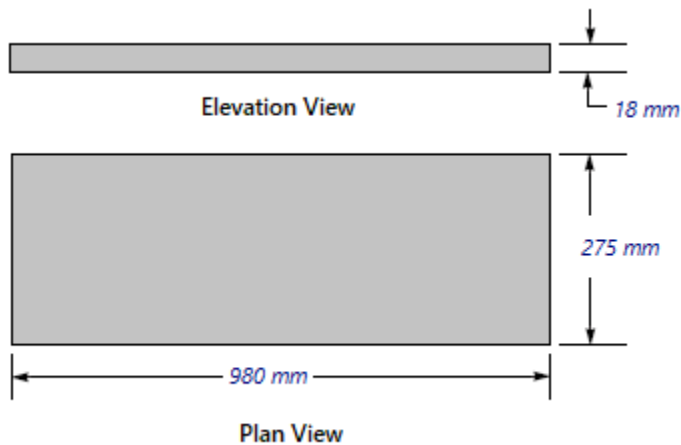
Saddle Width	250.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	990.000 mm
Wear Pad Width	370.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.



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 Vessel Design Summary: Step: 31 6:22pm Sep 8,2026

Distance from Saddle to Tangent	835.000 mm
Baseplate Length	980.000 mm
Baseplate Thickness	18.000 mm
Baseplate Width	275.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm
Web Thickness	10.000 mm
Height of Center Web	530.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	8228.16 Kg
Maximum Transverse Saddle Shear Load	2713.77 Kg
Maximum Longitudinal Saddle Shear Load	752.77 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	6615.79 Kg
Maximum Transverse Saddle Shear Load	92.18 Kg
Maximum Longitudinal Saddle Shear Load	61.39 Kg

Weights:

Fabricated - Bare w/o removable internals	8285.0 kg.
Shop Test - Fabricated + water (full)	12516.2 kg.
Shipping - Fab. + rem. intls.+ shipping app.	8285.0 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	8785.0 kg.
Empty - Fab. + intls. + details + wghts.	9499.7 kg.
Operating - Empty + operating liquid (no ca)	13145.1 kg.
Field Test - Empty weight + test medium (full)	13230.9 kg.

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 Nozzle Summary: Step: 29 6:22pm Sep 8,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	3045	OK	...	OK	1.76	OK	Passed
N3-Shell inlet	3118	OK	...	OK	4.29	OK	Passed
N4-Shell outlet	3098	OK	...	OK	4.32	OK	Passed
N2-Tube Outlet	3037	OK	...	OK	1.79	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 3098 Nozzle : N4-Shell outlet [Shellside]
 Minimum MAWP Nozzles : 3037 Nozzle : N2-Tube Outlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1-Tube Inlet	694.175	0.000	596.900
40	N3-Shell inlet	2584.175	0.000	596.900
40	N4-Shell outlet	4034.175	180.000	596.900
50	N2-Tube Outlet	5944.174	180.000	596.900

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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 Nozzle Schedule: Step: 28 6:22pm Sep 8,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Tube Inlet	12.000 in	80S	WNF	12.750	12.700	423.85	12.00	244.14	150
N3-Shell inlet	12.000 in	80S	WNF	12.750	12.700	423.85	12.00	244.14	300
N4-Shell outlet	12.000 in	80S	WNF	12.750	12.700	423.85	12.00	244.14	300
N2-Tube Outlet	12.000 in	80S	WNF	12.750	12.700	424.00	12.00	244.14	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Tube Inl	SA-312 TP304	...	10.000	10.000	12.000	...
N3-Shell in	SA-312 TP304	...	10.000	10.000	12.000	...
N4-Shell ou	SA-965 F304	...	10.000	10.000	12.000	...
N2-Tube Out	SA-312 TP304	12.000	10.000	10.000	12.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Tube Inlet	694.175	0.0	200.00	0.00	Ch Shell
N3-Shell inlet	2584.175	0.0	200.00	0.00	main shell
N4-Shell outlet	4034.175	180.0	200.00	0.00	main shell
N2-Tube Outlet	5944.174	180.0	200.00	0.00	RHS Ch shell

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Bill of Material:

Step: 27

6:22pm

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 56.0mm THK X 1110.0mm OD	SA-965 F304
2	WELD NECK FLANGE: 105.0mm THK X 1110.0mm OD	SA-965 F304
2	CYLINDER: 12.0mm THK X 900.0mm ID X 1000.0mm	SA-240 304
1	CYLINDER: 12.0mm THK X 900.0mm ID X 4170.0mm	SA-240 304
2	INSULATION: 40mm X 100mm THK	Hot Insulation
2	INSULATION: 80mm X 100mm THK	Hot Insulation
2	INSULATION: 1000mm X 100mm THK	Hot Insulation
15	BAFFLES: 893.0mm DIA X 8.0mm THK	SA-516 70
6	TIE RODS: 14mm DIA X 4250mm	SA-193 B7M
2	TUBESHEET: 924.0mm X 40.0mm THK	SA-965 F304
1010	TUBES: 4250.0mm X 19.0mm DIA X 1.6mm THK	SA-213 TP304
2	SADDLE: 250.0mm X 120 DEG	SA-516 70
1	INSULATION: 4170mm X 80mm THK	Hot Insulation
2	GASKET: 958.0mm OD X 932.0mm ID	...
48	BODY FLANGE BOLTS: 34.9mm DIA	SA-193 B7
96	NUTS FOR BODY FLANGE BOLTS: 34.9mm DIA	...
1	NAMEPLATE	...

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 MDMT Summary: Step: 30 6:22pm Sep 8,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
main shell [16]		-196						
N3-Shell inlet[1]	B	-26	-104	-29	0.127	11.113	1.00	No
N4-Shell outle[1]	B	-26	-104	-29	0.128	11.113	1.00	No
Warmest MDMT:		-26	-104					
Ch Shell [16]		-196						
RHS Ch shell [16]		-196						
N1-Tube Inlet [1]	B	-26	-104	-29	0.051	11.113	1.00	No
N2-Tube Outlet[1]	B	-26	-104	-29	0.052	11.113	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		-26	-104					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-104.0	-12.0	Pass
Channel/Tube	-48.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.



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Warnings and Errors: Step: 0 6:22pm Sep 8, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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 Input Echo: Step: 1 6:22pm Sep 8,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	1300	KPa.
Channel Side Design Pressure	520	KPa.
Shell Side Design Temperature	100.0	°C
Channel Side Design Temperature	130.0	°C
Radiography, Shell Side	RT-2	
Radiography, Channel Side	RT-2	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	10000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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 Input Echo: Step: 1 6:22pm Sep 8,2026

Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Blind Flange
Distance "FROM" to "TO"	56 mm
Flange Outside Diameter	1110 mm
Element Thickness	56 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	56 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	106.2 N./mm ²
Allowable Stress, Hydrotest	184.5 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	159.6 N./mm ²
External Pressure Chart Name	HA-1
UNS Number	S30400
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	40 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	2/7
Element From Node	20



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Element To Node	30
Element Type	Flange
Description	LHS CH Flang
Distance "FROM" to "TO"	135 mm
Flange Inside Diameter	900 mm
Element Thickness	105 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	105 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	900 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	80 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Ch Shell
Distance "FROM" to "TO"	1000 mm
Inside Diameter	900 mm
Element Thickness	12 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	133.2 N./mm ²
Allowable Stress, Hydrotest	184.5 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	159.6 N./mm ²
External Pressure Chart Name	HA-1
UNS Number	S30400
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No



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Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	900 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	30
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1000 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	N1-Tube Inlet
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	12 in.
Nozzle Schedule	80S
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	96.39 Kg
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-312 TP304

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	main shell
Distance "FROM" to "TO"	4170 mm
Inside Diameter	900 mm
Element Thickness	12 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1300 KPa.
Design Temperature Internal Pressure	100 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle flo
Dist. from "FROM" Node / Offset dist	835 mm
Width of Saddle	250 mm
Height of Saddle at Bottom	990 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	370 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm



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Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	40
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	3335 mm
Width of Saddle	250 mm
Height of Saddle at Bottom	990 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	370 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	900 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	40
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	4170 mm
Thickness of Insulation	80 mm
Density	0.000352 kg./cm³

Element From Node	40
Detail Type	Nozzle
Detail ID	N3-Shell inlet
Dist. from "FROM" Node / Offset dist	1350 mm
Nozzle Diameter	12 in.
Nozzle Schedule	80S
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	96.39 Kg
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-312 TP304

Element From Node	40
Detail Type	Nozzle
Detail ID	N4-Shell outlet
Dist. from "FROM" Node / Offset dist	2800 mm
Nozzle Diameter	12 in.
Nozzle Schedule	80S
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	96.39 Kg
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-965 F304

Element From Node	40
Detail Type	Weight



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Detail ID	Fireproofing we	
Dist. from "FROM" Node / Offset dist	3300	mm
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	990.6	mm

Element From Node	40	
Detail Type	Weight	
Detail ID	Wght:[2 of 2]	
Dist. from "FROM" Node / Offset dist	850	mm
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	990.6	mm

Element#	5/7	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	RHS Ch shell	
Distance "FROM" to "TO"	1000	mm
Inside Diameter	900	mm
Element Thickness	12	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	520	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-240 304	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	1	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	900	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1000	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N2-Tube Outlet	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	12	in.
Nozzle Schedule	80S	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	98.139	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-312 TP304	



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Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Flange
Description	RHS Ch Flg
Distance "FROM" to "TO"	135 mm
Flange Inside Diameter	900 mm
Element Thickness	105 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	105 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	106.2 N./mm ²
Allowable Stress, Hydrotest	184.5 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	159.6 N./mm ²
External Pressure Chart Name	HA-1
UNS Number	S30400
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	60
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	80 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Flange
Description	RHS Blind Flg
Distance "FROM" to "TO"	56 mm
Flange Outside Diameter	1110 mm
Element Thickness	56 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	56 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	70
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	40 mm
Thickness of Insulation	100 mm



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Input Echo:

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1

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Density

0.000352 kg./cm³

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XY Coordinate Calculations:

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Blind Flange		59.175	...	56	...
LHS CH Flang		194.175	...	135	...
Ch Shell		1194.17	...	1000	...
main shell		5404.17	...	4170	...
RHS Ch shell		6444.17	...	1000	...
RHS Ch Flg		6582.35	...	135	...
RHS Blind Flg		6638.35	...	56	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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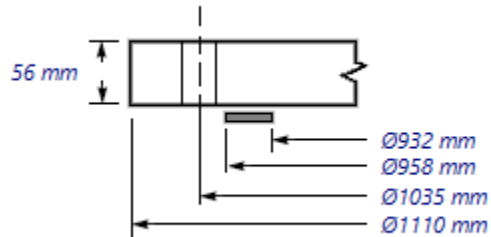
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 Flg Calc [Int P]: New Flange Flng: 9 6:22pm Sep 8,2026

Flange Input Data Values

Description: New Flange :

Blind Flange

Flange Type		Blind	
Design Pressure	P	520.00	KPa.
Design Temperature		130	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1110.000	mm
Flange Thickness	t	56.0000	mm
Flange Material		SA-965 F304	
Flange Material UNS number		S30400	
Flange Allowable Stress At Temperature	Sfo	106.20	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	3251.548	mm
Diameter of Bolt Circle	C	1035.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads		UNC Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	969.000	mm
Flange Face Inside Diameter	Fid	900.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	958.000	mm
Gasket Inside Diameter	Gi	932.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	945.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:



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 Flg Calc [Int P]: New Flange Flng: 9 6:22pm Sep 8,2026

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 945.1509^2 \cdot 520.0$$

$$= 37204.211 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 945.1509 \cdot 3.0 \cdot 520.0$$

$$= 6069.407 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(37204 + 6069 + 0, 0)$$

$$= 43273.617 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 945.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 134122.203 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((43274 + 0 + 4 \cdot 0 / 945.15) / 172, 134122 / 172)$$

$$= 76.473 \text{ cm}^2$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	76.473	163.200	
Radial Distance between Bolts and Edge:	34.925	37.500	
Circ. Spacing between the Bolts:	77.800	135.095	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 163.2 \cdot 172.0 / (68.95 \cdot \pi (958.0 + 932.0))$$

$$= 6.856 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 163.1997$$

$$= 286227.44 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 43273.62 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1035.0 - 945.1509) / 2$$

$$= 44.9245 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 43274 \cdot 44.9245 \cdot 1.0 / (56.0^2 \cdot 945.1509) + 0.3 \cdot 1.0 \cdot 520.0 \cdot 945.1509^2 / 56.0^2$$

$$= 56.66 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 286227 \cdot 44.9245 \cdot 1.0 / (56.0^2 \cdot 945.1509)$$

$$= 80.84 \text{ N./mm}^2$$



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 Flg Calc [Int P]: New Flange Flng: 9 6:22pm Sep 8,2026

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 43274/163.1997$$

$$= 26.00 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (134122/163.1997)$$

$$= 80.60 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	56.66	106.20	80.84	138.00
Bolting	26.00	172.00	80.60	172.00

Reqd. Blind Flange Thickness at Center 42.860 mm
 Reqd. Blind Flange Thickness at Gasket 42.860 mm

Estimated M.A.W.P. (Operating) 975 KPa.
 Estimated Finished Weight of Flange at given Thk. 435.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 435.0 kg.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	56.66	138.00	80.84	138.00
Bolting	26.00	172.00	80.60	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 42.875 mm

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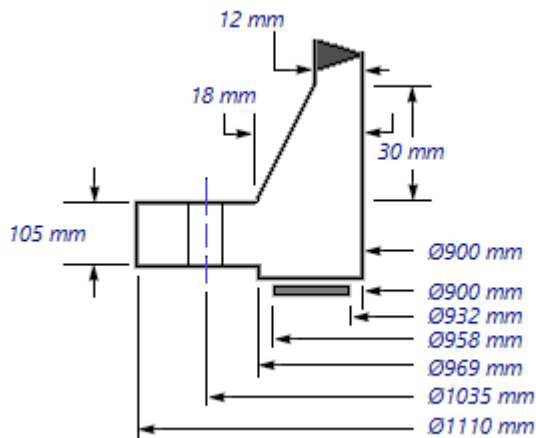
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 Flg Calc [Int P]: LHS CH Flang Flng: 10 6:22pm Sep 8,2026

Flange Input Data Values

Description: LHS CH Flang :

LHS CH Flang

Flange Type	Integral Weld Neck		
Design Pressure	P	528.82	KPa.
Design Temperature		130	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	900.000	mm
Flange Outside Diameter	A	1110.000	mm
Flange Thickness	t	105.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	g1	18.0000	mm
Length of Hub	h	30.0000	mm
Flange Material	SA-965 F304		
Flange Material UNS number	S30400		
Flange Allowable Stress At Temperature	Sfo	106.20	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1035.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	UNC Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	969.000	mm
Flange Face Inside Diameter	Fid	900.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	958.000	mm
Gasket Inside Diameter	Gi	932.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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 Flg Calc [Int P]: LHS CH Flang Flng: 10 6:22pm Sep 8,2026

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (528.82 \cdot (900.0/2 + 0.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 528.82) + Ca$$

$$= 1.7909 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 12.0) / (450.0 + 0.6 \cdot 12.0)$$

$$= 3495.847 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (450.0 + 0.6 \cdot 12.0)$$

$$= 3621.771 \text{ KPa.}$$

Corroded Flange ID,	Bcor = B+2*ci	900.000 mm
Corroded Large Hub,	g1Cor = g1-ci	18.000 mm
Corroded Small Hub,	g0Cor = go-ci	12.000 mm
Code R Dimension,	R = (C-B)/2 - g1	49.500 mm
Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm
Basic Gasket Width,	bo = N/2	6.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm
Gasket Reaction Diameter,	G = Go - 2 * b	945.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 945.1509^2 \cdot 528.822$$

$$= 37835.383 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 945.1509 \cdot 3.0 \cdot 528.82$$

$$= 6172.375 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 900.0^2 \cdot 528.8218$$

$$= 34306.848 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 37835 - 34307$$

$$= 3528.533 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(37835 + 6172 + 0, 0)$$

$$= 44007.758 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 945.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 134122.203 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((44008 + 0 + 4 \cdot 0 / 945.15) / 172, 134122 / 172)$$

$$= 76.473 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$



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 Flg Calc [Int P]: LHS CH Flang Flng: 10 6:22pm Sep 8,2026

$$= 2 \cdot 34.925 + 6 \cdot 105.0 / (3.0 + 0.5)$$

$$= 249.850 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1035.0 \cdot \sin(\pi / 24)$$

$$= 135.095 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{(Bs / (2a + t))}, 1)$$

$$= \max(\sqrt{(135.095 / (2 \cdot 34.925 + 105.0))}, 1)$$

$$= 1.0000$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	76.473	163.200	
Radial Distance between Hub and Bolts:	47.625	49.500	
Radial Distance between Bolts and Edge:	34.925	37.500	
Circ. Spacing between the Bolts:	77.800	135.095	249.850

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 163.2 \cdot 172.0 / (68.95 \cdot \pi(958.0 + 932.0))$$

$$= 6.856 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 163.1997$$

$$= 286227.44 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 44008 - 37835$$

$$= 6172.38 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1035.0 - 945.1509) / 2$$

$$= 44.9245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1035.0 - 900.0) / 2 + 44.9245)$$

$$= 56.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1035.0 - 900.0 - 18.0) / 2$$

$$= 58.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	34307.	58.5000	1.0000	2007.
Face Pressure, Mt	3529.	56.2123	1.0000	198.



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Gasket Load, Mg	6172.	44.9245	1.0000	277.
Gasket Seating, Matm	286227.	44.9245	1.0000	12859.

Total Moment for Operation, Mo 2483. Kg-m.
 Total Moment for Gasket seating, Mg 12859. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 103.923 mm
 Hub Ratio, h/h0 = HL / H0 0.289
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.500

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.884
 Factor V per 4.16.5 0.375
 Factor f per 4.16.5 1.207

Stress factor from Table 4.16.4 [K]:

= A / B
 = 1110.0/900.0
 = 1.233

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
 = (0.66845 + (5.7169(1.233² • log10(1.233))) / (1.233² - 1))/(1.233 - 1)
 = 9.379

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
 = (1.233²(1 + 8.55246 • log10(1.233)) - 1) / ((1.0472 + 1.9448 • 1.233²) • (1.233 - 1))
 = 1.825

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
 = (1.233²(1 + 8.55246 • log10(1.233)) - 1) / (1.36136(1.233² - 1) • (1.233 - 1))
 = 10.306

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
 = (1.233² + 1) / (1.233² - 1)
 = 4.838

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
 = 10.306 • 12.0² • 103.923/0.375
 = 411736.875 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.884/103.923
 = 0.009 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
 = (105.0 • 0.009 + 1) / 1.825 + 105.0³/411736.875
 = 3.849

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
 = (1.2074•2483/900.0)/(3.8486•18.0²)
 = 26.19 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
 = (1.2074•12859/900.0)/(3.8486•18.0²)



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= 135.67 N./mm²

Radial Flange Stress, Operating [SRo]:

= ($\beta \cdot Mop / Bcor$) / ($L \cdot t^2$)
 = (2.1906•2483/900.0)/(3.8486•105.0²)
 = 1.40 N./mm²

Radial Flange Stress, Seating [SRa]:

= ($\beta \cdot Matm/Bcor$) / ($L \cdot t^2$)
 = (2.1906•12859/900.0)/(3.8486•105.0²)
 = 7.23 N./mm²

Tangential Flange Stress, Operating [STo]:

= ($Y \cdot Mo / (t^2 \cdot Bcor)$) - $Z \cdot SRo$
 = (9.3787•2483/(105.0²•900.0))-4.838•1
 = 16.26 N./mm²

Tangential Flange Stress, Seating [STa]:

= ($y \cdot Matm / (t^2 \cdot Bcor)$) - $Z \cdot SRa$
 = (9.3787•12859/(105.0²•900.0))-4.838•7
 = 84.19 N./mm²

Average Flange Stress, Operating [SAo]:

= ($SHo + \max(SRo, STo)$) / 2
 = (26+max(1,16))/2
 = 21.22 N./mm²

Average Flange Stress, Seating [SAa]:

= ($SHa + \max(SRa, STa)$) / 2
 = (136+max(7,84))/2
 = 109.93 N./mm²

Bolt Stress, Operating [BSo]:

= $Wm1 / Ab$
 = 44008/163.1997
 = 26.44 N./mm²

Bolt Stress, Seating [BSa]:

= ($Wm2 / Ab$)
 = (134122/163.1997)
 = 80.60 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	26.19	159.30	135.67	207.00
Radial Flange	1.40	106.20	7.23	138.00
Tangential Flange	16.26	106.20	84.19	138.00
Maximum Average	21.22	106.20	109.93	138.00
Bolting	26.44	172.00	80.60	172.00

Minimum Required Flange Thickness 92.812 mm

Estimated M.A.W.P. (Operating) 2646 KPa.

Estimated Finished Weight of Flange at given Thk. 289.8 kg.

Estimated Unfinished Weight of Forging at given Thk 359.3 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

	Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
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Longitudinal Hub	Sh	364.80	364.80	1.000	364.80	409.99	0.890
Tangential Flange	St	228.21	273.60	0.834	228.21	307.49	0.742
Tangential Flange	Sto	99.53	182.40	0.546	99.53	205.00	0.486
Combined	Sh/f + St	532.77	547.20	0.974	532.77	614.99	0.866
Combined	Sr + St	247.81	547.20	0.453	247.81	614.99	0.403
Combined	Sh + Sto	464.33	547.20	0.849	464.33	614.99	0.755
Radial Flange	Sr	19.61	364.80	0.054	19.61	409.99	0.048
Maximum Stress Ratio				1.000			0.890

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:
 = max(1; F1num / F1den)
 = max(1; 1.4685/1.226)
 = 1.1978

Hub Stress Correction Factor [f2]:
 = max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)
 = max(0.25; 0.3784/1.158) if $h/h_0 < 0.35$, else max (0.25; 0.2673/0.7714)
 = 0.3268

In this case, the hub ratio h/h_0 , 0.2887 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



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Input Data:

Maximum permissible bolt stress	Sbmax	507.6 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	466.2 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	6.8000 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	205.0 N./mm ²
Flange Material Operating Yield Stress	Syo	159.6 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.594 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (958.0^2 - 932.0^2)$$

$$= 385.9 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207.0 \cdot 385.946 / (24 \cdot 6.8)$$

$$= 489.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(489.528; 507.6)$$

$$= 489.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(489.528; 290.0)$$

$$= 489.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(489.528; 466.193)$$

$$= 466.2 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 466.193 \geq 140.0(385.946 / (24 \cdot 6.8))$$

$$:: 466.2 \geq 331.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 466.193 \geq (97.0 \cdot 385.946 + \pi/4 \cdot 528.822 \cdot 932.0^2) / (0.7 \cdot 6.8 \cdot 24)$$

$$:: 466.2 \geq 359.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 466.193 \leq 276.0(385.946 / (24 \cdot 6.8))$$

$$:: 466.2 \leq 652.7 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetaamax)$$

$$:: 466.193 \leq 466.193(1.0 / 0.594)$$

$$:: 466.2 \leq 784.8 \text{ N./mm}^2 \text{ Check Passes}$$



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Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 466.193 \cdot 0.2 \cdot 6.8 \cdot 34.925$
 $= 225.8 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 107 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 12858.9 / 1.0 \cdot 9806.637 \cdot 0.375 / (2.922 \cdot 195129 \cdot 12.0^2 \cdot 103.923 \cdot 0.3)$
 $= 0.962 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 2482.6 / 1.0 \cdot 9806.637 \cdot 0.375 / (2.922 \cdot 187338 \cdot 12.0^2 \cdot 103.923 \cdot 0.3)$
 $= 0.193 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 12858.9 / 1.0 \cdot 9806.637 \cdot 0.375 / (3.849 \cdot 195129 \cdot 12.0^2 \cdot 103.923 \cdot 0.3)$
 $= 0.730 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 2482.6 / 1.0 \cdot 9806.637 \cdot 0.375 / (3.849 \cdot 187338 \cdot 12.0^2 \cdot 103.923 \cdot 0.3)$
 $= 0.146 \text{ (should be } \leq 1)$

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	26.19	207.00	135.67	207.00
Radial Flange	1.40	138.00	7.23	138.00
Tangential Flange	16.26	138.00	84.19	138.00
Maximum Average	21.22	138.00	109.93	138.00
Bolting	26.44	172.00	80.60	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 92.812 mm

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Flg Calc [Int P]: New Flange

Flng: 11

6:22pm

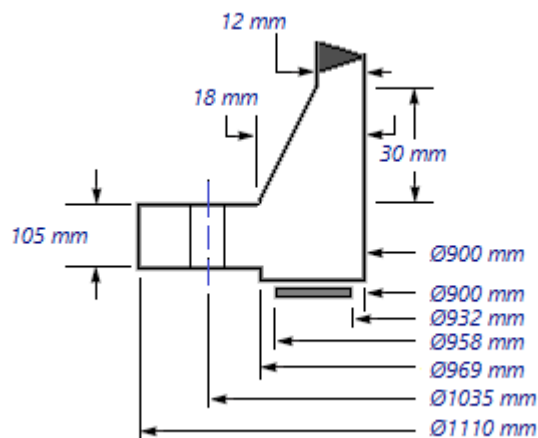
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Flange Input Data Values

Description: New Flange :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	520.00	KPa.
Design Temperature		130	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	900.000	mm
Flange Outside Diameter	A	1110.000	mm
Flange Thickness	t	105.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	g1	18.0000	mm
Length of Hub	h	30.0000	mm
Flange Material	SA-965 F304		
Flange Material UNS number	S30400		
Flange Allowable Stress At Temperature	Sfo	106.20	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1035.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	UNC Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	969.000	mm
Flange Face Inside Diameter	Fid	900.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	958.000	mm
Gasket Inside Diameter	Gi	932.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (520.0 \cdot (900.0/2 + 0.0)) / (133.2 \cdot 1.0 - 0.6 \cdot 520.0) + Ca$$

$$= 1.7610 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (133.2 \cdot 1.0 \cdot 12.0) / (450.0 + 0.6 \cdot 12.0)$$

$$= 3495.847 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (450.0 + 0.6 \cdot 12.0)$$

$$= 3621.771 \text{ KPa.}$$

Corroded Flange ID,	Bcor = B+2*ci	900.000 mm
Corroded Large Hub,	g1Cor = g1-ci	18.000 mm
Corroded Small Hub,	g0Cor = go-ci	12.000 mm
Code R Dimension,	R = (C-B)/2 - g1	49.500 mm
Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm
Basic Gasket Width,	bo = N/2	6.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm
Gasket Reaction Diameter,	G = Go - 2 * b	945.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 945.1509^2 \cdot 520.0$$

$$= 37204.211 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 945.1509 \cdot 3.0 \cdot 520.0$$

$$= 6069.407 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 900.0^2 \cdot 520.0$$

$$= 33734.543 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 37204 - 33735$$

$$= 3469.668 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(37204 + 6069 + 0, 0)$$

$$= 43273.617 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 945.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 134122.203 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((43274 + 0 + 4 \cdot 0 / 945.15) / 172, 134122 / 172)$$

$$= 76.473 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$



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$$= 2 \cdot 34.925 + 6 \cdot 105.0 / (3.0 + 0.5)$$

$$= 249.850 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1035.0 \cdot \sin(\pi / 24)$$

$$= 135.095 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{(Bs / (2a + t))}, 1)$$

$$= \max(\sqrt{(135.095 / (2 \cdot 34.925 + 105.0))}, 1)$$

$$= 1.0000$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	76.473	163.200	
Radial Distance between Hub and Bolts:	47.625	49.500	
Radial Distance between Bolts and Edge:	34.925	37.500	
Circ. Spacing between the Bolts:	77.800	135.095	249.850

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 163.2 \cdot 172.0 / (68.95 \cdot \pi(958.0 + 932.0))$$

$$= 6.856 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 163.1997$$

$$= 286227.44 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 43274 - 37204$$

$$= 6069.41 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1035.0 - 945.1509) / 2$$

$$= 44.9245 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1035.0 - 900.0) / 2 + 44.9245)$$

$$= 56.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1035.0 - 900.0 - 18.0) / 2$$

$$= 58.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	33735.	58.5000	1.0000	1974.
Face Pressure, Mt	3470.	56.2123	1.0000	195.



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Flg Calc [Int P]: New Flange

Flng: 11

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Gasket Load, Mg	6069.	44.9245	1.0000	273.
Gasket Seating, Matm	286227.	44.9245	1.0000	12859.

Total Moment for Operation, Mo 2441. Kg-m.
Total Moment for Gasket seating, Mg 12859. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 103.923 mm
Hub Ratio, h/h0 = HL / H0 0.289
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.500

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.884
Factor V per 4.16.5 0.375
Factor f per 4.16.5 1.207

Stress factor from Table 4.16.4 [K]:

= A / B
= 1110.0/900.0
= 1.233

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.233² • log10(1.233))) / (1.233² - 1))/(1.233 - 1)
= 9.379

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.233²(1 + 8.55246 • log10(1.233)) - 1) / ((1.0472 + 1.9448 • 1.233²) • (1.233 - 1))
= 1.825

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.233²(1 + 8.55246 • log10(1.233)) - 1) / (1.36136(1.233² - 1) • (1.233 - 1))
= 10.306

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.233² + 1) / (1.233² - 1)
= 4.838

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 10.306 • 12.0² • 103.923 / 0.375
= 411736.875 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.884 / 103.923
= 0.009 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (105.0 • 0.009 + 1) / 1.825 + 105.0³/411736.875
= 3.849

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.2074 • 2441 / 900.0) / (3.8486 • 18.0²)
= 25.76 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
= (1.2074 • 12859 / 900.0) / (3.8486 • 18.0²)



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 Flg Calc [Int P]: New Flange Flng: 11 6:22pm Sep 8,2026

= 135.67 N./mm²

Radial Flange Stress, Operating [SRo]:

= ($\beta \cdot Mop / Bcor$) / ($L \cdot t^2$)
 = (2.1906•2441/900.0)/(3.8486•105.0²)
 = 1.37 N./mm²

Radial Flange Stress, Seating [SRa]:

= ($\beta \cdot Matm/Bcor$) / ($L \cdot t^2$)
 = (2.1906•12859/900.0)/(3.8486•105.0²)
 = 7.23 N./mm²

Tangential Flange Stress, Operating [STo]:

= ($Y \cdot Mo / (t^2 \cdot Bcor)$) - $Z \cdot SRo$
 = (9.3787•2441/(105.0²•900.0))-4.838•1
 = 15.98 N./mm²

Tangential Flange Stress, Seating [STa]:

= ($y \cdot Matm / (t^2 \cdot Bcor)$) - $Z \cdot SRa$
 = (9.3787•12859/(105.0²•900.0))-4.838•7
 = 84.19 N./mm²

Average Flange Stress, Operating [SAo]:

= ($SHo + \max(SRo, STo)$) / 2
 = (26+max(1,16))/2
 = 20.87 N./mm²

Average Flange Stress, Seating [SAa]:

= ($SHa + \max(SRa, STa)$) / 2
 = (136+max(7,84))/2
 = 109.93 N./mm²

Bolt Stress, Operating [BSo]:

= $Wm1 / Ab$
 = 43274/163.1997
 = 26.00 N./mm²

Bolt Stress, Seating [BSa]:

= ($Wm2 / Ab$)
 = (134122/163.1997)
 = 80.60 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	25.76	159.30	135.67	207.00
Radial Flange	1.37	106.20	7.23	138.00
Tangential Flange	15.98	106.20	84.19	138.00
Maximum Average	20.87	106.20	109.93	138.00
Bolting	26.00	172.00	80.60	172.00

Minimum Required Flange Thickness 92.812 mm

Estimated M.A.W.P. (Operating) 2646 KPa.

Estimated Finished Weight of Flange at given Thk. 289.8 kg.

Estimated Unfinished Weight of Forging at given Thk 359.3 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

	Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
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 Flg Calc [Int P]: New Flange Flng: 11 6:22pm Sep 8,2026

Longitudinal Hub	Sh	364.80	364.80	1.000	364.80	409.99	0.890
Tangential Flange	St	228.21	273.60	0.834	228.21	307.49	0.742
Tangential Flange	Sto	99.53	182.40	0.546	99.53	205.00	0.486
Combined	Sh/f + St	532.77	547.20	0.974	532.77	614.99	0.866
Combined	Sr + St	247.81	547.20	0.453	247.81	614.99	0.403
Combined	Sh + Sto	464.33	547.20	0.849	464.33	614.99	0.755
Radial Flange	Sr	19.61	364.80	0.054	19.61	409.99	0.048
Maximum Stress Ratio				1.000			0.890

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:
 = max(1; F1num / F1den)
 = max(1; 1.4685/1.226)
 = 1.1978

Hub Stress Correction Factor [f2]:
 = max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.3784/1.158) if h/ho < 0.35, else max (0.25; 0.2673/0.7714)
 = 0.3268

In this case, the hub ratio h/ho, 0.2887 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:



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 Flg Calc [Int P]: New Flange Flng: 11 6:22pm Sep 8,2026

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	466.2 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	6.8000 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	205.0 N./mm ²
Flange Material Operating Yield Stress	Syo	159.6 N./mm ²
Single flange rotation at Sfmax	Thetaamax	0.594 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (958.0^2 - 932.0^2)$$

$$= 385.9 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208.0 \cdot 385.946 / (24 \cdot 6.8)$$

$$= 491.9 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(491.893; 507.5)$$

$$= 491.9 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(491.893; 290.0)$$

$$= 491.9 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(491.893; 466.193)$$

$$= 466.2 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 466.193 \geq 140.0 (385.946 / (24 \cdot 6.8))$$

$$:: 466.2 \geq 331.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 466.193 \geq (97.0 \cdot 385.946 + \pi/4 \cdot 520.0 \cdot 932.0^2) / (0.7 \cdot 6.8 \cdot 24)$$

$$:: 466.2 \geq 358.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 466.193 \leq 275.0 (385.946 / (24 \cdot 6.8))$$

$$:: 466.2 \leq 650.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetaamax)$$

$$:: 466.193 \leq 466.193 (1.0/0.594)$$

$$:: 466.2 \leq 784.8 \text{ N./mm}^2 \text{ Check Passes}$$



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Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 466.193 \cdot 0.2 \cdot 6.8 \cdot 34.925$
 $= 225.8 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 107 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 12858.9 / 1.0 \cdot 9806.637 \cdot 0.375 / (2.922 \cdot 195129 \cdot 12.0^2 \cdot 103.923 \cdot 0.3)$
 $= 0.962 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 2441.2 / 1.0 \cdot 9806.637 \cdot 0.375 / (2.922 \cdot 187338 \cdot 12.0^2 \cdot 103.923 \cdot 0.3)$
 $= 0.190 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 12858.9 / 1.0 \cdot 9806.637 \cdot 0.375 / (3.849 \cdot 195129 \cdot 12.0^2 \cdot 103.923 \cdot 0.3)$
 $= 0.730 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 2441.2 / 1.0 \cdot 9806.637 \cdot 0.375 / (3.849 \cdot 187338 \cdot 12.0^2 \cdot 103.923 \cdot 0.3)$
 $= 0.144 \text{ (should be } \leq 1)$

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	25.76	207.00	135.67	207.00
Radial Flange	1.37	138.00	7.23	138.00
Tangential Flange	15.98	138.00	84.19	138.00
Maximum Average	20.87	138.00	109.93	138.00
Bolting	26.00	172.00	80.60	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 92.812 mm

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Flg Calc [Int P]: New Flange

Fling: 12

6:22pm

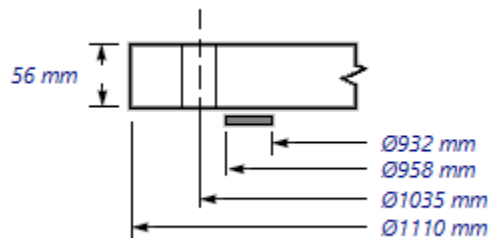
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Flange Input Data Values

Description: New Flange :

RHS Blind Flg

Flange Type		Blind	
Design Pressure	P	520.00	KPa.
Design Temperature		130	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1110.000	mm
Flange Thickness	t	56.0000	mm
Flange Material		SA-965 F304	
Flange Material UNS number		S30400	
Flange Allowable Stress At Temperature	Sfo	125.20	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	3251.548	mm
Diameter of Bolt Circle	C	1035.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	UNC	Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	969.000	mm
Flange Face Inside Diameter	Fid	900.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	958.000	mm
Gasket Inside Diameter	Gi	932.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width,	$b_o = N/2$	6.500 mm
Effective Gasket Width,	$b = C_b \cdot \text{sqrt}(b_o)$	6.425 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	945.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:



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$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 945.1509^2 \cdot 520.0$$

$$= 37204.211 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 945.1509 \cdot 3.0 \cdot 520.0$$

$$= 6069.407 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(37204 + 6069 + 0, 0)$$

$$= 43273.617 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 945.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 134122.203 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((43274 + 0 + 4 \cdot 0 / 945.15) / 172, 134122 / 172)$$

$$= 76.473 \text{ cm}^2$$

Bolting Information for UNC Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	76.473	163.200	
Radial Distance between Bolts and Edge:	34.925	37.500	
Circ. Spacing between the Bolts:	77.800	135.095	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 163.2 \cdot 172.0 / (68.95 \cdot \pi (958.0 + 932.0))$$

$$= 6.856 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 163.1997$$

$$= 286227.44 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 43273.62 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1035.0 - 945.1509) / 2$$

$$= 44.9245 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 43274 \cdot 44.9245 \cdot 1.0 / (56.0^2 \cdot 945.1509) + 0.3 \cdot 1.0 \cdot 520.0 \cdot 945.1509^2 / 56.0^2$$

$$= 56.66 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 286227 \cdot 44.9245 \cdot 1.0 / (56.0^2 \cdot 945.1509)$$

$$= 80.84 \text{ N./mm}^2$$



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Flg Calc [Int P]: New Flange Flng: 12 6:22pm Sep 8,2026

Bolt Stress, Operating [BS_o]:

$$\begin{aligned} &= Wm1 / Ab \\ &= 43274/163.1997 \\ &= 26.00 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Seating [BS_a]:

$$\begin{aligned} &= (Wm2 / Ab) \\ &= (134122/163.1997) \\ &= 80.60 \text{ N./mm}^2 \end{aligned}$$

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Tangential Flange	56.66	125.20	80.84	138.00
Bolting	26.00	172.00	80.60	172.00

Reqd. Blind Flange Thickness at Center 42.860 mm
Reqd. Blind Flange Thickness at Gasket 42.860 mm

Estimated M.A.W.P. (Operating) 1149 KPa.
Estimated Finished Weight of Flange at given Thk. 435.0 kg.
Estimated Unfinished Weight of Forging at given Thk 435.0 kg.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Tangential Flange	56.66	138.00	80.84	138.00
Bolting	26.00	172.00	80.60	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 42.875 mm

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 Internal Pressure Calculations: Step: 7 6:22pm Sep 8,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Blind Flange		520	56	...	1110	106.2	1
LHS CH Flang		528.82	105	...	900	106.2	2
Ch Shell		528.82	12	...	900	133.2	3
main shell		1308.8	12	...	900	137	4
RHS Ch shell		528.82	12	...	900	133.2	5
RHS Ch Flg		520	105	...	900	106.2	6
RHS Blind Flg		520	56	...	1110	125.2	7

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Blind Flange		520	974.621	1266.45	56	42.86	1
LHS CH Flang		520	2637.27	3438.36	105	92.8116	2
Ch Shell		520	3487.02	3621.77	12	1.79094	3
main shell		1300	3586.71	3621.77	12	4.32416	4
RHS Ch shell		520	3487.02	3621.77	12	1.79094	5
RHS Ch Flg		520	2646.09	3438.36	105	92.8116	6
RHS Blind Flg		520	1148.98	1266.45	56	42.86	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 3098 KPa.
 Shell Side MAPnc = 3622 KPa.
 Channel Side MAWP = 975 KPa.
 Channel Side MAPnc = 1266 KPa.

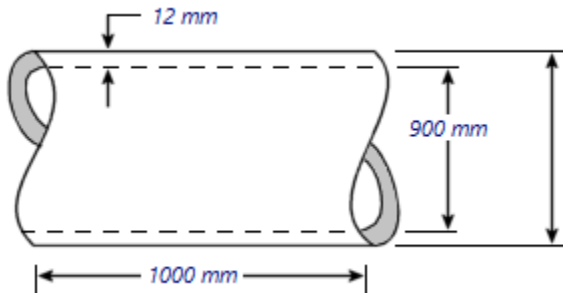
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	528.822 KPa.	Int. Design Temperature		130 °C
Material		SA-240 304	Ext. Design Temperature		130 °C
Length		1000.000 mm	External Pressure Chart		HA-1
			UNS Number		S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S	133.2 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	900.000 mm	Actual Stress		20.1 N./mm ²
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		29028.3 cm ²	SG of Contents		1.000
Element Volume		636286.6 cm ³	Weight of Contents		635.9 Kg
Empty Weight		276.0 Kg	Operating Weight		911.9 Kg
			Comp. All. Ope. Stress		88.8 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (528.822 \cdot 450.0) / (133.2 \cdot 1.0 - 0.6 \cdot 528.822) \\
 &= 1.7909 + 0.0000 = 1.7909 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 8.822 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (133.2 \cdot 1.0 \cdot 12.0) / (450.0 + 0.6 \cdot 12.0) \\
 &= 3496 - 9 = 3487 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (450.0 + 0.6 \cdot 12.0) \\
 &= 3622 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (528.822 \cdot (450.0 + 0.6 \cdot 12.0)) / (1.0 \cdot 12.0) \\
 &= 20.149 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 109-C-Solis SS-Original file Page 44 of 197
 Element: Ch Shell Step: 8 6:22pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 1.316 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: Ch Shell

Cylindrical Shell From 30 to Tsht Ext. Chart: HA-1 at 130 °C

Elastic Modulus from Chart: HA-1 at 130 °C: 185038288 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	924.00	999.97	77.00	1.0822	0.0018499	66.18

$$MAEP = (4*B)/(3*(Do/t)) = (4*66.1793)/(3*77.0) = 1145.8961 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.164	924.00	999.97	292.05	1.0822	0.0002449	22.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.6552)/(3*292.0476) = 103.4256 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	924.00	27608.78	77.00	29.8796	0.0001890	17.49

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.4906)/(3*77.0) = 302.8491 \text{ KPa.}$$

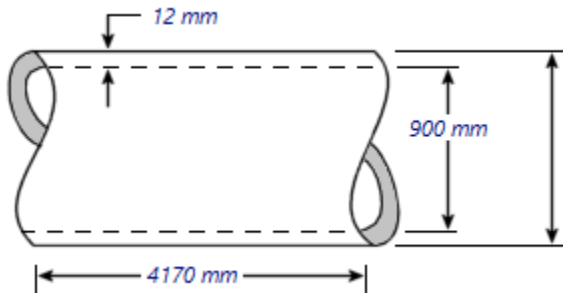
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 FileName : 109-C-Solis SS-Original file Page 45 of 197
 Element: main shell Step: 9 6:22pm Sep 8,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: main shell

Cylindrical Shell Design Information:			
Design Pressure P	1308.822 KPa.	Int. Design Temperature	100 °C
Material	SA-240 304	Ext. Design Temperature	100 °C
Length	4170.000 mm	External Pressure Chart	HA-1
		UNS Number	S30400
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	137.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	900.000 mm	Actual Stress	49.9 N./mm ²
Spec. Min. Thk t,e, or sa	12.0000 mm	Spec. Nominal Thk.	12.000 mm
Surface Area	121048.1 cm ²	SG of Contents	1.000
Element Volume	1452668.5 cm ³	Weight of Contents	1451.8 Kg
Empty Weight	1150.9 Kg	Operating Weight	2602.7 Kg
		Comp. All. Ope. Stress	93.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1308.822 \cdot 450.0) / (137.0 \cdot 1.0 - 0.6 \cdot 1308.822)$$

$$= 4.3242 + 0.0000 = 4.3242 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 8.822 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (137.0 \cdot 1.0 \cdot 12.0) / (450.0 + 0.6 \cdot 12.0)$$

$$= 3596 - 9 = 3587 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (450.0 + 0.6 \cdot 12.0)$$

$$= 3622 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1308.822 \cdot (450.0 + 0.6 \cdot 12.0)) / (1.0 \cdot 12.0)$$

$$= 49.869 \text{ N./mm}^2$$



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 FileName : 109-C-Solis SS-Original file Page 46 of 197
 Element: main shell Step: 9 6:22pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 1.316 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: main shell

Cylindrical Shell From 40 to Tsht Ext. Chart: HA-1 at 100 °C

Elastic Modulus from Chart: HA-1 at 100 °C: 187644240 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	924.00	4170.00	77.00	4.5130	0.0003951	37.08

$$MAEP = (4*B)/(3*(Do/t)) = (4*37.0758)/(3*77.0) = 641.9676 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.714	924.00	4170.00	161.70	4.5130	0.0001337	12.54

$$MAEP = (4*B)/(3*(Do/t)) = (4*12.5434)/(3*161.7018) = 103.4226 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	924.00	94597.84	77.00	50.0000	0.0001889	17.72

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.7232)/(3*77.0) = 306.8769 \text{ KPa.}$$

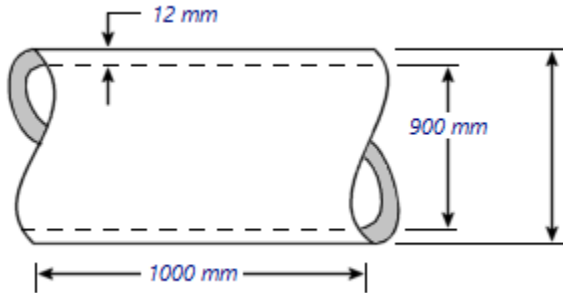
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 FileName : 109-C-Solis SS-Original file Page 47 of 197
 Element: RHS Ch shell Step: 10 6:22pm Sep 8,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: RHS Ch shell

Cylindrical Shell Design Information:			
Design Pressure P	528.822 KPa.	Int. Design Temperature	130 °C
Material	SA-240 304	Ext. Design Temperature	130 °C
Length	1000.000 mm	External Pressure Chart	HA-1
		UNS Number	S30400
Int. Corr. All. c	0.0000 mm	Allowable Stress (ope) S	133.2 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	900.000 mm	Actual Stress	20.1 N./mm ²
Spec. Min. Thk t,e, or sa	12.0000 mm	Spec. Nominal Thk.	12.000 mm
Surface Area	29028.3 cm ²	SG of Contents	1.000
Element Volume	636286.6 cm ³	Weight of Contents	635.9 Kg
Empty Weight	276.0 Kg	Operating Weight	911.9 Kg
		Comp. All. Ope. Stress	93.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (528.822 \cdot 450.0) / (133.2 \cdot 1.0 - 0.6 \cdot 528.822) \\
 &= 1.7909 + 0.0000 = 1.7909 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 8.822 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (133.2 \cdot 1.0 \cdot 12.0) / (450.0 + 0.6 \cdot 12.0) \\
 &= 3496 - 9 = 3487 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (450.0 + 0.6 \cdot 12.0) \\
 &= 3622 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (528.822 \cdot (450.0 + 0.6 \cdot 12.0)) / (1.0 \cdot 12.0) \\
 &= 20.149 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 109-C-Solis SS-Original file Page 48 of 197
 Element: RHS Ch shell Step: 10 6:22pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 1.316 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: RHS Ch shell

Cylindrical Shell From 50 to 60 Ext. Chart: HA-1 at 130 °C

Elastic Modulus from Chart: HA-1 at 130 °C: 185038288 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	924.00	1000.03	77.00	1.0823	0.0018498	66.18

$$MAEP = (4*B)/(3*(Do/t)) = (4*66.1784)/(3*77.0) = 1145.8792 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.164	924.00	1000.03	292.04	1.0823	0.0002449	22.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*22.6547)/(3*292.0405) = 103.4256 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
12.000	924.00	27609.66	77.00	29.8806	0.0001890	17.49

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.4905)/(3*77.0) = 302.8488 \text{ KPa.}$$

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4027.787 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1690.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	4697.422 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	3408.127 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1859.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1859.000 KPa.
App 27-4	= M.A.W.P.	3098.297 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1267.007 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1635.504 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1072.083 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	974.621 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 1300.0 • 1.0
 = 1690.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 520.0 • 1.0
 = 676.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch Shell	26.1	184.5	0.141	684.82
main shell	64.7	184.5	0.351	1698.82
RHS Ch shell	26.1	184.5	0.141	684.82

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	138.00	133.20	1.036
N1-Tube Inlet	Pad	138.00	138.00	1.000
N3-Shell inlet	Nozzle	138.00	133.20	1.036
N3-Shell inlet	Pad	138.00	138.00	1.000
N4-Shell outlet	Nozzle	138.00	130.00	1.062
N4-Shell outlet	Pad	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	138.00	133.20	1.036
N2-Tube Outlet	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



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Shop Pressure Test Results:

Step: 11

6:22pm

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Description	Ambient	Operating	Ratio
Blind Flange	138.00	106.20	1.299
LHS CH Flang	138.00	106.20	1.299
Ch Shell	138.00	133.20	1.036
main shell	138.00	137.00	1.007
RHS Ch shell	138.00	133.20	1.036
RHS Ch Flg	138.00	106.20	1.299
RHS Blind Flg	138.00	125.20	1.102

Minimum

1.007

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	138.00	133.20	1.036
Tubesheet Material	138.00	106.20	1.299

Minimum

1.036

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	8.46	184.50	0.046
N3-Shell inlet	20.98	184.50	0.114
N4-Shell outlet	20.98	184.50	0.114
N2-Tube Outlet	8.46	184.50	0.046

Elements Suitable for Test Pressure.

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 Field Pressure Test Results: Step: 12 6:22pm Sep 8,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4027.787 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1690.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	4699.481 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	3408.127 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1859.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1859.000 KPa.
App 27-4	= M.A.W.P.	3098.297 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1267.007 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1635.503 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1072.083 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	974.621 KPa.

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch Shell	48.6	184.5	0.263	1275.83
main shell	153.8	184.5	0.834	4036.61
RHS Ch shell	48.6	184.5	0.263	1275.83

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	138.00	133.20	1.036
N1-Tube Inlet	Pad	138.00	138.00	1.000
N3-Shell inlet	Nozzle	138.00	133.20	1.036
N3-Shell inlet	Pad	138.00	138.00	1.000
N4-Shell outlet	Nozzle	138.00	130.00	1.062
N4-Shell outlet	Pad	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	138.00	133.20	1.036
N2-Tube Outlet	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Blind Flange	138.00	106.20	1.299
LHS CH Flang	138.00	106.20	1.299
Ch Shell	138.00	133.20	1.036
main shell	138.00	137.00	1.007
RHS Ch shell	138.00	133.20	1.036
RHS Ch Flg	138.00	106.20	1.299
RHS Blind Flg	138.00	125.20	1.102



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Field Pressure Test Results: Step: 12 6:22pm Sep 8, 2026

Minimum 1.007

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	138.00	133.20	1.036
Tubesheet Material	138.00	106.20	1.299

Minimum 1.036

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	15.76	184.50	0.085
N3-Shell inlet	49.86	184.50	0.270
N4-Shell outlet	49.86	184.50	0.270
N2-Tube Outlet	15.76	184.50	0.085

Elements Suitable for Test Pressure.

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 ASME TS Calc: Pressure Test

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 Case: 1 6:22p Sep 8,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main shell

Shell Test Pressure		4027.79 KPa.
Shell Thickness	ts	12.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	900.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-240 304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		S30400
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		1267.01 KPa.
Channel Thickness	tc	12.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	900.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-240 304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		S30400
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1010
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	4250.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	4170.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		138.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	424.264 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	5397.3 cm ²



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 FileName : 109-C-Solis SS-Original file Page 54 of 197
 ASME TS Calc: Pressure Test Case: 1 6:22p Sep 8,2026

Perimeter of Tube Layout Cp 2735.9 mm

Tubesheet Data:

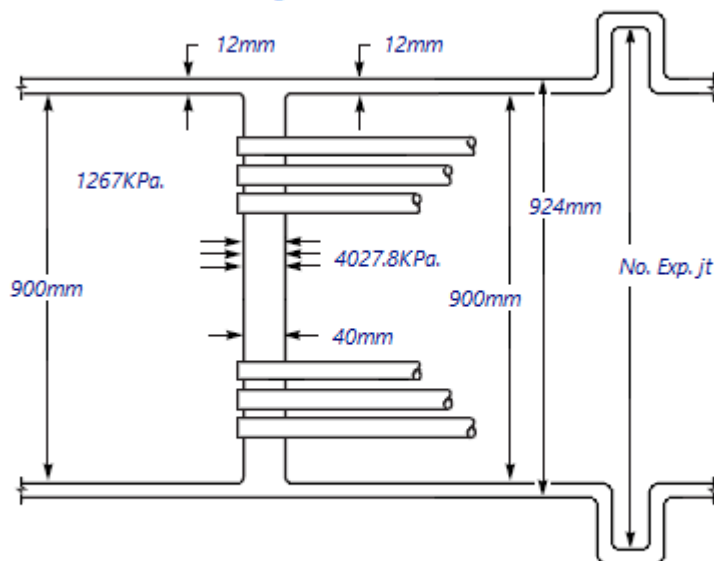
Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-965 F304
Tubesheet Material UNS Number		S30400
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	40.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	924.000 mm
Area of the Untubed Lanes	AL	0.0 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	280.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Ps _{d,min}	-103.4190 KPa.
Channel Side Min. Design Pressure	P _t _{d,min}	-103.4190 KPa.
Mean Shell Metal Temp. along Shell len.	T _{sm}	21.11 °C
Mean Tube Metal Temp. along Tube length	T _{tm}	21.11 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No

Tubesheet Integral w/Shell and Channel



Configuration a:



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Elasticity/Expansion Material Properties:

Shell - TE-1 Austenitic Group 3 Steels
 Shell - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Channel - TE-1 Austenitic Group 3 Steels
 Channel - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels
 Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TE-1 Austenitic Group 3 Steels
 TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1370.43 \cdot (19.05/2 - 0)) / (138.0 \cdot 1.0 + 0.4 \cdot 1370.43)$
 $= 0.0942 + 0.0000 = 0.0942 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-1 at 21.11 °C
 Elastic Modulus for Material 193048800.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	19.05	4170.00	11.55	50.0000	0.0082522	98.41
EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 98.4122) / (3 \cdot 11.5455) = 11364.5254 \text{ KPa.}$						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7739	19.05	4170.00	24.62	50.0000	0.0018153	76.28
EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 76.2834) / (3 \cdot 24.616) = 4131.667 \text{ KPa.}$						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all. 0.7739 mm
 Allowable Internal Pressure at Corroded thickness 25683.32 KPa.



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Required Internal Design Pressure	1370.43 KPa.
Allowable External Pressure at Corroded thickness	11364.53 KPa.
Required External Design Pressure	4131.21 KPa.
Required Thickness due to Shell Side pressure	0.7739 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

= l_{tx} / h (modified for corrosion if present)

= 0/40.0

= 0.0000 (must be $0 \leq p \leq 1$)

Effective tube hole diameter [d*]:

= $\max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$

= $\max(19.05 - 2 \cdot 1.65 (0.19512E+09/.19512E+09) \cdot$

$(137/137) \cdot (0); 19.05 - 2 \cdot 1.65)$

= 19.0500 mm

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot r_o + dt$

= $2 \cdot 424.264 + 19.05$

= 867.578 mm

Determine the basic ligament efficiency for shear [μ]:

= $(p - dt) / p$

= $(25.4 - 19.05) / 25.4$

= 0.250

Equivalent outer tube limit radius [ao]:

= $Do / 2$

= $867.5781/2$

= 433.7890 mm

Effective tube pitch [p*]:

= $p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$

= $25.4 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 867.578 \cdot 25.4) / (\pi \cdot 867.578^2)) }$

= 25.4000 mm

Effective ligament efficiency for bending [μ*]:

= $(p* - d*) / p*$

= $(25.4 - 19.05) / 25.4$

= 0.2500

Ratio [ps]:

= as / ao

= $450.0/433.789$

= 1.0374

Ratio [pc]:

= ac / ao

= $450.0/433.789$

= 1.0374

Compute parameter [xt]:

= $1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$

= $1 - 1010 ((19.05 - 2 \cdot 1.65) / (2 \cdot 433.789))^2$



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= 0.6671

Determine parameter [xs]:

= $1 - Nt(dt / (2 \cdot ao))^2$
 = $1 - 1010 (19.05 / (2 \cdot 433.789))^2$
 = 0.5130

Determine the value [h'g]:

= $\max((hg - CATC); 0)$ (For pressure only cases)
 = $\max((0 - 0); 0)$
 = 0.000 mm

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

= $\pi \cdot ts(Ds + ts)Es/L$
 = $\pi \cdot 12.0 (900.0 + 12.0) \cdot 1.9512E+09 / 4170.0$
 = 164052.1719 Kgf/mm

Axial Tube Stiffness [Kt]:

= $\pi \cdot tt(Dt - tt)Et/L$
 = $\pi \cdot 1.65 (19.05 - 1.65) \cdot 1.9512E+09 / 4170.0$
 = 430.3671 Kgf/mm

Stiffness Factor [Ks,t]:

= $Ks / (Nt \cdot Kt)$
 = $164052.172 / (1010 \cdot 430.367)$
 = 0.37742

Rigidity Ratio [J]:

= $1 / (1 + Ks/Kt)$
 = $1 / (1 + 164052.172 / 0)$
 = 1.0000 (= 1 if No Exp. Jt.)

Shell coefficient [β_s]:

= $((12(1 - vs^2))^{1/4}) / ((Ds + ts)ts)$
 = $((12(1 - 0.31^2))^{1/4}) / ((900.0 + 12.0) 12.0)$
 = 0.0173 1/mm

Shell coefficient [ks]:

= $\beta_s \cdot Es \cdot ts^3 / (6(1 - vs^2))$
 = $0.017 \cdot 0.19512E+09 \cdot 12.0^3 / (6(1 - 0.31^2))$
 = 1078462464.0000 KPa.·mm²

Shell coefficient [λ_s]:

= $6(Ds + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$
 = $6(900.0 + 12.0) \cdot 0.10785E+1 / 40.0^3 \cdot (1 + 40.0 \cdot 0.017 + (40.0 \cdot 0.017)^2 / 2)$
 = 178391248.0000 KPa.

Shell coefficient [δ_s]:

= $Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot vs / (2(Ds + ts)))$
 = $900.0^2 / (4 \cdot 0.19512E+09 \cdot 12.0) \cdot (1 - 900.0 \cdot 0.31 / (2(900.0 + 12.0)))$
 = 0.73253E-01 mm/N./mm²

Channel coefficient [β_c]:

= $((12(1 - vc^2))^{1/4}) / ((Dc + tc) tc)$
 = $((12(1 - 0.31^2))^{1/4}) / ((900.0 + 12.0) 12.0)$
 = 0.0173 1/mm

Channel coefficient [kc]:

= $\beta_c \cdot Ec \cdot tc^3 / (6(1 - vc^2))$
 = $0.017 \cdot 0.19512E+09 \cdot 12.0^3 / (6(1 - 0.31^2))$
 = 1078462464.0000 KPa.·mm²

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:



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$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned} &= 6(Dc + tc/\cos(\theta)) \cdot kc/h^3 \cdot (CmTheta + \\ &\quad h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta) \\ &= 6(900.0 + 12.0/\cos(0)) \cdot 0.10785E+1/40.0^3 \cdot (1.0 + \\ &\quad 40.0 \cdot 0.017 \cdot (1.0 + 1.0)/2 + (40.0 \cdot 0.017)^2/2 \cdot 1.0) \\ &= 178391248.0 \text{ KPa.} \end{aligned}$$

Channel coefficient [δ_c]:

$$\begin{aligned} &= Dc^2/(4 \cdot Ec \cdot Tc) \cdot (1 - Dc \cdot vc/(2(Dc + tc))) \\ &= 900.0^2/(4 \cdot 0.19512E+09 \cdot 12.0) \cdot (1 - 900.0 \cdot 0.31/(2(900.0 + 12.0))) \\ &= 0.73253E-01 \text{ mm/N./mm}^2 \end{aligned}$$

Channel coefficient [ζ_{tac}]:

$$\zeta_{tac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 1.574803 ; \mu^* = 0.250000 ; E^* = 45058828 \text{ KPa.}$$

$$E^*/E = 0.230932 ; \nu^* = 0.378304$$

Tube Bundle Stiffness Factor [X_a]:

$$\begin{aligned} &= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2)/(E^* \cdot L \cdot H^3))^{1/4} \\ &= ((24(1 - 0.378^2)1010 \cdot 0.19512E+09 \cdot 1.65 \\ &\quad (19.05 - 1.65)433.789^2)/(45058828 \cdot 4170.0 \cdot 40.0^3))^{1/4} \\ &= 6.5322 \end{aligned}$$

Values from Table 4.18.3

$$Z_d = 0.005359 ; Z_v = 0.023593 ; Z_m = 0.220402$$

$$Z_a = 0.160579E+03 ; Z_w = 0.023593$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / Do \\ &= 924.0/867.5781 \\ &= 1.0650 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*)/ E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.38)/45058828 \cdot (0.17839E+09 + 0.17839E+09 + \\ &\quad 0.19512E+09 \cdot \ln(1.07)) \\ &= 5.0923 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu^*)F \\ &= (1 + 0.3783)5.0923 \\ &= 7.0187 \end{aligned}$$

Compute parameter [Q_1]:

$$\begin{aligned} &= (\rho_s - 1 - \phi \cdot Z_v)/(1 + \phi \cdot Z_m) \\ &= (1.0374 - 1 - 7.0187 \cdot 0.0236)/(1 + 7.0187 \cdot 0.2204) \\ &= -0.050344 \end{aligned}$$

Compute Parameter [Q_{z1}]:

$$\begin{aligned} &= (Z_d + Q_1 \cdot Z_w)/2 \cdot X_a^4 \\ &= (0.00536 + -0.05034 \cdot 0.02359)/2 \cdot 6.53223^4 \\ &= 3.7973 \end{aligned}$$

Compute parameter [Q_{z2}]:

$$\begin{aligned} &= (Z_v + Q_1 \cdot Z_m)/2 \cdot X_a^4 \\ &= (0.02359 + -0.05034 \cdot 0.2204)/2 \cdot 6.53223^4 \\ &= 11.3770 \end{aligned}$$



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Compute parameter [U]:

$$= (Zw + (ps - 1)Zm)Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.0236 + (1.0374 - 1)0.2204)6.53223^4 / (1 + 7.0187 \cdot 0.2204)$$

$$= 22.7541$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [v]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= ps \cdot ks \cdot \beta s \cdot \delta S (1 + h \cdot \beta s)$$

$$= 1.0374 \cdot 0.10785E+1 \cdot 0.0173 \cdot 0.073253 (1 + 40.0 \cdot 0.0173)$$

$$= 2408.3315 \text{ mm}^2$$

Calculate parameter [ωs]:

$$= ao^2 (ps^2 - 1) \cdot (ps - 1) / 4 - \omega s$$

$$= 433.789^2 (1.037^2 - 1) \cdot (1.037 - 1) / 4 - 2408.332$$

$$= -2274.4780 \text{ mm}^2$$

Parameter [ωc]:

$$= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.037 \cdot 0.10785E+1 [0.073253 (1.0 + 0.017 \cdot 1.0 \cdot 40.0) - 0 (1.0/40.0 + 1.0 \cdot 0.017/2)]$$

$$= 2408.3315 \text{ mm}^2$$

Parameter [ωc]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega c$$

$$= 433.78903^2 [(1.03737^2 + 1) \cdot (1.03737 - 1) / 4 - (1.03737 - 1) / 2] - 2408.33154$$

$$= -2274.4778 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs)vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 4027.787 \{ 0.513 + 2(1 - 0.513)0.31 + [2/0.377 \cdot (900.0/867.578)^2] 0.31 - [(1.037^2 - 1)/(1.0 \cdot 0.377)] - [(1 - 1.0)/(2 \cdot 1.0 \cdot 0.377)] [(0^2 - (900.0)^2)/867.578^2] \}$$

$$= 9590.3252 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt)vt + 1/(J \cdot Kst)] Pt$$

$$= [0.667 + 2(1 - 0.667)0.31 + 1/(1.0 \cdot 0.377)] 1267.007$$

$$= 4463.7925 \text{ KPa.}$$

Pressure [Pv]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 1010 \cdot 430.367 \cdot 0 / (3.142 \cdot 433.789^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:

$$= -yb \cdot U \cdot W^* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 22.754 \cdot 0 / (2 \cdot \pi \cdot 433.789^2)$$

$$= -0.0000 \text{ KPa.}$$



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Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt)$$

$$= -(22.754/433.789^2) \cdot (-2274.478 \cdot 4027.787 - -2274.478 \cdot 1267.007)$$

$$= 759.3044 \text{ KPa.}$$

Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)$$

$$= (22.754/433.789^2) \cdot (2408.3315 \cdot 0 - 2408.3315 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (Ps - Pt + Py + Pw + Prim)$$

$$= 0.1000E+01 \cdot 0.377 / (1 + 1.0 \cdot 0.377 (3.797 + (1.037 - 1)11.377)) \cdot (9590.325 - 4463.792 + 0 + 0 + 0 + 759.304)$$

$$= 856.4887 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + W \cdot yb / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-2274.478 \cdot 4027.787 - -2274.478 \cdot 1267.007) - (2408.332 \cdot 0 - 2408.332 \cdot 0)) \cdot 0 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 7.01874 \cdot 0.2204)$$

$$= -251.4147 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.05 + 2 \cdot -251.415 / (856.489 \cdot 433.789^2)$$

$$= -0.08094$$

Fm Value from Table 4.18.3 = 0.040469

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.0405 / 0.25) \cdot (2 \cdot 433.789 / (40.0 - 0))^2 \cdot 856.49$$

$$= 97.8412 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 205.0 = 292.12 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.0276 / 0.25) \cdot (2 \cdot 433.789 / (17.034 - 0))^2 \cdot 679.16$$

$$= 292.1086 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 17.0337 + 0 + 0$$

$$= 17.0337 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy)$$

$$= (2 \cdot 0.25 \cdot 40.0 / 433.789) \cdot 155.797$$

$$= 7.1831 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 71.8649 \text{ N./mm}^2$$



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Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :
 = max(HreqB; HreqS)
 = max(17.0337 ; 3.81)
 = 17.0337 mm

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:
 Ftmin = -0.2594, Ftmax = 3.1402

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:
 = ((Ps • xs - Pt • xt) - Pe • Ftmin) / (Xt - Xs)
 = ((4027.79 • 0.513 - 1267.01 • 0.6671) - (856.489) • -0.259) /
 (0.6671 - 0.513)
 = 9.3668 N./mm²

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:
 = ((Ps • xs - Pt • xt) - Pe • Ftmax) / (Xt - Xs)
 = ((4027.79 • 0.513 - 1267.01 • 0.6671) - (856.489) • 3.14) /
 (0.6671 - 0.513)
 = -9.5294 N./mm²

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
 = max(| $\sigma_{t,1}$ | ; | $\sigma_{t,2}$ |) = 9.529 N./mm²

Allowable Tube Stress [$\sigma_{t,A}$]:
 = $\beta_T \cdot S_{y,t} = 194.7465$ N./mm²

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 = max((3.25 - 0.25(Zd + Q3 • Zw)Xa⁴); 1.25)
 = max((3.25 - 0.25(0.005 + -0.081 • 0.024)6.532⁴); 1.25)
 = 1.6857 (Should be ≤ 2)

Determine the factor [rt]:
 = ($\sqrt{(dt^2 + (dt - 2 \cdot tt)^2)}$) / 4
 = ($\sqrt{(19.05^2 + (19.05 - 2 \cdot 1.65)^2)}$) / 4
 = 6.1794 mm

Determine the factor [Ct]:
 = $\sqrt{(2 \cdot \pi^2 \cdot Et / Syt)}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.19512E+09 / 204)}$
 = 152.2161

Determine the factor [Ft]:
 = k • L/r
 = 1.0 • 280.0 / 6.179
 = 45.3117

Buckling Allowable Stress [Stb]:
 = (Sy,t / Fs) • (1 - Ft / (2 • Ct))
 = (204 / 1.69) • (1 - 45.312 / (2 • 152.216))
 = 194.746 N./mm² (Never greater than Sot)

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
 = $\sigma_{t,max} \cdot \text{Tube Area}$
 = 9.53 • 0.902
 = 87.64 Kg

Tube Metal Area [At]:



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$$\begin{aligned}
 &= \pi (d_o - t) t \\
 &= \pi (19.05 - 1.65) 1.65 \\
 &= 0.902 \text{ cm}^2
 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned}
 &= \pi \cdot t (d_o - t) S \\
 &= \pi \cdot 1.65 (19.05 - 1.65) 194.746 \\
 &= 1791.117 \text{ Kg}
 \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned}
 &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\
 &= 0.85 \cdot \pi \cdot 1.65 (19.05 + 0.67 \cdot 1.65) 194.75 \\
 &= 1763.5480 \text{ Kg}
 \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned}
 &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\
 &= 194.746 / 194.746 \\
 &= 1.000
 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned}
 &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\
 &= 0.55 \cdot \pi \cdot 1.65 (19.05 + 0.67 \cdot 1.65) 194.75 \\
 &= 1141.1193 \text{ Kg}
 \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned}
 &= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right) \\
 &= 2 \left(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 1.65 \cdot (19.05 - 1.65) 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right) \\
 &= 2.075 \text{ mm}
 \end{aligned}$$

Minimum Required Fillet Weld Leg [afr]: 1.0374 mm
 Minimum Required Groove Weld Leg [agr]: 1.0374 mm

Tube-Tubesheet Jt allowable, 1791.12 is $\geq$ tube strength 1791.12 Kg

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 205.0 = 292.12 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s]:

$$\begin{aligned}
 &= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s) \\
 &= 433.789^2 / ((900.0 + 12.0) 12.0) \cdot [856.49 + (1.037^2 - 1) \\
 &\quad (4027.79 - 1267.01)] + 450.0^2 \cdot 1267.01 / ((900.0 + 12.0) 12.0) \\
 &= 41.7870 \text{ N./mm}^2
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 12.0 mm, the minimum Shell length adjacent to the tubesheet should be 187.061 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	97.8 <=	292.1 N./mm ²	Pass



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Tubesheet Shear Stress	16.9	<=	155.8 N./mm ²	Pass
Maximum Tube Stress	9.5	<=	194.7 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-9.5	<=	-194.7 N./mm ²	Pass
Maximum Force on any one Tube	87.6	<=	1791.1 Kgf	Pass
Axial Membrane Stress in Shell	41.8	<=	292.1 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	17.034	40.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.037	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.037	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Shear	Allwd	Case Type	Pass/ Fail
D1uc	13.990 ...	71	292	15	156	Ps+Pt-Th	D1	Ok
D2uc	27.390 ...	170	292	32	156	Ps+Pt-Th	D2	Ok
D3uc	17.034 ...	98	292	17	156	Ps+Pt-Th	D3	Ok
Max:	27.3896 ... mm	0.582		0.204	(Str. Ratio)			

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses	----	Shell Band Stress	----	Pass
	Ten Allwd Cmp Allwd	:	Ten Allwd Cmp Allwd	:	Fail
D1uc	9 292 ...				Ok
D2uc	31 292 ...				Ok
D3uc	42 292 ...				Ok
Max RATIO	0.143 ...				

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses	Tube Loads	Shell Stress	Channel Stress	Pass
	Ten Allwd Cmp Allwd	Ld Allwd	Stress Allwd	Stress Allwd	Fail



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D1uc	11	194	-7	-194	97	1791	...	292	...	292	Ok
D2uc	20	194	-20	-194	188	1791	...	292	...	292	Ok
D3uc	10	194	-10	-194	88	1791	...	292	...	292	Ok
Max RATIO	0.105	0.105	0.105	...	...	...	...	...	...	...	...

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	27.390	40.000	Ok
Tube thickness :	0.774	1.650	Ok
Tube-tubesheet fillet weld leg :	1.037	1.650	Ok
Tube-tubesheet groove weld leg :	1.037	1.650	Ok

Min Shell length of thk, (12.000) adj. to tubesheet: 187.061 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.124 mm
 Due to Pressure + Thermal Shell Compresses by : -0.124 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure UnCorroded	Due to Pressure Corroded	Due to Thermal Exp. UnCorroded	Due to Thermal Exp. Corroded	Computed Delta Ms UnCorroded	Computed Delta Ms Corroded
1	0.0188	...	...	...	...	...
2	-0.1242	...	...	...	...	...
3	-0.1073	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	56	No Calc	No Calc
20	30	No Calc	...	105	No Calc	No Calc
30	Tsht	999.97	924	12	0.0018499	66.1793
Tsht	40	No Calc	...	12	No Calc	No Calc
40	Tsht	4170	924	12	0.00039515	37.0758
Tsht	50	No Calc	...	12	No Calc	No Calc
50	60	1000.03	924	12	0.0018498	66.1784
60	70	No Calc	...	105	No Calc	No Calc
70	80	No Calc	...	56	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	56	42.8752	103.419	No Calc
20	30	105	92.8116	103.419	No Calc
30	Tsht	12	3.16387	103.419	1145.9
Tsht	40	12	No Calc	103.419	No Calc
40	Tsht	12	5.71422	103.419	641.968
Tsht	50	12	No Calc	103.419	No Calc
50	60	12	3.16394	103.419	1145.88
60	70	105	92.8116	103.419	No Calc
70	80	56	42.8752	103.419	No Calc

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	999.97	27608.8	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	4170	94597.8	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	1000.03	27609.7	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	435.005	...	435.005	...	43.5005
20	30	289.808	85898.7	289.808	85898.7	28.9808
30	40	275.991	636287	275.991	636287	27.5991
40	50	1150.88	1452668	1150.88	1452668	115.088
50	60	275.991	636287	275.991	636287	27.5991
60	70	289.808	85898.7	289.808	85898.7	28.9808
70	80	435.005	...	435.005	...	43.5005
Total		3152	2897038.75	3152	2897038.75	315



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Ins1	42.5317	20	...	...	Insulation
20	Liqd	85.8463	67.5	...	...	Liquid: 20
20	Ins1	16.0478	40	...	...	Insulation
30	Liqd	635.898	500	...	...	Liquid: 30
30	Ins1	113.371	500	...	...	Insulation
30	Noz1	106.029	500	599.225	...	N1-Tube Inlet
40	Sad1	171.46	835	709	...	Left Saddle flo
40	Sad1	171.46	3335	709	...	New Saddle
40	Liqd	1451.78	2085	...	...	Liquid: 40
40	Ins1	370.818	2085	...	...	Insulation
40	Noz1	106.029	1350	599.225	...	N3-Shell inlet
40	Noz1	106.029	2800	-599.225	...	N4-Shell outlet
40	Wght	250	3300	-990.6	...	Fireproofing we
40	Wght	250	850	-990.6	...	Wght:[2 of 2]
50	Liqd	635.898	500	...	...	1
50	Ins1	113.371	500	...	...	Insulation
50	Noz1	107.953	500	-599.225	...	N2-Tube Outlet
60	Ins1	16.0478	40	...	...	Insulation
70	Ins1	42.5317	20	...	...	Insulation
30	FTsh	135.163	1020	...	...	Fixed Tubesheet
30	Tube	3418.66	3125	...	...	Tubes
30	RTsh	135.163	5230	...	...	Tubesheet
40	Baf1	23.9326	280	...	...	Baffles
40	Baf1	23.9326	560	...	...	Baffles
40	Baf1	23.9326	840	...	...	Baffles
40	Baf1	23.9326	1120	...	...	Baffles
40	Baf1	23.9326	1400	...	...	Baffles
40	Baf1	23.9326	1680	...	...	Baffles
40	Baf1	23.9326	1960	...	...	Baffles
40	Baf1	23.9326	2240	...	...	Baffles
40	Baf1	23.9326	2520	...	...	Baffles
40	Baf1	23.9326	2800	...	...	Baffles
40	Baf1	23.9326	3080	...	...	Baffles
40	Baf1	23.9326	3360	...	...	Baffles
40	Baf1	23.9326	3640	...	...	Baffles
40	Baf1	23.9326	3920	...	...	Baffles
40	Baf1	24.251	4200	...	...	Baffles



Total Weight of Each Detail Type:

Saddles	342.9
Liquid	2809.4
Insulation	714.7
Nozzles	426.0
Weights	500.0
Exchanger Components	4048.3
Liquid in Tubes	835.9
Sum of the Detail Weights	9677.3 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	3467.7	3467.7	3467.7	3467.7	3467.7	3467.7
Saddles	342.9	342.9	342.9	342.9	342.9	342.9
Nozzles	426.0	426.0	426.0	426.0	426.0	426.0
Exchanger	4048.3	4048.3	4048.3	4048.3	4048.3	4048.3
Insulation	...	...	...	...	714.7	714.7
Empty Weights	...	...	...	500.0	500.0	...
Test Weights	...	500.0	...	...	...	...
Ope Weights	...	...	...	...	...	500.0
Ope. Liquid	...	...	...	...	...	2809.4
Tube Ope Lqd	...	...	...	...	...	835.9
Test Liquid	...	2895.3	...	...	...	...
Tube Tst Lqd	...	835.9	...	...	...	...
Totals	8285.0	12516.2	8285.0	8785.0	9499.7	13145.1

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	8285.0 kg.
Shop Test Wt.	- Fabricated weight + water (full)	12516.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	8285.0 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	8785.0 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	9499.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	13145.1 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	13145.1 kg.
Field Test Wt.	- Empty weight + test medium (full)	13230.9 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	836448.3 cm ³
Weight of Ope Liq in tubes :	835.9 kg.
Weight of Water in tubes :	835.9 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	11629.7	...



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Element and Detail Weights:

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20	30	7858.85	3817.03
30	40	29028.3	28274.3
40	50	121048	117904
50	60	29028.3	28274.3
60	70	7858.85	3817.03
70	80	11629.7	...

Total	218081.8	182086.7 cm ²
Total	234.7	196.0 ft ²

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Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1-Tube Inlet	1476.0	1910.0	130	150	GR 2.1	2214.1	1033.2	1476.0	...
N3-Shell inlet	4080.2	4970.0	100	300	GR 2.1	2214.1	3906.1	4080.2	3906.1
N4-Shell outlet	4080.2	4970.0	100	300	GR 2.1	2214.1	3906.1	4080.2	3906.1
N2-Tube Outlet	3852.1	4970.0	130	300	GR 2.1	2214.1	3564.0	3852.1	3564.0

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 3906.139 KPa.
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 4970.005 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1033.183 KPa.
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1910.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Tube Inlet	357.251	1.20	1816	2295	974.6	1033.2	Pass
N3-Shell inlet	357.251	0.50	1816	2295	3098.3	3906.1	Pass
N4-Shell outlet	357.251	0.50	1816	2295	3098.3	3906.1	Pass
N2-Tube Outlet	357.251	0.50	1816	2295	974.6	3564.0	Pass

Evaluating Flange on Nozzle [N1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1816) + 4 \cdot \text{abs}(2295) \cdot 357.251 \leq \pi \cdot 357.251^3 \cdot ((1476.034 - 974.621) + 1.200 \cdot 1476.034)$$

$$= 0.32342E+05 \leq 0.33198E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1476.034, 1.200 \cdot 1476.034 + 1476.034 - 32342.3 / (\pi \cdot 357.251^3))$$

$$= \min(1476.034, 1033.183)$$

$$= 1033.183 \text{ KPa.}$$

Evaluating Flange on Nozzle [N3-Shell inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1816) + 4 \cdot \text{abs}(2295) \cdot 357.251 \leq \pi \cdot 357.251^3 \cdot ((4080.154 - 3098.297) + 0.500 \cdot 4080.154)$$

$$= 0.32342E+05 \leq 0.44143E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Shell inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4080.154, 0.500 \cdot 4080.154 + 4080.154 - 32342.3 / (\pi \cdot 357.251^3))$$

$$= \min(4080.154, 3906.139)$$

$$= 3906.139 \text{ KPa.}$$

Evaluating Flange on Nozzle [N4-Shell outlet]:



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Nozzle Flange MAWP:

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$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1816) + 4 \cdot \text{abs}(2295) 357.251 \leq \pi \cdot 357.251^3 [(4080.154 - 3098.297) + .500 \cdot 4080.154] \\ &= 0.32342E+05 \leq 0.44143E+05 \quad \text{Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4-Shell outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4080.154, 0.500 \cdot 4080.154 + 4080.154 - 32342.3 / (\pi \cdot 357.251^3)) \\ &= \min(4080.154, 3906.139) \\ &= 3906.139 \text{ KPa.} \end{aligned}$$

Evaluating Flange on Nozzle [N2-Tube Outlet]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1816) + 4 \cdot \text{abs}(2295) 357.251 \leq \pi \cdot 357.251^3 [(3852.081 - 974.621) + .500 \cdot 3852.081] \\ &= 0.32342E+05 \leq 0.70167E+05 \quad \text{Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N2-Tube Outlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(3852.081, 0.500 \cdot 3852.081 + 3852.081 - 32342.3 / (\pi \cdot 357.251^3)) \\ &= \min(3852.081, 3564.033) \\ &= 3564.033 \text{ KPa.} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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 Wind Load Calculation: Step: 16 6:22pm Sep 8,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	10000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.218 \cdot 0.941)/(1 + 1.7 \cdot 3.4 \cdot 0.218)))$
 $= \min(0.85, 0.895)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.586
Structure Height to Diameter ratio	6.134

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 990.0 + 10000.0 = 10990.0$ mm
 $= 36.056$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (36.056 ft.) > 15 ft.
 $= 2.01(z / Zg)^{2/\alpha}$
 $= 2.01(36.056/900.0)^{2/9.5}$
 $= 1.021$

Type of Hill: No Hill



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Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 $= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 = 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 $= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 1.021 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$
 = 16.0 psf [78.12] Kgs/m²

Force on the first element [F]:
 $= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 16.0 \cdot 0.85 \cdot 0.586 \cdot 0.906$
 = 7.2 lbs. [3.3] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Blind Flange	10990.0	0.000	0.000	0.000	1.021	1.000	78.120
LHS CH Flang	10990.0	0.000	0.000	0.000	1.021	1.000	78.120
Ch Shell	10990.0	0.000	0.000	0.000	1.021	1.000	78.120
main shell	10990.0	0.000	0.000	0.000	1.021	1.000	78.120
RHS Ch shell	10990.0	0.000	0.000	0.000	1.021	1.000	78.120
RHS Ch Flg	10990.0	0.000	0.000	0.000	1.021	1.000	78.120
RHS Blind Flg	10990.0	0.000	0.000	0.000	1.021	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Fireproofing	0.00	10000.00	78.12	0.00
Wght:[2 of 2]	0.00	10000.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	10990	1503.43	841.92	78.12	3.60104
20	30	10990	1222.22	1650	78.12	7.05734
30	40	10990	1348.8	13488	78.12	57.6905
40	50	10990	1300.8	54243.4	78.12	232.008
50	60	10990	1348.8	13488	78.12	57.6905
60	70	10990	1222.22	1650	78.12	7.05734
70	80	10990	1503.43	841.92	78.12	3.60104

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(5.06890.75)$
 = 0.068 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.068$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.157 \cdot 12802 \cdot 1.25$
 = 752.767 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 12802$
 = 401.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	555	1422.46	98.0717



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Earthquake Load Calculation:

Step: 17

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20	30	450	1422.46	79.5176
30	40	450	1422.46	79.5176
40	Sad1	450	1422.46	79.5176
Sad1	50	450	1422.46	79.5176
40	50	450	1422.46	79.5176
50	60	450	1422.46	79.5176
60	70	450	1422.46	79.5176
70	80	555	1422.46	98.0717

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 18 6:22pm Sep 8,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3319.175 mm
Center of Gravity of the Liquid	3221.623 mm
Center of Gravity of the Insulation	3317.059 mm
Center of Gravity of the Nozzles	3326.052 mm
Center of Gravity of the Added Weights (Operating)	3309.175 mm
Center of Gravity of the Added Weights (Empty)	3309.175 mm
Center of Gravity of the Tubesheet(s)	3319.175 mm
Center of Gravity of the Tubes	3319.175 mm
Center of Gravity of the Baffles	3475.912 mm
Center of Gravity of Bare Shell New and Cold	3319.905 mm
Center of Gravity of Bare Shell Corroded	3319.905 mm
Vessel CG in the Operating Condition	3302.530 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3325.637 mm
Vessel CG in the Test Condition	3323.733 mm

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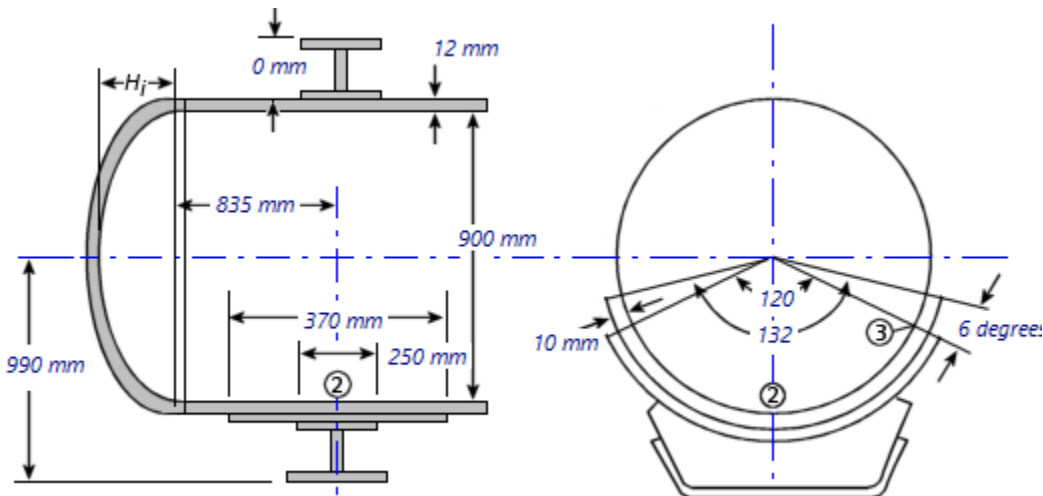
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	456.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	4170.00	mm
Distance from Saddle to Vessel tangent	a1 or a	835.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	370.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		106.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	990.00	mm
Coefficient of Friction	mu	0.30	

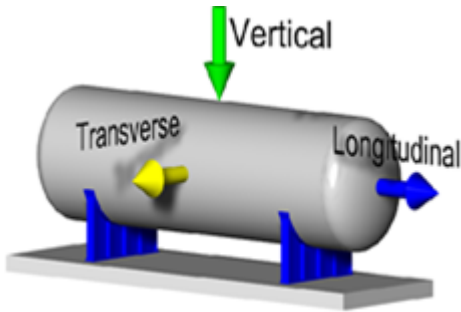


Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		7694.53	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	333.7	Km/hr



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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-2.24	-184.50
Long. Stress at Top of Midspan	2.24	137.00
Long. Stress at Bottom of Midspan	2.24	137.00
Long. Stress at Top of Saddles	12.85	137.00
Long. Stress at Bottom of Saddles	-7.12	-184.50
Long. Stress at Bottom of Saddles	7.12	137.00
Tangential Shear in Shell	9.68	90.61
Circ. Stress at Horn of Saddle	14.71	205.50
Circ. Compressive Stress in Shell	0.71	137.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:
 = Transverse Area • 0.00256 • wv² • wlsf
 = 92.79 • 0.00256 • 207.37² • 1.0
 = 10214.6 lbf
 = 4633.323 Kgf

Longitudinal Force due to Wind during Transport [F_{lw}]:
 = π • D²/4 • 0.00256 • wv² • wlsf • Wind Dia. Mult
 = π • 3.688²/4 • 0.00256 • 207.37² • 1.0 • 1.2
 = 1410.9 lbf
 = 640.0 Kgf

Transverse Force due to Lateral Acceleration [F_t]:
 = g_z • Shipping Weight • g_w + F_{tw}
 = 0.1 • 7942.08 • 1.0 + 4633.32
 = 5427.5 Kgf

Longitudinal Force due to Axial Acceleration [F_l]:
 = g_x • Shipping Weight • g_w + F_{lw}
 = 0.1 • 7942.08 • 1.0 + 639.99
 = 1434.2 Kgf

Saddle Reaction Force due to Transverse Load [F_{wt}]:
 = F_t/Num of Saddles • B / E
 = 5427.5/2 • 990.0/817.528
 = 3286.3 Kgf

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:
 = F_l • B / L_s
 = 1434 • 990.0/2500.0
 = 567.9 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:
 = Saddle Load • g_y
 = 4198.3 • 0.05



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= 209.916 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 4198 + 210 + max(568, 3286)
 = 7694.5 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7865.99	Kgf
Transverse Shear Load	2713.77	Kgf
Longitudinal Shear Load	1434.20	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
 = min(2(900.0/2 + 12.0 + 10.0)sin(120.0/2), 2•456.0)
 = 817.528 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -7695•835.0[1-(1-835.0/4170.0+(456.0²-0.0²)/
 (2•835.0•4170.0))/(1+(4•0.0)/(3•4170.0))]
 = -1094.7 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 7695•4170/4((1+2(456²-0²)/(4170²))/(1+(4•0)/
 (3•4170))-4•835/4170)
 = 1788.5 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)
 = 0.0 • 456.0/(2•12.0) - 1788.5/(π•456.0²•12.0)
 = -2.24 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0.0 • 456.0/(2 • 12.0) + 1788.5/(π • 456.0² • 12.0)
 = 2.24 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 0.0•456.0/(2•12.0)--1094.7/(0.1066•π•456.0²•12.0)
 = 12.85 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0.0•456.0/(2•12.0)+-1094.7/(0.1923•π•456.0²•12.0)
 = -7.12 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 7695(4170.0 - 2 • 835.0)/(4170.0 + (4 • 0.0/3))
 = 4613.0 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 4613.03 / (456.0 \cdot 12.0)$$

$$= 9.68 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(456.0 \cdot 12.0)}$$

$$= 57.699 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(456.0 \cdot 12.0)}, 2 \cdot 835.0)$$

$$= 365.40 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 7695 \cdot 0.1 / (365.398(12.0 + 1.0 \cdot 10.0))$$

$$= -0.71 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -7695 / (4(12.0 + 1.0 \cdot 10.0)365.398) -$$

$$3 \cdot 0.053 \cdot 7695 / (2(12.0 + 1.0 \cdot 10.0)^2)$$

$$= -14.71 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	530.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(450.0 \cdot 12.0)} + 370.0)12.0^3) / 12$$

$$= 6.979 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 6.979 + 58.156(209.921 - 6.0)^2$$

$$= 24190.533 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	484.6	12.0	6.0	58.2	34893.8	0.242E+05
Wearplate	370.0	10.0	17.0	37.0	62900.0	0.138E+05
Web	10.0	500.0	272.0	50.0	1359999.8	0.123E+05
BasePlate	275.0	18.0	531.0	49.5	2628449.8	0.510E+05
Totals	...	...	...	194.7	4086243.2	0.101E+06

Distance to Centroid [C1]:

$$= AY / A$$



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$$= 4086243/194.656$$

$$= 209.921 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 7694.525$$

$$= 1566.0028 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 1.1251 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 990.0 - 450.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 617.853 \text{ mm}$$

Bending Moment, M = Fh * d = 967.5786 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 1.9653 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(7695 + 171)275.0) / (4 \cdot 980.0 \cdot 173.33)}$$

$$= 9.678 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (450.0 + 12.0 + 10.0)$$

$$= 817.528 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 817.528 / (3 - 1)$$

$$= 408.764 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 408.764 \cdot 275.0/2$$

$$= 562.051 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 7865.985 / (980.0 \cdot 275.0)$$

$$= 2.919 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 562.1 \cdot 2.92$$

$$= 1640.475 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 24.0 + 20.438$$

$$= 44.438 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 1640.5/44.4382$$

$$= 3.620 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	25.0	...	0.130E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 250.0 - 10.0$$

$$= 240.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/44.438$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1302.1/44.438)}$$

$$= 54.130 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 990.0 - \cos(120.0/2) (450.0 + 12.0 + 10.0) - 18.0$$

$$= 736.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 736.0/54.13$$

$$= 13.597$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 1434.2 / (2 \cdot 980.0) \cdot 408.764 \cdot 736.0/2$$

$$= 110.073 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($13.5968 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (13.6)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (13.6) / (8 \cdot 123.21) - (13.6^3) / (8 \cdot 123.21^3))$$

$$= 151.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 3.62/151.31 + (110.07 \cdot 125.0/13020833) / 173.33$$



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= 0.084

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	24.0	0.0	0.130E+04
Web	408.8	10.0	0.0	40.9	0.0	3.41
Totals	...	...	...	64.9	...	0.131E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/64.876
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 250.0/2
 = 125.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 990.0 - cos(472.0/2)(18.0 + 0.0 + 0.0) - 0.0
 = 500.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{1305.4/64.876}$
 = 44.857 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 500.0/44.857
 = 11.147

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 1434.197/(2 • 980.0)
 = 0.732 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 0.732 • 408.764 • 500.0
 = 149.556 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus
 = 149.556/104432.492
 = 14.044 N./mm²

Compressive Allowable, KL/r < Cc (11.1466 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$
 = $(1 - (11.15)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (11.15) / (8 \cdot 123.21) - (11.15^3) / (8 \cdot 123.21^3))$
 = 152.3 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 4.96/152.27 + (149.56 • 125.0/1305.405)/173.33
 = 0.114

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	456.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	4170.00	mm
Distance from Saddle to Vessel tangent	a1 or a	835.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	370.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		106.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	990.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		7742.22	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	333.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-2.25	-184.50	
Long. Stress at Top of Midspan	2.25	137.00	
Long. Stress at Bottom of Midspan	2.25	137.00	
Long. Stress at Top of Saddles	12.93	137.00	
Long. Stress at Bottom of Saddles	-7.16	-184.50	
Long. Stress at Bottom of Saddles	7.16	137.00	
Tangential Shear in Shell	9.74	90.61	
Circ. Stress at Horn of Saddle	14.80	205.50	
Circ. Compressive Stress in Shell	0.72	137.00	

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:
 = Transverse Area • 0.00256 • wv² • wlsf
 = 92.79 • 0.00256 • 207.37² • 1.0
 = 10214.6 lbf
 = 4633.323 Kgf

Longitudinal Force due to Wind during Transport [Flw]:
 = $\pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$
 = $\pi \cdot 3.688^2/4 \cdot 0.00256 \cdot 207.37^2 \cdot 1.0 \cdot 1.2$
 = 1410.9 lbf
 = 640.0 Kgf

Transverse Force due to Lateral Acceleration [Ft]:
 = gz • Shipping Weight • gw + Ftw
 = 0.1 • 7942.08 • 1.0 + 4633.32



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= 5427.5 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$
 = $0.1 \cdot 7942.08 \cdot 1.0 + 639.99$
 = 1434.2 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$
 = $5427.5/2 \cdot 990.0/817.528$
 = 3286.3 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$
 = $1434 \cdot 990.0/2500.0$
 = 567.9 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$
 = $4243.7 \cdot 0.05$
 = 212.187 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = $4244 + 212 + \max(568, 3286)$
 = 7742.2 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	7913.68	Kgf
Transverse Shear Load	2713.77	Kgf
Longitudinal Shear Load	1434.20	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 = $\min(2(900.0/2 + 12.0 + 10.0) \sin(120.0/2), 2 \cdot 456.0)$
 = 817.528 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 = $-7742 \cdot 835.0 [1 - (1 - 835.0/4170.0 + (456.0^2 - 0.0^2) / (2 \cdot 835.0 \cdot 4170.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 4170.0))]$
 = -1101.5 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 \{ (1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/(3L)) - 4a/L \}$
 = $7742 \cdot 4170/4 \{ (1 + 2(456^2 - 0^2)/(4170^2)) / (1 + (4 \cdot 0)/(3 \cdot 4170)) - 4 \cdot 835/4170 \}$
 = 1799.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 = $0.0 \cdot 456.0 / (2 \cdot 12.0) - 1799.6 / (\pi \cdot 456.0^2 \cdot 12.0)$
 = -2.25 N./mm²



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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$
 $= 0.0 \cdot 456.0 / (2 \cdot 12.0) + 1799.6 / (\pi \cdot 456.0^2 \cdot 12.0)$
 $= 2.25 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:
 $= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$
 $= 0.0 \cdot 456.0 / (2 \cdot 12.0) - 1101.5 / (0.1066 \cdot \pi \cdot 456.0^2 \cdot 12.0)$
 $= 12.93 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:
 $= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= 0.0 \cdot 456.0 / (2 \cdot 12.0) + 1101.5 / (0.1923 \cdot \pi \cdot 456.0^2 \cdot 12.0)$
 $= -7.16 \text{ N./mm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a) / (L + (4 \cdot h_m / 3))$
 $= 7742(4170.0 - 2 \cdot 835.0) / (4170.0 + (4 \cdot 0.0 / 3))$
 $= 4641.6 \text{ Kgf}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K_2 \cdot T / (R_m \cdot t)$
 $= 1.1707 \cdot 4641.62 / (456.0 \cdot 12.0)$
 $= 9.74 \text{ N./mm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(456.0 \cdot 12.0)}$
 $= 57.699 \text{ mm}$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(250.0 + 1.56 \cdot \sqrt{(456.0 \cdot 12.0)}, 2 \cdot 835.0)$
 $= 365.40 \text{ mm}$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 $= 1.0000$ Materials are the same, transport case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 $= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$
 $= -0.7603 \cdot 7742 \cdot 0.1 / (365.398(12.0 + 1.0 \cdot 10.0))$
 $= -0.72 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:
 $= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$
 $= -7742 / (4(12.0 + 1.0 \cdot 10.0)365.398) -$
 $3 \cdot 0.053 \cdot 7742 / (2(12.0 + 1.0 \cdot 10.0)^2)$
 $= -14.80 \text{ N./mm}^2$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	530.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



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Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(450.0 \cdot 12.0)} + 370.0)12.0^3)/12$$

$$= 6.979 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 6.979 + 58.156(209.921 - 6.0)^2$$

$$= 24190.533 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	484.6	12.0	6.0	58.2	34893.8	0.242E+05
Wearplate	370.0	10.0	17.0	37.0	62900.0	0.138E+05
Web	10.0	500.0	272.0	50.0	1359999.8	0.123E+05
BasePlate	275.0	18.0	531.0	49.5	2628449.8	0.510E+05
Totals	...	...	...	194.7	4086243.2	0.101E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4086243/194.656$$

$$= 209.921 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 7742.218$$

$$= 1575.7092 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 1.1321 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 990.0 - 450.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 617.853 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 973.5759 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.9775 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(7742 + 171)275.0 / (4 \cdot 980.0 \cdot 173.33))}$$

$$= 9.707 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (450.0 + 12.0 + 10.0)$$

$$= 817.528 \text{ mm}$$

Distance between Ribs [e]:



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$$\begin{aligned}
 &= \text{Web Length} / (\text{Nribs} - 1) \\
 &= 817.528 / (3 - 1) \\
 &= 408.764 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 408.764 \cdot 275.0 / 2 \\
 &= 562.051 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 7913.678 / (980.0 \cdot 275.0) \\
 &= 2.936 \text{ Kg/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= \text{Ap} \cdot \text{Bp} \\
 &= 562.1 \cdot 2.94 \\
 &= 1650.422 \text{ Kg}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 24.0 + 20.438 \\
 &= 44.438 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / \text{Ar} \\
 &= 1650.4 / 44.4382 \\
 &= 3.642 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	25.0	...	0.130E+04

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 250.0 - 10.0 \\
 &= 240.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= \text{AY} / \text{A} \\
 &= 0.0 / 44.438 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 250.0 / 2 \\
 &= 125.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 1302.1 / 44.438 } \\
 &= 54.130 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 990.0 - \cos(120.0 / 2) (450.0 + 12.0 + 10.0) - 18.0 \\
 &= 736.000 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} } \\
 &= \sqrt{ 2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0 }
 \end{aligned}$$



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$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 736.0/54.13$$

$$= 13.597$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 1434.2 / (2 \cdot 980.0) \cdot 408.764 \cdot 736.0/2$$

$$= 110.073 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (13.5968 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.6)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (13.6) / (8 \cdot 123.21) - (13.6^3) / (8 \cdot 123.21^3))$$

$$= 151.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 3.64/151.31 + (110.07 \cdot 125.0/13020833) / 173.33$$

$$= 0.084$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	24.0	0.0	0.130E+04
Web	408.8	10.0	0.0	40.9	0.0	3.41
Totals	...	...	...	64.9	...	0.131E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/64.876$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 990.0 - \cos(472.0/2) (18.0 + 0.0 + 0.0) - 0.0$$

$$= 500.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1305.4/64.876) }$$

$$= 44.857 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 500.0/44.857$$

$$= 11.147$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 1434.197 / (2 \cdot 980.0)$$

$$= 0.732 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.732 \cdot 408.764 \cdot 500.0$$



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= 149.556 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 149.556/104432.492

= 14.044 N./mm²

Compressive Allowable, $KL/r < C_c$ (11.1466 < 123.2105) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(11.15)²/(2 • 123.21²))260/

(5/3+3•(11.15)/(8• 123.21)-(11.15³)/(8•123.21³)

= 152.3 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C1/I)/S_{ba}$

= 4.99/152.27 + (149.56 • 125.0/1305.405)/173.33

= 0.114

Review notes about nozzle loadings and supports in the Warnings report.

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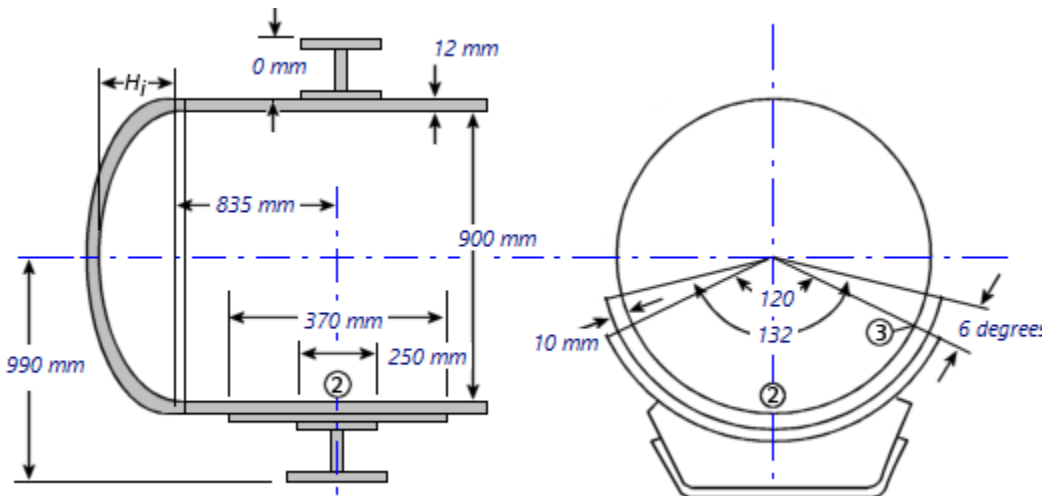
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	456.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	4170.00	mm
Distance from Saddle to Vessel tangent	a1 or a	835.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	370.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		106.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	990.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	8056.70	Kgf
Pressure used in Analysis	P	1304.411 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	22.44	137.00
Long. Stress at Bottom of Midspan	27.13	137.00
Long. Stress at Top of Saddles	38.24	137.00
Long. Stress at Bottom of Saddles	17.33	137.00
Tangential Shear in Shell	10.13	90.61
Circ. Stress at Horn of Saddle	15.40	171.25
Circ. Compressive Stress in Shell	0.75	137.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(368.7/2 + 0) \cdot 990.0/817.528$
 $= 669.7 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(122.78, 0) \cdot 990.0/2500.0$
 $= 48.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(752.77, 0) \cdot 990.0/2500.0$
 $= 298.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(753/2 + 0) \cdot 990.0/817.528$
 $= 1367.4 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 6689 + \max(49, 670, 298, 1367)$
 $= 8056.7 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08(0.0 + \pi/4(1.124 \cdot 1.2)^2)$
 $= 1204.079 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8228.16	Kgf
Transverse Shear Load	376.38	Kgf
Longitudinal Shear Load	752.77	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(900.0/2 + 12.0 + 10.0)\sin(120.0/2), 2 \cdot 456.0)$
 $= 817.528 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -8057 \cdot 835.0 [1 - (1 - 835.0/4170.0 + (456.0^2 - 0.0^2)/(2 \cdot 835.0 \cdot 4170.0))/(1 + (4 \cdot 0.0)/(3 \cdot 4170.0))]$
 $= -1146.2 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 8057 \cdot 4170/4((1 + 2(456^2 - 0^2)/(4170^2))/(1 + (4 \cdot 0)/(3 \cdot 4170)) - 4 \cdot 835/4170)$



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$$(3 \cdot 4170) - 4 \cdot 835 / 4170 \\ = 1872.7 \text{ Kg-m.}$$

$$\text{Longitudinal Stress at Top of Shell (4.15.4) } [\sigma_1]: \\ = P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 1304.411 \cdot 456.0 / (2 \cdot 12.0) - 1872.7 / (\pi \cdot 456.0^2 \cdot 12.0) \\ = 22.44 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell (4.15.5) } [\sigma_2]: \\ = P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1304.411 \cdot 456.0 / (2 \cdot 12.0) + 1872.7 / (\pi \cdot 456.0^2 \cdot 12.0) \\ = 27.13 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Top of Shell at Support (4.15.8) } [\sigma^*3]: \\ = P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 1304.411 \cdot 456.0 / (2 \cdot 12.0) - 1146.2 / (0.1066 \cdot \pi \cdot 456.0^2 \cdot 12.0) \\ = 38.24 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell at Support (4.15.9) } [\sigma^*4]: \\ = P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 1304.411 \cdot 456.0 / (2 \cdot 12.0) + 1146.2 / (0.1923 \cdot \pi \cdot 456.0^2 \cdot 12.0) \\ = 17.33 \text{ N./mm}^2$$

$$\text{Maximum Shear Force in the Saddle (4.15.3) } [T]: \\ = Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 8057(4170.0 - 2 \cdot 835.0) / (4170.0 + (4 \cdot 0.0 / 3)) \\ = 4830.2 \text{ Kg}$$

$$\text{Shear Stress in the shell no rings, not stiffened (4.15.12) } [\tau_2]: \\ = K_2 \cdot T / (R_m \cdot t) \\ = 1.1707 \cdot 4830.16 / (456.0 \cdot 12.0) \\ = 10.13 \text{ N./mm}^2$$

$$\text{Decay Length (4.15.20) } [x_1, x_2]: \\ = 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{456.0 \cdot 12.0} \\ = 57.699 \text{ mm}$$

$$\text{Effective reinforcing plate width (4.15.24) } [B_1]: \\ = \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(250.0 + 1.56 \cdot \sqrt{456.0 \cdot 12.0}, 2 \cdot 835.0) \\ = 365.40 \text{ mm}$$

$$\text{Wear Plate/Shell Stress ratio (4.15.26) } [\eta]: \\ = \min(S_r / S, 1) \\ = \min(136.998 / 136.998, 1) \\ = 1.0000$$

$$\text{Circumferential Stress at Saddle Base with Wear Plate (4.15.25) } [\sigma_6, r]: \\ = -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.7603 \cdot 8057 \cdot 0.1 / (365.398(12.0 + 1.0 \cdot 10.0)) \\ = -0.75 \text{ N./mm}^2$$

$$\text{Circ. Comp. Stress at Horn of Saddle, } L \geq 8R_m \text{ (4.15.28) } [\sigma_7, r]: \\ = -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -8057 / (4(12.0 + 1.0 \cdot 10.0)365.398) - 3 \cdot 0.053 \cdot 8057 / (2(12.0 + 1.0 \cdot 10.0)^2) \\ = -15.40 \text{ N./mm}^2$$

$$\text{Distance between Saddle Supports } [L_s]: \\ = 2500.0 \text{ mm}$$

$$\text{Free Un-Restrained Thermal Expansion between the Saddles } [\text{Exp}]: \\ = \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.000016 \cdot 2500.0(100.0 - 21.1)$$



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= 3.176 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	530.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(450.0 \cdot 12.0)} + 370.0)^3) / 12$$

$$= 6.979 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 6.979 + 58.156 (209.921 - 6.0)^2$$

$$= 24190.533 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	484.6	12.0	6.0	58.2	34893.8	0.242E+05
Wearplate	370.0	10.0	17.0	37.0	62900.0	0.138E+05
Web	10.0	500.0	272.0	50.0	1359999.8	0.123E+05
BasePlate	275.0	18.0	531.0	49.5	2628449.8	0.510E+05
Totals	...	...	...	194.7	4086243.2	0.101E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4086243 / 194.656$$

$$= 209.921 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 8056.698$$

$$= 1639.7126 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.1781 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 990.0 - 450.0 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 617.853 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 1013.1213 \text{ Kg-m.}$$



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Bending Stress, $S_b = (M * C_1 / I) = 2.0578 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{ (3(8057 + 171)275.0 / (4 * 980.0 * 174.663))}$
 $= 9.860 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2)(\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120.0/2)(450.0 + 12.0 + 10.0)$
 $= 817.528 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 817.528 / (3 - 1)$
 $= 408.764 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 408.764 * 275.0 / 2$
 $= 562.051 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 8228.157 / (980.0 * 275.0)$
 $= 3.053 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 562.1 * 3.05$
 $= 1716.008 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 24.0 + 20.438$
 $= 44.438 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 1716.0 / 44.4382$
 $= 3.787 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	25.0	...	0.130E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 250.0 - 10.0$
 $= 240.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 44.438$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
 = 250.0/2
 = 125.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{1302.1/44.438}$
 = 54.130 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 990.0 - cos(120.0/2)(450.0 + 12.0 + 10.0) - 18.0
 = 736.000 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0}$
 = 122.739

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 736.0/54.13
 = 13.597

Bending Moment [Rm]:
 $= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$
 $= 752.8 / (2 \cdot 980.0) \cdot 408.764 \cdot 736.0/2$
 = 57.774 Kg-m.

Compressive Allowable, $KL/r < Cc$ (13.5968 < 122.7394) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (13.6)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (13.6) / (8 \cdot 122.74) - (13.6^3) / (8 \cdot 122.74^3))$
 = 152.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 3.79/152.45 + (57.77 \cdot 125.0/13020833) / 174.66$
 = 0.056

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	24.0	0.0	0.130E+04
Web	408.8	10.0	0.0	40.9	0.0	3.41
Totals	...	...	...	64.9	...	0.131E+04

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0/64.876$
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 250.0/2
 = 125.000 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 990.0 - cos(472.0/2)(18.0 + 0.0 + 0.0) - 0.0
 = 500.000 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(1305.4/64.876)}$$

$$= 44.857 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 500.0/44.857$$

$$= 11.147$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 752.767 / (2 \cdot 980.0)$$

$$= 0.384 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.384 \cdot 408.764 \cdot 500.0$$

$$= 78.497 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 78.497/104432.492$$

$$= 7.371 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($11.1466 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.15)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (11.15) / (8 \cdot 122.74) - (11.15)^3 / (8 \cdot 122.74^3))$$

$$= 153.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$$

$$= 5.19/153.42 + (78.5 \cdot 125.0/1305.405)/174.66$$

$$= 0.076$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	0.9592 cm ²
Saddle Load Q0 (Weight)	Q0	6860.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	298.1 Kg
Maximum Transverse Force	Ft	376.4 Kg
Maximum Longitudinal Force	F1	752.8 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 752.8 / (0.959 \cdot 4)$$

$$= 19.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 376.4 / (0.959 \cdot 4)$$

$$= 9.6 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (BoltShearAllowable \cdot Nbolts)$
 = $752.77 / (82.8 \cdot 4.0)$
 = 0.2229 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $990.0 \cdot 376.38 + 0.0$
 = 372.63 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = $372.63 / 6860.79$
 = 54.31 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.2229, 0.0]
 = 0.2229 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	456.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	4170.00	mm
Distance from Saddle to Vessel tangent	a1 or a	835.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	370.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		106.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	990.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		7881.66	Kgf
Pressure used in Analysis	P	1304.411	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	22.49	137.00
Long. Stress at Bottom	of Midspan	27.08	137.00
Long. Stress at Top	of Saddles	37.94	137.00



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Long. Stress at Bottom of Saddles	17.49	137.00
Tangential Shear in Shell	9.91	90.61
Circ. Stress at Horn of Saddle	15.06	171.25
Circ. Compressive Stress in Shell	0.73	137.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (368.7/2 + 0) \cdot 990.0/817.528$
 $= 669.7 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(122.78, 0) \cdot 990.0/2500.0$
 $= 48.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(752.77, 0) \cdot 990.0/2500.0$
 $= 298.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (753/2 + 0) \cdot 990.0/817.528$
 $= 1367.4 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 6514 + \max(49, 670, 298, 1367)$
 $= 7881.7 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08 (0.0 + \pi/4 (1.124 \cdot 1.2)^2)$
 $= 1204.079 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8053.12	Kgf
Transverse Shear Load	376.38	Kgf
Longitudinal Shear Load	752.77	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(900.0/2 + 12.0 + 10.0) \sin(120.0/2), 2 \cdot 456.0)$
 $= 817.528 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -7882 \cdot 835.0 [1 - (1 - 835.0/4170.0 + (456.0^2 - 0.0^2)/(2 \cdot 835.0 \cdot 4170.0))/(1 + (4 \cdot 0.0)/(3 \cdot 4170.0))]$



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= -1121.3 Kg-m.

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L \right)$$

$$= 7882 \cdot 4170 / 4 \left((1 + 2(456^2 - 0^2) / (4170^2)) / (1 + (4 \cdot 0) / (3 \cdot 4170)) - 4 \cdot 835 / 4170 \right)$$

$$= 1832.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 1304.411 \cdot 456.0 / (2 \cdot 12.0) - 1832.0 / (\pi \cdot 456.0^2 \cdot 12.0)$$

$$= 22.49 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 1304.411 \cdot 456.0 / (2 \cdot 12.0) + 1832.0 / (\pi \cdot 456.0^2 \cdot 12.0)$$

$$= 27.08 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 1304.411 \cdot 456.0 / (2 \cdot 12.0) - 1121.3 / (0.1066 \cdot \pi \cdot 456.0^2 \cdot 12.0)$$

$$= 37.94 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 1304.411 \cdot 456.0 / (2 \cdot 12.0) + 1121.3 / (0.1923 \cdot \pi \cdot 456.0^2 \cdot 12.0)$$

$$= 17.49 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 7882(4170.0 - 2 \cdot 835.0) / (4170.0 + (4 \cdot 0.0 / 3))$$

$$= 4725.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 4725.22 / (456.0 \cdot 12.0)$$

$$= 9.91 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(456.0 \cdot 12.0)}$$

$$= 57.699 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(456.0 \cdot 12.0)}, 2 \cdot 835.0)$$

$$= 365.40 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(136.998/136.998, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 7882 \cdot 0.1 / (365.398(12.0 + 1.0 \cdot 10.0))$$

$$= -0.73 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [$\sigma_{7,r}$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -7882 / (4(12.0 + 1.0 \cdot 10.0)365.398) - 3 \cdot 0.053 \cdot 7882 / (2(12.0 + 1.0 \cdot 10.0)^2)$$

$$= -15.06 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:



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Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	530.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$
 $= ((1.56 \cdot \sqrt{(450.0 \cdot 12.0)} + 370.0) 12.0^3) / 12$
 $= 6.979 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$
 $= 6.979 + 58.156 (209.921 - 6.0)^2$
 $= 24190.533 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	484.6	12.0	6.0	58.2	34893.8	0.242E+05
Wearplate	370.0	10.0	17.0	37.0	62900.0	0.138E+05
Web	10.0	500.0	272.0	50.0	1359999.8	0.123E+05
BasePlate	275.0	18.0	531.0	49.5	2628449.8	0.510E+05
Totals	...	...	...	194.7	4086243.2	0.101E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 4086243 / 194.656$
 $= 209.921 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120.0 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 7881.661$
 $= 1604.0889 \text{ Kgf}$

Tension Stress, St = (Fh/As) = 1.1525 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$
 $= 990.0 - 450.0 \cdot \sin(120.0 / 2) / 1.0472$
 $= 617.853 \text{ mm}$

Bending Moment, M = Fh * d = 991.1107 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 2.0131 N./mm²



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Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{\frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$
 $= \sqrt{\frac{3(7882 + 171)275.0}{4 \cdot 980.0 \cdot 174.663}}$
 $= 9.755 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) (450.0 + 12.0 + 10.0)$
 $= 817.528 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 817.528 / (3 - 1)$
 $= 408.764 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 408.764 \cdot 275.0 / 2$
 $= 562.051 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 8053.121 / (980.0 \cdot 275.0)$
 $= 2.988 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 562.1 \cdot 2.99$
 $= 1679.503 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 24.0 + 20.438$
 $= 44.438 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 1679.5 / 44.4382$
 $= 3.706 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	25.0	...	0.130E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 250.0 - 10.0$
 $= 240.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 44.438$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 250.0 / 2$



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= 125.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{1302.1/44.438}$
 = 54.130 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 990.0 - cos(120.0/2)(450.0 + 12.0 + 10.0) - 18.0
 = 736.000 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0}$
 = 122.739

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 736.0/54.13
 = 13.597

Bending Moment [Rm]:
 $= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
 $= 752.8 / (2 \cdot 980.0) \cdot 408.764 \cdot 736.0/2$
 = 57.774 Kg-m.

Compressive Allowable, KL/r < Cc (13.5968 < 122.7394) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (13.6)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (13.6) / (8 \cdot 122.74) - (13.6^3) / (8 \cdot 122.74^3))$
 = 152.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= \text{Sc}/\text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$
 $= 3.71/152.45 + (57.77 \cdot 125.0/13020833) / 174.66$
 = 0.055

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	24.0	0.0	0.130E+04
Web	408.8	10.0	0.0	40.9	0.0	3.41
Totals	...	...	...	64.9	...	0.131E+04

Distance to Centroid from Datum [ytot]:
 $= \text{AY} / \text{A}$
 $= 0.0/64.876$
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 250.0/2
 = 125.000 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 990.0 - cos(472.0/2)(18.0 + 0.0 + 0.0) - 0.0
 = 500.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{1305.4/64.876}$



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= 44.857 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 500.0/44.857
 = 11.147

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 752.767/(2 • 980.0)
 = 0.384 Kgf/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 0.384 • 408.764 • 500.0
 = 78.497 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 78.497/104432.492
 = 7.371 N./mm²

Compressive Allowable, KL/r < Cc (11.1466 < 122.7394) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³)/(8•Cc³)
 = (1-(11.15)²/(2 • 122.74²))262/
 (5/3+3•(11.15)/(8• 122.74)-(11.15³)/(8•122.74³)
 = 153.4 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 5.08/153.42 + (78.5 • 125.0/1305.405)/174.66
 = 0.075

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	0.9592 cm ²
Saddle Load Q0 (Weight)	Q0	6685.8 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	298.1 Kgf
Maximum Transverse Force	Ft	376.4 Kgf
Maximum Longitudinal Force	F1	752.8 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

= 2058.2 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= max(Ff, F1) / (Bolt Area • Number of Bolts)
 = 2058.2/(0.959 • 4)
 = 52.6 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)
 = 376.4/(0.959 • 4)
 = 9.6 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 752.77/(82.8 • 4.0)
= 0.2229 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 990.0 • 376.38 + 0.0
= 372.63 Kg-m.

Eccentricity (e):
= Rmom / QO
= 372.63/6685.76
= 55.73 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.2229, 0.0]
= 0.2229 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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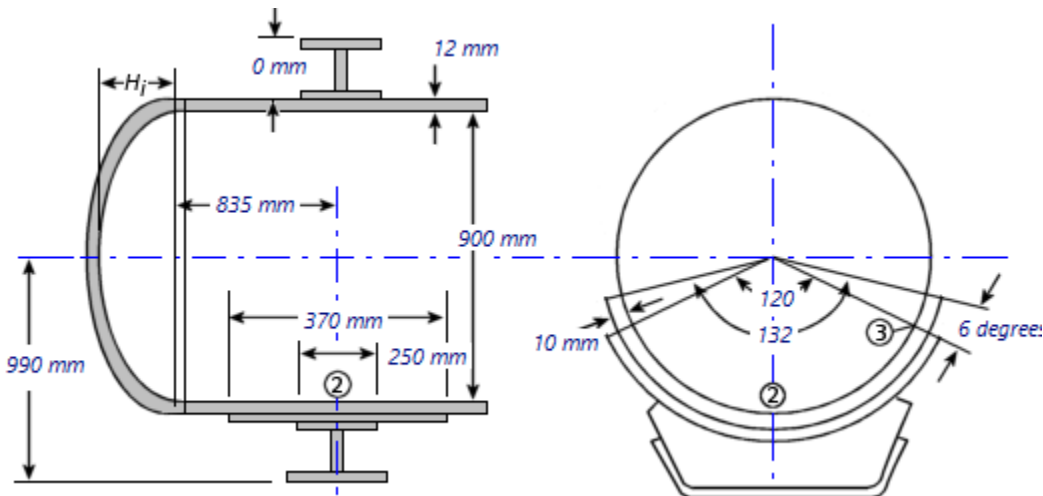
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	456.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	4170.00	mm
Distance from Saddle to Vessel tangent	a1 or a	835.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	370.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Shell Allowable Stress used in Calculation		184.50	N./mm ²
Head Allowable Stress used in Calculation		184.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	990.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	6398.69	Kgf
Pressure used in Analysis	P	4032.256 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	74.76	184.50
Long. Stress at Bottom of Midspan	78.48	184.50
Long. Stress at Top of Saddles	87.30	184.50
Long. Stress at Bottom of Saddles	70.70	184.50
Tangential Shear in Shell	8.05	109.26
Circ. Stress at Horn of Saddle	12.23	276.74
Circ. Compressive Stress in Shell	0.59	184.50



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(184.4/2 + 0) \cdot 990.0/817.528$$

$$= 334.9 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(61.39, 0) \cdot 990.0/2500.0$$

$$= 24.3 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 6064 + \max(24, 335, 0, 0)$$

$$= 6398.7 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(1.124 \cdot 1.2)^2)$$

$$= 1204.079 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	6570.15	Kgf
Transverse Shear Load	92.18	Kgf
Longitudinal Shear Load	61.39	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(900.0/2 + 12.0 + 10.0)\sin(120.0/2), 2 \cdot 456.0)$$

$$= 817.528 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -6399 \cdot 835.0 [1 - (1 - 835.0/4170.0 + (456.0^2 - 0.0^2)/(2 \cdot 835.0 \cdot 4170.0))/(1 + (4 \cdot 0.0)/(3 \cdot 4170.0))]$$

$$= -910.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 6399 \cdot 4170/4((1 + 2(456^2 - 0^2)/(4170^2))/(1 + (4 \cdot 0)/(3 \cdot 4170)) - 4 \cdot 835/4170)$$

$$= 1487.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 4032.256 \cdot 456.0/(2 \cdot 12.0) - 1487.3/(\pi \cdot 456.0^2 \cdot 12.0)$$

$$= 74.76 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$



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$$= 4032.256 \cdot 456.0 / (2 \cdot 12.0) + 1487.3 / (\pi \cdot 456.0^2 \cdot 12.0)$$

$$= 78.48 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 4032.256 \cdot 456.0 / (2 \cdot 12.0) - 910.3 / (0.1066 \cdot \pi \cdot 456.0^2 \cdot 12.0)$$

$$= 87.30 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4032.256 \cdot 456.0 / (2 \cdot 12.0) + 910.3 / (0.1923 \cdot \pi \cdot 456.0^2 \cdot 12.0)$$

$$= 70.70 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 6399(4170.0 - 2 \cdot 835.0) / (4170.0 + (4 \cdot 0.0 / 3))$$

$$= 3836.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3836.15 / (456.0 \cdot 12.0)$$

$$= 8.05 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(456.0 \cdot 12.0)}$$

$$= 57.699 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(456.0 \cdot 12.0)}, 2 \cdot 835.0)$$

$$= 365.40 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot t_r))$$

$$= -0.7603 \cdot 6399 \cdot 0.1 / (365.398(12.0 + 1.0 \cdot 10.0))$$

$$= -0.59 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot t_r)^2)$$

$$= -6399 / (4(12.0 + 1.0 \cdot 10.0) 365.398) - 3 \cdot 0.053 \cdot 6399 / (2(12.0 + 1.0 \cdot 10.0)^2)$$

$$= -12.23 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	530.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(450.0 \cdot 12.0)} + 370.0)^3) / 12$$

$$= 6.979 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 6.979 + 58.156(209.921 - 6.0)^2$$

$$= 24190.533 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	484.6	12.0	6.0	58.2	34893.8	0.242E+05
Wearplate	370.0	10.0	17.0	37.0	62900.0	0.138E+05
Web	10.0	500.0	272.0	50.0	1359999.8	0.123E+05
BasePlate	275.0	18.0	531.0	49.5	2628449.8	0.510E+05
Totals	...	...	...	194.7	4086243.2	0.101E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 4086243/194.656$$

$$= 209.921 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 6398.69$$

$$= 1302.2723 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.9356 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 990.0 - 450.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 617.853 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 804.6287 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.6343 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(3(6399 + 171) \cdot 275.0) / (4 \cdot 980.0 \cdot 173.33)}$$

$$= 8.845 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (450.0 + 12.0 + 10.0)$$

$$= 817.528 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 817.528 / (3 - 1)$$

$$= 408.764 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 408.764 \cdot 275.0 / 2$$

$$= 562.051 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 6570.15 / (980.0 \cdot 275.0)$$

$$= 2.438 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 562.1 \cdot 2.44$$

$$= 1370.225 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 24.0 + 20.438$$

$$= 44.438 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 1370.2 / 44.4382$$

$$= 3.024 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	25.0	...	0.130E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 250.0 - 10.0$$

$$= 240.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 44.438$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 250.0 / 2$$

$$= 125.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (1302.1 / 44.438) }$$

$$= 54.130 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 990.0 - \cos(120.0 / 2) (450.0 + 12.0 + 10.0) - 18.0$$

$$= 736.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994E+09 / 260.0) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 736.0/54.13$$

$$= 13.597$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 61.4 / (2 \cdot 980.0) \cdot 408.764 \cdot 736.0/2$$

$$= 4.712 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ (13.5968 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (13.6)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (13.6) / (8 \cdot 123.21) - (13.6^3) / (8 \cdot 123.21^3))$$

$$= 151.3 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 3.02/151.31 + (4.71 \cdot 125.0/13020833) / 173.33$$

$$= 0.023$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	24.0	0.0	0.130E+04
Web	408.8	10.0	0.0	40.9	0.0	3.41
Totals	...	...	...	64.9	...	0.131E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/64.876$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 990.0 - \cos(472.0/2) (18.0 + 0.0 + 0.0) - 0.0$$

$$= 500.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1305.4/64.876) }$$

$$= 44.857 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 500.0/44.857$$

$$= 11.147$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 61.392 / (2 \cdot 980.0)$$

$$= 0.031 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.031 \cdot 408.764 \cdot 500.0$$

$$= 6.402 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σB ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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= 6.402/104432.492
 = 0.601 N./mm²

Compressive Allowable, $KL/r < C_c$ (11.1466 < 123.2105) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 = $(1 - (11.15)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (11.15) / (8 \cdot 123.21) - (11.15^3) / (8 \cdot 123.21^3))$
 = 152.3 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 = $4.15 / 152.27 + (6.4 \cdot 125.0 / 1305.405) / 173.33$
 = 0.031

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	0.9592 cm ²
Saddle Load QO (Weight)	QO	6235.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	24.3 Kg
Maximum Transverse Force	Ft	92.2 Kg
Maximum Longitudinal Force	F1	61.4 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $F1 / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $61.4 / (0.959 \cdot 4)$
 = 1.6 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $92.2 / (0.959 \cdot 4)$
 = 2.4 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (BoltShearAllowable \cdot Nbolts)$
 = $61.39 / (28.98 \cdot 4.0)$
 = 0.0519 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + Sum\ of\ X\ Moments$
 = $990.0 \cdot 92.18 + 0.0$
 = 91.26 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = $91.26 / 6235.28$
 = 14.64 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)



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Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.0519, 0.0]
 = 0.0519 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	456.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	4170.00	mm
Distance from Saddle to Vessel tangent	a1 or a	835.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	370.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Shell Allowable Stress used in Calculation		184.50	N./mm ²
Head Allowable Stress used in Calculation		184.50	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	990.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		6444.33	Kgf
Pressure used in Analysis	P	4032.256	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	74.74	184.50
Long. Stress at Bottom of Midspan	78.49	184.50
Long. Stress at Top of Saddles	87.38	184.50
Long. Stress at Bottom of Saddles	70.65	184.50
Tangential Shear in Shell	8.11	109.26
Circ. Stress at Horn of Saddle	12.32	276.74
Circ. Compressive Stress in Shell	0.60	184.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3.0(184.4/2 + 0) • 990.0/817.528
 = 334.9 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(61.39, 0) • 990.0/2500.0
 = 24.3 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 6109 + max(24, 335, 0, 0)



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= 6444.3 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress (Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766.08(0.0 + $\pi/4$ (1.124 • 1.2)²)
 = 1204.079 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	6615.79	Kgf
Transverse Shear Load	92.18	Kgf
Longitudinal Shear Load	61.39	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(900.0/2 + 12.0 + 10.0)sin(120.0/2), 2•456.0)
 = 817.528 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -6444•835.0 [1 - (1 - 835.0/4170.0 + (456.0²-0.0²)/(2•835.0•4170.0))/(1+(4•0.0)/(3•4170.0))]
 = -916.8 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)
 = 6444•4170/4((1+2(456²-0²)/(4170²))/(1+(4•0)/(3•4170)) - 4•835/4170)
 = 1497.9 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = 4032.256 • 456.0/(2•12.0) - 1497.9/(π •456.0²•12.0)
 = 74.74 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 4032.256 • 456.0/(2 • 12.0) + 1497.9/(π • 456.0² • 12.0)
 = 78.49 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 4032.256•456.0/(2•12.0)-916.8/(0.1066• π •456.0²•12.0)
 = 87.38 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 4032.256•456.0/(2•12.0)+916.8/(0.1923• π •456.0²•12.0)
 = 70.65 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 6444(4170.0 - 2 • 835.0)/(4170.0 + (4 • 0.0/3))
 = 3863.5 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3863.51 / (456.0 \cdot 12.0)$$

$$= 8.11 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(456.0 \cdot 12.0)}$$

$$= 57.699 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(250.0 + 1.56 \cdot \sqrt{(456.0 \cdot 12.0)}, 2 \cdot 835.0)$$

$$= 365.40 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 6444 \cdot 0.1 / (365.398(12.0 + 1.0 \cdot 10.0))$$

$$= -0.60 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -6444 / (4(12.0 + 1.0 \cdot 10.0)365.398) -$$

$$3 \cdot 0.053 \cdot 6444 / (2(12.0 + 1.0 \cdot 10.0)^2)$$

$$= -12.32 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	530.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(450.0 \cdot 12.0)} + 370.0)12.0^3) / 12$$

$$= 6.979 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 6.979 + 58.156(209.921 - 6.0)^2$$

$$= 24190.533 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	484.6	12.0	6.0	58.2	34893.8	0.242E+05
Wearplate	370.0	10.0	17.0	37.0	62900.0	0.138E+05
Web	10.0	500.0	272.0	50.0	1359999.8	0.123E+05
BasePlate	275.0	18.0	531.0	49.5	2628449.8	0.510E+05
Totals	...	...	...	194.7	4086243.2	0.101E+06

Distance to Centroid [C1]:



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$$= AY / A$$

$$= 4086243/194.656$$

$$= 209.921 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 6444.33$$

$$= 1311.5608 \text{ Kgf}$$

Tension Stress, $St = (Fh/As) = 0.9423 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 990.0 - 450.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 617.853 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 810.3677 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 1.6460 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(6444 + 171) \cdot 275.0 / (4 \cdot 980.0 \cdot 173.33)}$$

$$= 8.876 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (450.0 + 12.0 + 10.0)$$

$$= 817.528 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 817.528 / (3 - 1)$$

$$= 408.764 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 408.764 \cdot 275.0/2$$

$$= 562.051 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 6615.79 / (980.0 \cdot 275.0)$$

$$= 2.455 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 562.1 \cdot 2.45$$

$$= 1379.743 \text{ Kgf}$$

Area of the Rib and Web [Ar]:



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= Rib Area + Web Area
 = 24.0 + 20.438
 = 44.438 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 1379.7/44.4382
 = 3.045 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	25.0	...	0.130E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 250.0 - 10.0
 = 240.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0.0/44.438
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 250.0/2
 = 125.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{(1302.1/44.438)}$
 = 54.130 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 990.0 - cos(120.0/2)(450.0 + 12.0 + 10.0) - 18.0
 = 736.000 mm

Intermediate Term [Cc]:
 = $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$
 = 123.211

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 736.0/54.13
 = 13.597

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 61.4/(2 • 980.0) • 408.764 • 736.0/2
 = 4.712 Kg-m.

Compressive Allowable, KL/r < Cc (13.5968 < 123.2105) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (13.6)^2 / (2 \cdot 123.21^2)) 260 /$
 = $(5/3 + 3 \cdot (13.6) / (8 \cdot 123.21) - (13.6^3) / (8 \cdot 123.21^3))$
 = 151.3 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba



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$$= 3.04/151.31 + (4.71 \cdot 125.0/13020833)/173.33$$

$$= 0.023$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	24.0	0.0	0.130E+04
Web	408.8	10.0	0.0	40.9	0.0	3.41
Totals	...	...	...	64.9	...	0.131E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/64.876$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 990.0 - \cos(472.0/2)(18.0 + 0.0 + 0.0) - 0.0$$

$$= 500.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1305.4/64.876)}$$

$$= 44.857 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 500.0/44.857$$

$$= 11.147$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 61.392/(2 \cdot 980.0)$$

$$= 0.031 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.031 \cdot 408.764 \cdot 500.0$$

$$= 6.402 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 6.402/104432.492$$

$$= 0.061 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (11.1466 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.15)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (11.15) / (8 \cdot 123.21) - (11.15^3) / (8 \cdot 123.21^3))$$

$$= 152.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 4.17/152.27 + (6.4 \cdot 125.0/1305.405)/173.33$$

$$= 0.031$$

Input Data for Base Plate Bolting Calculations:



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Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	1.5500 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	0.9592 cm ²
Saddle Load QO (Weight)	QO	6280.9 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	24.3 Kgf
Maximum Transverse Force	Ft	92.2 Kgf
Maximum Longitudinal Force	Fl	61.4 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 = $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $61.4 / (0.959 \cdot 4)$
 = 1.6 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $92.2 / (0.959 \cdot 4)$
 = 2.4 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $61.39 / (28.98 \cdot 4.0)$
 = 0.0519 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $990.0 \cdot 92.18 + 0.0$
 = 91.26 Kg-m.

Eccentricity (e):
 = R_{mom} / QO
 = $91.26 / 6280.92$
 = 14.53 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.0519, 0.0]
 = 0.0519 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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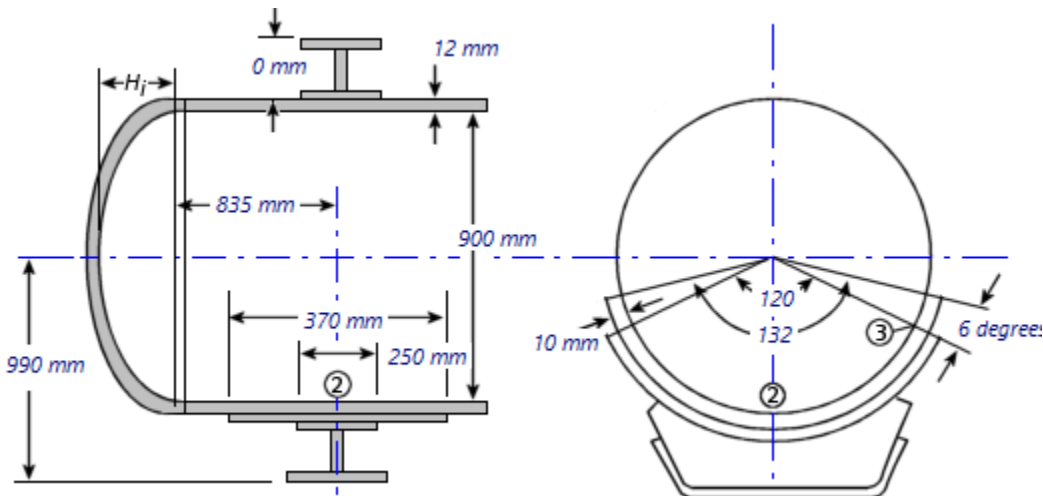
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	456.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	4170.00	mm
Distance from Saddle to Vessel tangent	a1 or a	835.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	370.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		106.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	990.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	8056.70	Kgf
Pressure used in Analysis	P	-103.419 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-4.31	-153.00
Long. Stress at Top of Midspan	4.31	137.00
Long. Stress at Bottom of Midspan	0.38	137.00
Long. Stress at Top of Saddles	11.49	137.00
Long. Stress at Bottom of Saddles	-9.42	-153.00
Long. Stress at Bottom of Saddles	9.42	137.00
Tangential Shear in Shell	10.13	90.61



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Circ. Stress at Horn of Saddle	15.40	171.25
Circ. Compressive Stress in Shell	0.75	137.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(368.7/2 + 0) \cdot 990.0/817.528$
 $= 669.7\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(122.78, 0) \cdot 990.0/2500.0$
 $= 48.6\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(752.77, 0) \cdot 990.0/2500.0$
 $= 298.1\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(753/2 + 0) \cdot 990.0/817.528$
 $= 1367.4\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 6689 + \max(49, 670, 298, 1367)$
 $= 8056.7\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766.08(0.0 + \pi/4(1.124 \cdot 1.2)^2)$
 $= 1204.079\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8228.16	Kgf
Transverse Shear Load	376.38	Kgf
Longitudinal Shear Load	752.77	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(900.0/2 + 12.0 + 10.0) \sin(120.0/2), 2 \cdot 456.0)$
 $= 817.528\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -8057 \cdot 835.0 [1 - (1 - 835.0/4170.0 + (456.0^2 - 0.0^2) / (2 \cdot 835.0 \cdot 4170.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 4170.0))]$
 $= -1146.2\ Kg \cdot m$

Moment per Equation 4.15.2 [M2]:



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$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 8057 \cdot 4170 / 4 \cdot ((1 + 2(456^2 - 0^2) / (4170^2)) / (1 + (4 \cdot 0) / (3 \cdot 4170)) - 4 \cdot 835 / 4170) \\
 &= 1872.7 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -103.419 \cdot 456.0 / (2 \cdot 12.0) - 1872.7 / (\pi \cdot 456.0^2 \cdot 12.0) \\
 &= -4.31 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -103.419 \cdot 456.0 / (2 \cdot 12.0) + 1872.7 / (\pi \cdot 456.0^2 \cdot 12.0) \\
 &= 0.38 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -103.419 \cdot 456.0 / (2 \cdot 12.0) - 1146.2 / (0.1066 \cdot \pi \cdot 456.0^2 \cdot 12.0) \\
 &= 11.49 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -103.419 \cdot 456.0 / (2 \cdot 12.0) + 1146.2 / (0.1923 \cdot \pi \cdot 456.0^2 \cdot 12.0) \\
 &= -9.42 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 8057(4170.0 - 2 \cdot 835.0) / (4170.0 + (4 \cdot 0.0 / 3)) \\
 &= 4830.2 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 4830.16 / (456.0 \cdot 12.0) \\
 &= 10.13 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(456.0 \cdot 12.0)} \\
 &= 57.699 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(250.0 + 1.56 \cdot \sqrt{(456.0 \cdot 12.0)}, 2 \cdot 835.0) \\
 &= 365.40 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr / S, 1) \\
 &= \min(136.998 / 136.998, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.7603 \cdot 8057 \cdot 0.1 / (365.398(12.0 + 1.0 \cdot 10.0)) \\
 &= -0.75 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\
 &= -8057 / (4(12.0 + 1.0 \cdot 10.0)365.398) - 3 \cdot 0.053 \cdot 8057 / (2(12.0 + 1.0 \cdot 10.0)^2) \\
 &= -15.40 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	275.0000	mm



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Number of Ribs (inc. outside ribs)	Nribs	3
Rib Thickness	Ribtk	10.0000 mm
Web Thickness	Webtk	10.0000 mm
Web Location	Webloc	Center
Saddle Design Temperature		21.1 °C
Saddle Yield Stress	Sy	262.0 N./mm²
Height of Web at Center	Hw,c	530.0 mm
Friction Coefficient	mu	0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(450.0 \cdot 12.0)} + 370.0)12.0^3)/12$
 $= 6.979 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 6.979 + 58.156(209.921 - 6.0)^2$
 $= 24190.533 \text{ cm}^4$

	B mm	D mm	Y mm	A cm²	AY mm³	I + AD^(2) cm⁴
Shell	484.6	12.0	6.0	58.2	34893.8	0.242E+05
Wearplate	370.0	10.0	17.0	37.0	62900.0	0.138E+05
Web	10.0	500.0	272.0	50.0	1359999.8	0.123E+05
BasePlate	275.0	18.0	531.0	49.5	2628449.8	0.510E+05
Totals	...	...	...	194.7	4086243.2	0.101E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 4086243/194.656$
 $= 209.921 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120.0/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 8056.698$
 $= 1639.7126 \text{ Kgf}$

Tension Stress, St = (Fh/As) = 1.1781 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 990.0 - 450.0 \cdot \sin(120.0/2) / 1.0472$
 $= 617.853 \text{ mm}$

Bending Moment, M = Fh * d = 1013.1213 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 2.0578 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$



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$$= \sqrt{3(8057 + 171)275.0 / (4 \cdot 980.0 \cdot 174.663)}$$

$$= 9.860 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (450.0 + 12.0 + 10.0)$$

$$= 817.528 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 817.528 / (3 - 1)$$

$$= 408.764 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 408.764 \cdot 275.0 / 2$$

$$= 562.051 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 8228.157 / (980.0 \cdot 275.0)$$

$$= 3.053 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 562.1 \cdot 3.05$$

$$= 1716.008 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 24.0 + 20.438$$

$$= 44.438 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 1716.0 / 44.4382$$

$$= 3.787 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	25.0	...	0.130E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 250.0 - 10.0$$

$$= 240.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 44.438$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0 / 2$$

$$= 125.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$



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$$= \sqrt{(1302.1/44.438)}$$

$$= 54.130 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 990.0 - \cos(120.0/2) (450.0 + 12.0 + 10.0) - 18.0$$

$$= 736.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 736.0/54.13$$

$$= 13.597$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 752.8 / (2 \cdot 980.0) \cdot 408.764 \cdot 736.0/2$$

$$= 57.774 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($13.5968 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.6)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (13.6) / (8 \cdot 122.74) - (13.6^3) / (8 \cdot 122.74^3))$$

$$= 152.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 3.79/152.45 + (57.77 \cdot 125.0/13020833) / 174.66$$

$$= 0.056$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	24.0	0.0	0.130E+04
Web	408.8	10.0	0.0	40.9	0.0	3.41
Totals	...	...	...	64.9	...	0.131E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/64.876$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 250.0/2$$

$$= 125.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 990.0 - \cos(472.0/2) (18.0 + 0.0 + 0.0) - 0.0$$

$$= 500.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1305.4/64.876)}$$

$$= 44.857 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



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$$= 1 \cdot 500.0 / 44.857$$

$$= 11.147$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 752.767 / (2 \cdot 980.0)$$

$$= 0.384 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.384 \cdot 408.764 \cdot 500.0$$

$$= 78.497 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 78.497 / 104432.492$$

$$= 7.371 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($11.1466 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.15)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (11.15) / (8 \cdot 122.74) - (11.15^3) / (8 \cdot 122.74^3))$$

$$= 153.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 5.19 / 153.42 + (78.5 \cdot 125.0 / 1305.405) / 174.66$$

$$= 0.076$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	456.00	mm
Shell Thickness used in this Case	t	12.000	mm
Stiffened Vessel Length per 4.15.6	L	4170.00	mm
Distance from Saddle to Vessel tangent	a1 or a	835.00	mm
Saddle Width	b1 or b	250.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	370.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm
Wear Plate Allowable Stress	fw or Sr	137.00	N./mm ²
Shell Allowable Stress used in Calculation		137.00	N./mm ²
Head Allowable Stress used in Calculation		106.20	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	990.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		7881.66	Kgf
Pressure used in Analysis	P	-103.419	KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	-4.26	-153.00
Long. Stress at Top	of Midspan	4.26	137.00
Long. Stress at Bottom	of Midspan	0.33	137.00



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Long. Stress at Top of Saddles	11.19	137.00
Long. Stress at Bottom of Saddles	-9.26	-153.00
Long. Stress at Bottom of Saddles	9.26	137.00
Tangential Shear in Shell	9.91	90.61
Circ. Stress at Horn of Saddle	15.06	171.25
Circ. Compressive Stress in Shell	0.73	137.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(368.7/2 + 0) \cdot 990.0/817.528$
 $= 669.7\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(122.78, 0) \cdot 990.0/2500.0$
 $= 48.6\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(752.77, 0) \cdot 990.0/2500.0$
 $= 298.1\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(753/2 + 0) \cdot 990.0/817.528$
 $= 1367.4\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 6514 + \max(49, 670, 298, 1367)$
 $= 7881.7\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766.08(0.0 + \pi/4(1.124 \cdot 1.2)^2)$
 $= 1204.079\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8053.12	Kgf
Transverse Shear Load	376.38	Kgf
Longitudinal Shear Load	752.77	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(900.0/2 + 12.0 + 10.0) \sin(120.0/2), 2 \cdot 456.0)$
 $= 817.528\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$



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$$= -7882 \cdot 835.0 [1 - (1 - 835.0/4170.0 + (456.0^2 - 0.0^2) / (2 \cdot 835.0 \cdot 4170.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 4170.0))] \\ = -1121.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ = 7882 \cdot 4170 / 4 \cdot ((1 + 2(456^2 - 0^2) / (4170^2)) / (1 + (4 \cdot 0) / (3 \cdot 4170)) - 4 \cdot 835 / 4170) \\ = 1832.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\ = -103.419 \cdot 456.0 / (2 \cdot 12.0) - 1832.0 / (\pi \cdot 456.0^2 \cdot 12.0) \\ = -4.26 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ = -103.419 \cdot 456.0 / (2 \cdot 12.0) + 1832.0 / (\pi \cdot 456.0^2 \cdot 12.0) \\ = 0.33 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = -103.419 \cdot 456.0 / (2 \cdot 12.0) - 1121.3 / (0.1066 \cdot \pi \cdot 456.0^2 \cdot 12.0) \\ = 11.19 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = -103.419 \cdot 456.0 / (2 \cdot 12.0) + 1121.3 / (0.1923 \cdot \pi \cdot 456.0^2 \cdot 12.0) \\ = -9.26 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ = 7882(4170.0 - 2 \cdot 835.0) / (4170.0 + (4 \cdot 0.0 / 3)) \\ = 4725.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 1.1707 \cdot 4725.22 / (456.0 \cdot 12.0) \\ = 9.91 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ = 0.78 \cdot \sqrt{(456.0 \cdot 12.0)} \\ = 57.699 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ = \min(250.0 + 1.56 \cdot \sqrt{(456.0 \cdot 12.0)}, 2 \cdot 835.0) \\ = 365.40 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1) \\ = \min(136.998/136.998, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\ = -0.7603 \cdot 7882 \cdot 0.1 / (365.398(12.0 + 1.0 \cdot 10.0)) \\ = -0.73 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -7882 / (4(12.0 + 1.0 \cdot 10.0)365.398) - 3 \cdot 0.053 \cdot 7882 / (2(12.0 + 1.0 \cdot 10.0)^2) \\ = -15.06 \text{ N./mm}^2$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	979.9999	mm
Baseplate Thickness	Bpthk	18.0000	mm
Baseplate Width	Bpwid	275.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	530.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(450.0 \cdot 12.0)} + 370.0)^3) / 12$
 $= 6.979 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 6.979 + 58.156 (209.921 - 6.0)^2$
 $= 24190.533 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	484.6	12.0	6.0	58.2	34893.8	0.242E+05
Wearplate	370.0	10.0	17.0	37.0	62900.0	0.138E+05
Web	10.0	500.0	272.0	50.0	1359999.8	0.123E+05
BasePlate	275.0	18.0	531.0	49.5	2628449.8	0.510E+05
Totals	...	...	...	194.7	4086243.2	0.101E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 4086243 / 194.656$
 $= 209.921 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120.0 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 7881.661$
 $= 1604.0889 \text{ Kg}$

Tension Stress, St = (Fh/As) = 1.1525 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$
 $= 990.0 - 450.0 \cdot \sin(120.0 / 2) / 1.0472$
 $= 617.853 \text{ mm}$

Bending Moment, M = Fh * d = 991.1107 Kg-m.



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Bending Stress, $S_b = (M * C_1 / I) = 2.0131 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(7882 + 171)275.0 / (4 * 980.0 * 174.663))}$
 $= 9.755 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120.0/2) (450.0 + 12.0 + 10.0)$
 $= 817.528 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 817.528 / (3 - 1)$
 $= 408.764 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 408.764 * 275.0 / 2$
 $= 562.051 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 8053.121 / (980.0 * 275.0)$
 $= 2.988 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 562.1 * 2.99$
 $= 1679.503 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 24.0 + 20.438$
 $= 44.438 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 1679.5 / 44.4382$
 $= 3.706 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	250.0	...	25.0	...	0.130E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 250.0 - 10.0$
 $= 240.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 44.438$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
 = 250.0/2
 = 125.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{1302.1/44.438}$
 = 54.130 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 990.0 - cos(120.0/2)(450.0 + 12.0 + 10.0) - 18.0
 = 736.000 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0}$
 = 122.739

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 736.0/54.13
 = 13.597

Bending Moment [Rm]:
 $= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 $= 752.8 / (2 \cdot 980.0) \cdot 408.764 \cdot 736.0/2$
 = 57.774 Kg-m.

Compressive Allowable, $KL/r < C_c$ (13.5968 < 122.7394) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (13.6)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (13.6) / (8 \cdot 122.74) - (13.6^3) / (8 \cdot 122.74^3))$
 = 152.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$
 $= 3.71/152.45 + (57.77 \cdot 125.0/13020833) / 174.66$
 = 0.055

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	240.0	0.0	24.0	0.0	0.130E+04
Web	408.8	10.0	0.0	40.9	0.0	3.41
Totals	...	...	...	64.9	...	0.131E+04

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0/64.876$
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 250.0/2
 = 125.000 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 990.0 - cos(472.0/2)(18.0 + 0.0 + 0.0) - 0.0
 = 500.000 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{(1305.4 / 64.876)}$$

$$= 44.857 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 500.0 / 44.857$$

$$= 11.147$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 752.767 / (2 \cdot 980.0)$$

$$= 0.384 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.384 \cdot 408.764 \cdot 500.0$$

$$= 78.497 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 78.497 / 104432.492$$

$$= 7.371 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($11.1466 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.15)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (11.15) / (8 \cdot 122.74) - (11.15)^3 / (8 \cdot 122.74^3))$$

$$= 153.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 5.08 / 153.42 + (78.5 \cdot 125.0 / 1305.405) / 174.66$$

$$= 0.075$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1-Tube Inlet

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Input, Nozzle Desc: N1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	520.000	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	133.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	900.00	mm
Design Length of Section	L	999.9695	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		694.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

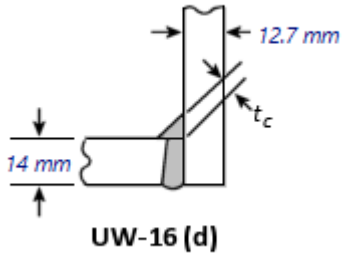
Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	133.20	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80S	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	423.8500	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
ASME Code Weld Type per UW-16		d	
Flange Class		150	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: N1-Tube Inlet Nozl: 9 6:22pm Sep 8,2026



Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 11.750 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 520.0 \cdot 450.0 / (133 \cdot 1.0 - 0.6 \cdot 520.0)$
 $= 1.7610 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 520.0 \cdot 149.22 / (133 \cdot 1.0 - 0.6 \cdot 520.0)$
 $= 0.5840 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8745 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	596.9000	mm
Parallel to Vessel Wall, opening length	d	298.4500	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		35.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 133.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 133.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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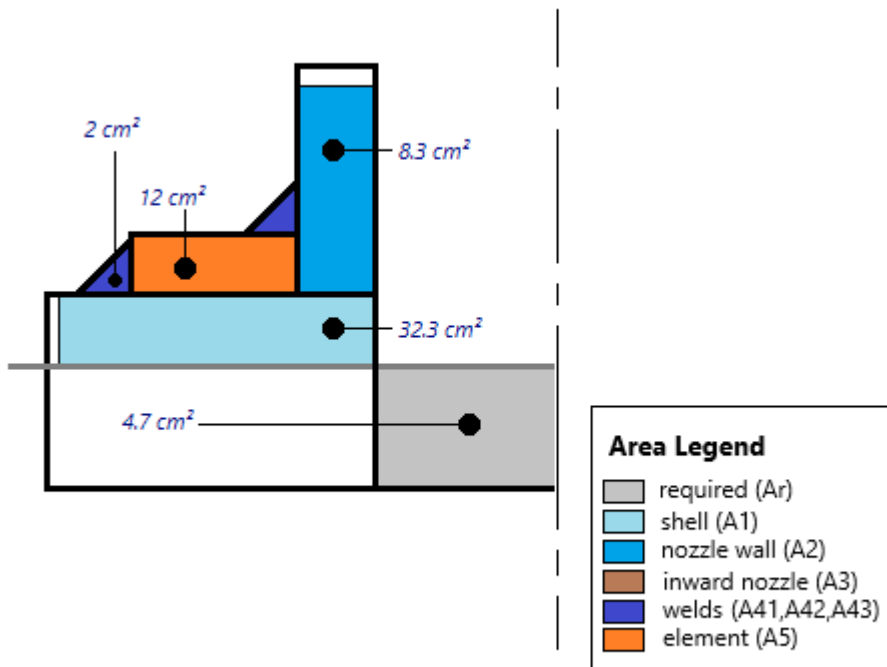


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5.256	4.721	NA
Area in Shell	A1	36.527	32.340	NA
Area in Nozzle Wall	A2	8.481	8.278	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.000	2.000	NA
Area in Element	A5	12.000	12.000	NA
TOTAL AREA AVAILABLE	Atot	59.009	54.618	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1 - fr1)) \text{ per UG-37(d)}$$

$$= 0.5(298.45 \cdot 3.1639 \cdot 1 + 2 \cdot 12.7 \cdot 3.1639 \cdot 1(1 - 1.0))$$

$$= 4.721 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 298.45(1.0 \cdot 14.0 - 1.0 \cdot 3.164) - 2 \cdot 12.7$$

$$(1.0 \cdot 14.0 - 1.0 \cdot 3.1639) \cdot (1 - 1.0)$$

$$= 32.340 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 35.0) \cdot (12.7 - 0.87) \cdot 1.0$$



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= 8.278 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 10.0^2 \cdot 1.0$
 = 2.000 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(423.85 - 323.85) \cdot 12.0 \cdot 1.0$
 = 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 0.8745 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 1.7610 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 1.7610 mm
Wall Thickness, shell/head, external pressure	trb2 = 0.3496 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[8.331, max(1.761, 1.5)]
 = 1.7610 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(0.8745, 1.761)
 = 1.7610 mm

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 20.3,	Allowable	: 133.2 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 318.7 N./mm ²	Passed
Occasional	: 2.9,	Allowable	: 177.2 N./mm ²	Passed
Shear	: 13.7,	Allowable	: 93.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11.113, tr = 0.564, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.051$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = -1.0$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code



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Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	6.0000 = 0.5*TminPad	7.0700 = 0.7 * Wp mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3045 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 30.1425 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 244.1426 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	900.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 133.20	N./mm ²

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	298.450	mm
Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 133.20	N./mm ²

Diameter of Reinforcing Pad	423.850	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	520.00	KPa.
Radial Load	P	-2294.51	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	ML	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



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Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{463.0(14.0 - 0.0)}$
 $= 589.536 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 926.000 mm
 Nozzle Outside Diameter (d) = 323.850 mm
 Vessel Thickness used (T) = 26.000 mm
 Nozzle Thickness used (t) = 12.700 mm
 $T / t = 2.047$
 $d / t = 25.500$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.087$$

Nr/P = 0.079
 Mr/P = 0.078
 M0/P = 0.047
 N0/P = 0.151
 MrD/Mc = 0.186
 NrDL/Mc = 0.117
 M0d/Mc = 0.129
 N0DL/Mc = 0.171
 MrD/Ml = 0.114
 NrDL/Ml = 0.069
 M0D/Ml = 0.072
 N0DL/Ml = 0.267

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-15	-15	15	15
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	8	9	8	9
Outplane Stress Summary	-16	14	63	-4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-39	39	39	-39
Inplane Pressure Stress	4	4	4	4
Inplane Stress Summary	-20	26	65	-43

Vessel Stresses (N./mm²):



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LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			4	4	4	4
Shear Stress Summary			5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress			24	28	67	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)			5	5	5	5
Outplane Bending (P)			9	-9	9	-9
Outplane Membrane (Mc)			-9	-9	9	9
Outplane Bending (Mc)			-44	44	44	-44
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			4	4	4	4
Outplane Stress Summary			-35	34	73	-34

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)			2	2	2	2
Inplane Bending (P)			15	-15	15	-15
Inplane Membrane (Mc)			-6	-6	6	6
Inplane Bending (Mc)			-64	64	64	-64
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			8	9	8	9
Inplane Stress Summary			-44	53	97	-61

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			4	4	4	4
Shear Stress Summary			3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress			45	54	99	62



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-15	-15	15	15
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	7	7	7
Hoop Stress Summary	-3	-2	27	28

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13
Axial Bending (ML)	-114	114	114	-114
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-90	74	165	-127

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	10	10	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	91	79	165	155

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	7	7	7
Hoop Stress Summary	2	2	21	22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-184	184	184	-184
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-160	144	235	-197
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8

Shear Stress Summary	6	6	10	10
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	163	144	235	221
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(457.0(14.0 - 0.0))}$$

$$= 587.809 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	914.000 mm
Nozzle Outside Diameter	(d) =	423.850 mm
Vessel Thickness used	(T) =	14.000 mm
Nozzle Thickness used	(t) =	62.700 mm
T / t	=	0.223
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.747$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.087
M0/P	=	0.027
N0/P	=	0.057
MrD/Mc	=	0.257
NrDL/Mc	=	0.090
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.072
NrDL/Ml	=	0.051
M0D/Ml	=	0.021
N0DL/Ml	=	0.062

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



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Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary	12	14	70	-5
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	8	8	8	8

Inplane Stress Summary	1	11	146	-103
------------------------	---	----	-----	------

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	6	6	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	17	19	146	103
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-70	70	70	-70
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	8	8	8	8

Outplane Stress Summary	-49	53	116	-61
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	5	5	5	5
-----------------------	---	---	---	---



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Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-234	234	234	-234
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	16	16	16	16

Inplane Stress Summary	-165	183	330	-257
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	165	183	330	257
------------------------	-----	-----	-----	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	9	8	9	8	9	8	9
Circ. Pl (SUS)	-10	-10	20	20	-4	-4	8	8
Circ. Q (SUS)	-15	15	33	-33	-49	49	79	-79
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-1	-1	5	5	-4	-4	14	14
Long. Q (SUS)	-24	24	54	-54	-35	35	53	-53
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	8.0	9.0	8.0	9.0	8.0	9.0	8.0	9.0
Pm+Pl (SUS)	5.4	4.5	28.1	29.0	4.5	5.4	18.4	18.6
Pm+Pl+Q (Total)	24.4	28.7	65.2	45.2	45.8	54.5	96.0	62.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	133.20	Passed
Pm+Pl (SUS)	29.05	199.80	Passed
Pm+Pl+Q (TOTAL)	96.00	399.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	16	16	16	16	16	16	16
Circ. Pl (SUS)	-3	-3	15	15	-8	-8	18	18
Circ. Q (SUS)	-1	1	37	-37	-175	175	293	-293
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-2	-2	12	12	-6	-6	18	18
Long. Q (SUS)	-6	6	124	-124	-52	52	88	-88
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	16.0	16.0	16.0	16.0	16.0	16.0	16.0	16.0
Pm+Pl (SUS)	13.1	13.1	31.1	31.1	8.2	8.2	34.1	34.1
Pm+Pl+Q (Total)	15.6	18.1	144.1	104.1	167.1	183.1	327.1	259.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	133.20	Passed
Pm+Pl (SUS)	34.12	199.80	Passed
Pm+Pl+Q (TOTAL)	327.12	399.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	7	7	7	7	7	7	7
Circ. Pl (SUS)	-10	-10	20	20	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-12	-12	14	14	-12	-12	14	14
Long. Q (SUS)	-83	83	145	-145	-153	153	215	-215
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0
Pm+Pl (SUS)	9.6	9.6	27.4	27.4	12.6	12.6	21.8	21.8
Pm+Pl+Q (Total)	93.1	79.6	162.3	155.5	165.4	144.3	232.5	219.9

Nozzle Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	133.20	Passed
Pm+Pl (SUS)	27.39	199.80	Passed
Pm+Pl+Q (TOTAL)	232.47	399.60	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N3-Shell inlet From: 40

Pressure for Reinforcement Calculations	P	1300.000	KPa.
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	137.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	900.00	mm
Design Length of Section	L	4169.9995	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2584.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

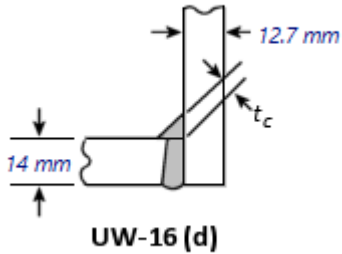
Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	133.20	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80S	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	423.8500	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
ASME Code Weld Type per UW-16		d	
Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N3-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 11.750 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1300.0 \cdot 450.0 / (137 \cdot 1.0 - 0.6 \cdot 1300.0)$
 $= 4.2948 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1300.0 \cdot 149.22 / (133 \cdot 1.0 - 0.6 \cdot 1300.0)$
 $= 1.4651 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8698 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	596.9000	mm
Parallel to Vessel Wall, opening length	d	298.4500	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		35.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 137.0)$
 $= 0.972$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 137.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.972, 1.0)$
 $= 0.972$



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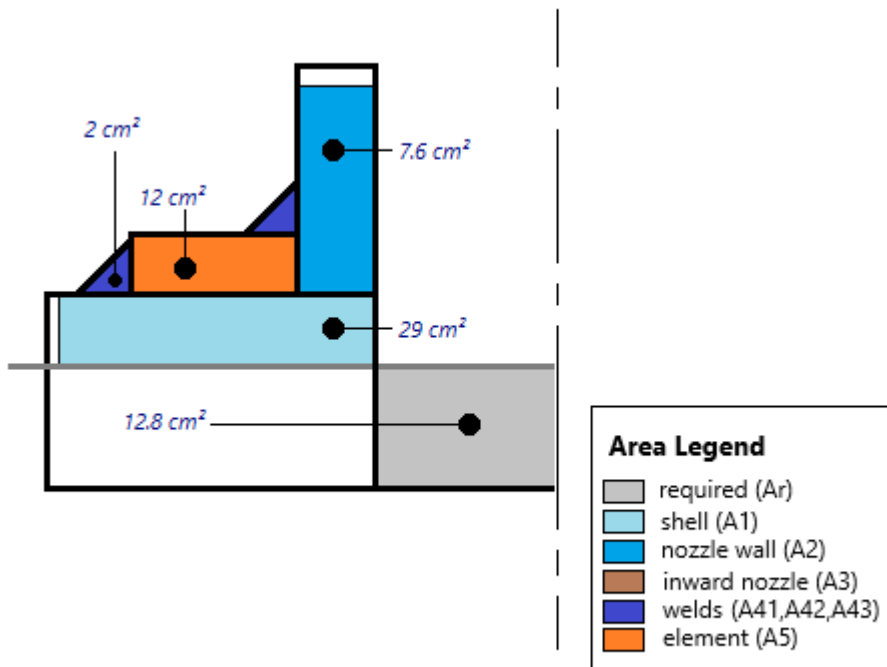


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.818	8.527	NA
Area in Shell	A1	28.965	24.729	NA
Area in Nozzle Wall	A2	7.646	8.052	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.972	1.972	NA
Area in Element	A5	12.000	12.000	NA
TOTAL AREA AVAILABLE	Atot	50.584	46.753	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (298.45 \cdot 4.2948 \cdot 1.0 + 2 \cdot 12.7 \cdot 4.2948 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 12.818 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 298.45(1.0 \cdot 14.0 - 1.0 \cdot 4.295) - 2 \cdot 12.7$$

$$(1.0 \cdot 14.0 - 1.0 \cdot 4.2948) \cdot (1 - 1.0)$$

$$= 28.965 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 35.0) \cdot (12.7 - 1.47) \cdot 0.9723$$



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= 7.646 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 0.97 + (0.0)^2 \cdot 0.97 + 10.0^2 \cdot 1.0$
 = 1.972 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$
 = $(423.85 - 323.85) \cdot 12.0 \cdot 1.0$
 = 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.4651 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 4.2948 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 4.2948 mm
Wall Thickness, shell/head, external pressure	trb2 = 0.3399 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[8.331, max(4.2948, 1.5)]
 = 4.2948 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.4651, 4.2948)
 = 4.2948 mm

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	24.7,	Allowable	:	133.2 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	314.3 N./mm ²	Passed
Occasional	:	7.3,	Allowable	:	177.2 N./mm ²	Passed
Shear	:	13.7,	Allowable	:	93.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11.113, tr = 1.414, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.127$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = -1.0$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code



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Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N3-Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	6.0000 = 0.5*TminPad	7.0700 = 0.7 * Wp mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3118 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 30.1425 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 244.1426 mm

Input Echo, WRC297 Item 1, Description: N3-Shell inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	900.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.00	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	298.450	mm
Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 133.20	N./mm ²
Diameter of Reinforcing Pad	423.850	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	1300.00	KPa.
Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	ML	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No



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Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{463.0(14.0 - 0.0)}$
 $= 589.536 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 926.000 mm
 Nozzle Outside Diameter (d) = 323.850 mm
 Vessel Thickness used (T) = 26.000 mm
 Nozzle Thickness used (t) = 12.700 mm
 $T / t = 2.047$
 $d / t = 25.500$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.087$$

Nr/P = 0.079
 Mr/P = 0.078
 M0/P = 0.047
 N0/P = 0.151
 MrD/Mc = 0.186
 NrDL/Mc = 0.117
 M0d/Mc = 0.129
 N0DL/Mc = 0.171
 MrD/Ml = 0.114
 NrDL/Ml = 0.069
 M0D/Ml = 0.072
 N0DL/Ml = 0.267

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-15	-15	15	15
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	21	23	21	23
Outplane Stress Summary	-3	28	76	9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-39	39	39	-39
Inplane Pressure Stress	10	10	10	10
Inplane Stress Summary	-14	33	72	-37

Vessel Stresses (N./mm²):



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LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			4	4	4	4
Shear Stress Summary			5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			16	36	78	46

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			5	5	5	5
Outplane Bending (P)			9	-9	9	-9
Outplane Membrane (Mc)			-9	-9	9	9
Outplane Bending (Mc)			-44	44	44	-44
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			10	10	10	10
Outplane Stress Summary			-28	41	79	-28

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			2	2	2	2
Inplane Bending (P)			15	-15	15	-15
Inplane Membrane (Mc)			-6	-6	6	6
Inplane Bending (Mc)			-64	64	64	-64
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			21	23	21	23
Inplane Stress Summary			-30	67	111	-47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			4	4	4	4
Shear Stress Summary			3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			33	68	111	48



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-15	-15	15	15
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	19	17	19
Hoop Stress Summary	7	8	37	39

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13
Axial Bending (ML)	-114	114	114	-114
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-85	79	170	-121

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	10	10	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	94	80	170	162

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	19	17	19
Hoop Stress Summary	12	14	32	33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------



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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-184	184	184	-184
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-155	149	240	-192
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8
Shear Stress Summary	6	6	10	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	168	149	241	227

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(457.0(14.0 - 0.0))}$$

$$= 587.809 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	914.000 mm
Nozzle Outside Diameter	(d) =	423.850 mm
Vessel Thickness used	(T) =	14.000 mm
Nozzle Thickness used	(t) =	62.700 mm
T / t	=	0.223
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.747$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.087
M0/P	=	0.027
N0/P	=	0.057
MrD/Mc	=	0.257
NrDL/Mc	=	0.090
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.072
NrDL/Ml	=	0.051
M0D/Ml	=	0.021
N0DL/Ml	=	0.062

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



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Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	41	42	41	42

Outplane Stress Summary	37	40	95	20
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	20	20	20	20

Inplane Stress Summary	13	24	159	-90
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	6	6	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	39	42	159	111
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-70	70	70	-70
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20

Outplane Stress Summary	-37	65	128	-48
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	5	5	5	5
-----------------------	---	---	---	---



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Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-234	234	234	-234
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	41	42	41	42
Inplane Stress Summary	-141	208	354	-231

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1	1
Torsional Shear (Mt)	4	4	4	4	4
Shear Stress Summary	2	2	6	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	141	209	354	232	232

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	23	21	23	21	23	21	23
Circ. Pl (SUS)	-10	-10	20	20	-4	-4	8	8
Circ. Q (SUS)	-15	15	33	-33	-49	49	79	-79
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-1	-1	5	5	-4	-4	14	14
Long. Q (SUS)	-24	24	54	-54	-35	35	53	-53
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	21.0	23.0	21.0	23.0	21.0	23.0	21.0	23.0
Pm+Pl (SUS)	11.4	13.2	41.0	43.0	17.1	19.1	29.2	31.1
Pm+Pl+Q (Total)	16.9	36.1	75.4	49.4	33.9	68.3	108.8	49.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23.00	137.00	Passed
Pm+Pl (SUS)	43.04	205.50	Passed
Pm+Pl+Q (TOTAL)	108.79	410.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	41	42	41	42	41	42	41	42
Circ. Pl (SUS)	-3	-3	15	15	-8	-8	18	18
Circ. Q (SUS)	-1	1	37	-37	-175	175	293	-293
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-2	-2	12	12	-6	-6	18	18
Long. Q (SUS)	-6	6	124	-124	-52	52	88	-88
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	41.0	42.0	41.0	42.0	41.0	42.0	41.0	42.0
Pm+Pl (SUS)	38.0	39.0	56.0	57.0	33.1	34.0	59.0	60.0
Pm+Pl+Q (Total)	38.0	41.4	156.1	112.2	142.1	209.1	352.1	233.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	42.00	137.00	Passed
Pm+Pl (SUS)	60.05	205.50	Passed
Pm+Pl+Q (TOTAL)	352.11	410.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	19	17	19	17	19	17	19
Circ. Pl (SUS)	-10	-10	20	20	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-12	-12	14	14	-12	-12	14	14
Long. Q (SUS)	-83	83	145	-145	-153	153	215	-215
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	17.0	19.0	17.0	19.0	17.0	19.0	17.0	19.0
Pm+Pl (SUS)	11.7	13.6	37.3	39.2	17.5	19.4	31.4	33.4
Pm+Pl+Q (Total)	96.1	80.4	167.3	162.4	170.4	149.3	237.5	226.9

Nozzle Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	133.20	Passed
Pm+Pl (SUS)	39.23	199.80	Passed
Pm+Pl+Q (TOTAL)	237.48	399.60	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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 Nozzle Calcs.: N4-Shell outlet Nozl: 11 6:22pm Sep 8,2026

Input, Nozzle Desc: N4-Shell outlet

From: 40

Pressure for Reinforcement Calculations	P	1308.822	KPa.
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-240	304
Parent Allowable Stress at Temperature	Sv	137.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	900.00	mm
Design Length of Section	L	4169.9995	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4034.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

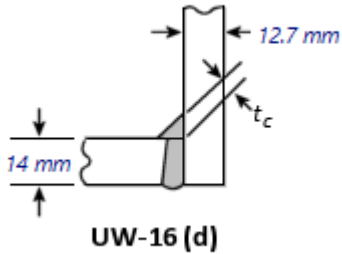
Type of Element Connected to the Parent : Nozzle

Material		SA-965	F304
Material UNS Number		S30400	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	130.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80S	
Flange Material		SA-182	F304
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm
Pad Material		SA-516	70
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	423.8500	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
ASME Code Weld Type per UW-16		d	
Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: N4-Shell outlet Nozl: 11 6:22pm Sep 8,2026



Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 11.750 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1308.82 \cdot 450.0 / (137 \cdot 1.0 - 0.6 \cdot 1308.82)$
 $= 4.3242 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1308.82 \cdot 149.22 / (130 \cdot 1.0 - 0.6 \cdot 1308.82)$
 $= 1.5116 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8698 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	596.9000	mm
Parallel to Vessel Wall, opening length	d	298.4500	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		35.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 130.0 / 137.0)$
 $= 0.949$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 137.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.949, 1.0)$
 $= 0.949$



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Nozzle Calcs.: N4-Shell outlet

Nozl:

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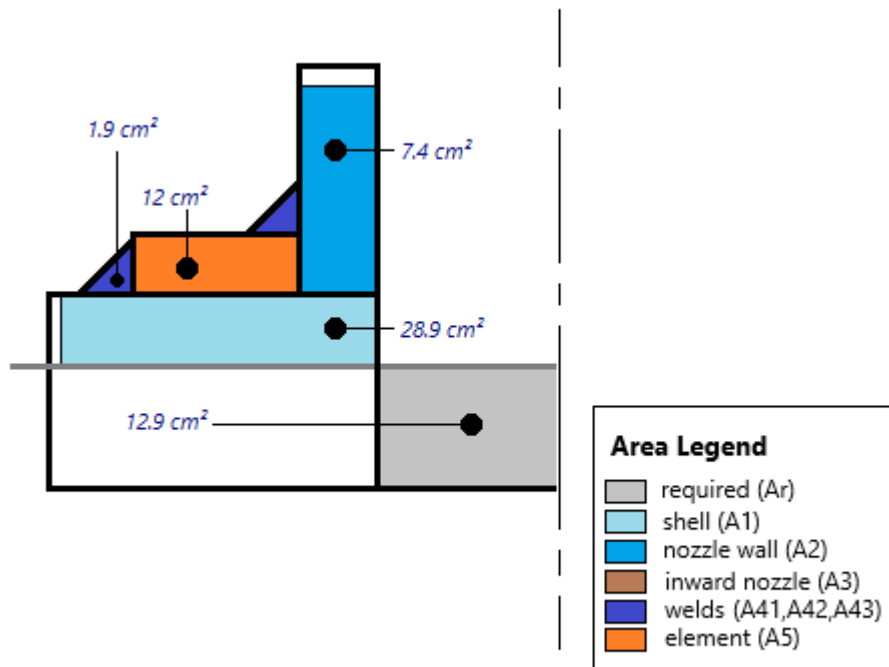


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	12.905	8.527	NA
Area in Shell	A1	28.878	24.729	NA
Area in Nozzle Wall	A2	7.432	7.858	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.949	1.949	NA
Area in Element	A5	12.000	12.000	NA
TOTAL AREA AVAILABLE	Atot	50.258	46.536	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (298.45 \cdot 4.3242 \cdot 1.0 + 2 \cdot 12.7 \cdot 4.3242 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 12.905 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 298.45(1.0 \cdot 14.0 - 1.0 \cdot 4.324) - 2 \cdot 12.7 \\
 &\quad (1.0 \cdot 14.0 - 1.0 \cdot 4.3242) \cdot (1 - 1.0) \\
 &= 28.878 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 35.0) \cdot (12.7 - 1.51) \cdot 0.9489
 \end{aligned}$$



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= 7.432 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 0.95 + (0.0)^2 \cdot 0.95 + 10.0^2 \cdot 1.0$
 = 1.949 cm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$
 = $(423.85 - 323.85) \cdot 12.0 \cdot 1.0$
 = 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.5116 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 4.3242 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 4.3242 mm
Wall Thickness, shell/head, external pressure	trb2 = 0.3399 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[8.331, max(4.3242, 1.5)]
 = 4.3242 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.5116, 4.3242)
 = 4.3242 mm

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	24.7,	Allowable	:	130.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	310.3 N./mm ²	Passed
Occasional	:	7.4,	Allowable	:	172.9 N./mm ²	Passed
Shear	:	13.7,	Allowable	:	91.0 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 11.113, tr = 1.423, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.128$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = -1.0$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code



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 Nozzle Calcs.: N4-Shell outlet Nozl: 11 6:22pm Sep 8,2026

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N4-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	6.0000 = 0.5*TminPad	7.0700 = 0.7 * Wp mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3098 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 30.1425 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 244.1426 mm

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	900.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 137.00	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	298.450	mm
Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-965 F304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 130.00	N./mm ²
Diameter of Reinforcing Pad	423.850	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	1308.82	KPa.
Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	ML	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No



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 Nozzle Calcs.: N4-Shell outlet Nozl: 11 6:22pm Sep 8,2026

Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{463.0(14.0 - 0.0)}$
 $= 589.536 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 926.000 mm
 Nozzle Outside Diameter (d) = 323.850 mm
 Vessel Thickness used (T) = 26.000 mm
 Nozzle Thickness used (t) = 12.700 mm
 $T / t = 2.047$
 $d / t = 25.500$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.087$$

Nr/P = 0.079
 Mr/P = 0.078
 M0/P = 0.047
 N0/P = 0.151
 MrD/Mc = 0.186
 NrDL/Mc = 0.117
 M0d/Mc = 0.129
 N0DL/Mc = 0.171
 MrD/Ml = 0.114
 NrDL/Ml = 0.069
 M0D/Ml = 0.072
 N0DL/Ml = 0.267

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-15	-15	15	15
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	22	23	22	23
Outplane Stress Summary	-3	28	76	9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-39	39	39	-39
Inplane Pressure Stress	11	11	11	11
Inplane Stress Summary	-13	33	72	-37

Vessel Stresses (N./mm²):



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Nozzle Calcs.: N4-Shell outlet

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LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			4	4	4	4
Shear Stress Summary			5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress			16	36	78	47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)			5	5	5	5
Outplane Bending (P)			9	-9	9	-9
Outplane Membrane (Mc)			-9	-9	9	9
Outplane Bending (Mc)			-44	44	44	-44
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			11	11	11	11
Outplane Stress Summary			-28	41	79	-27

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)			2	2	2	2
Inplane Bending (P)			15	-15	15	-15
Inplane Membrane (Mc)			-6	-6	6	6
Inplane Bending (Mc)			-64	64	64	-64
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			22	23	22	23
Inplane Stress Summary			-30	67	111	-47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			4	4	4	4
Shear Stress Summary			3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress			33	68	112	48



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 Nozzle Calcs.: N4-Shell outlet Nozl: 11 6:22pm Sep 8,2026

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-15	-15	15	15
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	19	17	19
Hoop Stress Summary	7	8	38	39

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13
Axial Bending (ML)	-114	114	114	-114
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-85	79	170	-121

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	10	10	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	95	81	170	162

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	19	17	19
Hoop Stress Summary	12	14	32	34

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N4-Shell outlet Nozl: 11 6:22pm Sep 8,2026

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-184	184	184	-184
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-155	149	240	-192
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8
Shear Stress Summary	6	6	10	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	168	149	241	227

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(457.0(14.0 - 0.0))}$$

$$= 587.809 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	914.000 mm
Nozzle Outside Diameter (d)	=	423.850 mm
Vessel Thickness used (T)	=	14.000 mm
Nozzle Thickness used (t)	=	62.700 mm
T / t	=	0.223
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.747$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.087
M0/P	=	0.027
N0/P	=	0.057
MrD/Mc	=	0.257
NrDL/Mc	=	0.090
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.072
NrDL/Ml	=	0.051
M0D/Ml	=	0.021
N0DL/Ml	=	0.062

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside



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Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	41	42	41	42

Outplane Stress Summary	37	40	95	20
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	20	20	20	20

Inplane Stress Summary	13	24	159	-90
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	6	6	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	39	42	159	111
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-70	70	70	-70
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20

Outplane Stress Summary	-37	65	129	-48
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	5	5	5	5
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Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-234	234	234	-234
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	41	42	41	42
Inplane Stress Summary	-140	209	355	-231

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1	1
Torsional Shear (Mt)	4	4	4	4	4
Shear Stress Summary	2	2	6	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	140	209	355	231	231

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	23	22	23	22	23	22	23
Circ. Pl (SUS)	-10	-10	20	20	-4	-4	8	8
Circ. Q (SUS)	-15	15	33	-33	-49	49	79	-79
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-1	-1	5	5	-4	-4	14	14
Long. Q (SUS)	-24	24	54	-54	-35	35	53	-53
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	22.0	23.0	22.0	23.0	22.0	23.0	22.0	23.0
Pm+Pl (SUS)	12.4	13.3	42.0	43.0	18.1	19.1	30.2	31.2
Pm+Pl+Q (Total)	15.9	36.8	76.4	48.4	32.9	68.3	109.8	49.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	23.00	137.00	Passed
Pm+Pl (SUS)	43.04	205.50	Passed
Pm+Pl+Q (TOTAL)	109.79	410.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	41	42	41	42	41	42	41	42
Circ. Pl (SUS)	-3	-3	15	15	-8	-8	18	18
Circ. Q (SUS)	-1	1	37	-37	-175	175	293	-293
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-2	-2	12	12	-6	-6	18	18
Long. Q (SUS)	-6	6	124	-124	-52	52	88	-88
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	41.0	42.0	41.0	42.0	41.0	42.0	41.0	42.0
Pm+Pl (SUS)	38.0	39.0	56.0	57.0	33.1	34.0	59.0	60.0
Pm+Pl+Q (Total)	38.0	41.4	156.1	112.2	142.1	209.1	352.1	233.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	42.00	137.00	Passed
Pm+Pl (SUS)	60.05	205.50	Passed
Pm+Pl+Q (TOTAL)	352.11	410.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	19	17	19	17	19	17	19
Circ. Pl (SUS)	-10	-10	20	20	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-12	-12	14	14	-12	-12	14	14
Long. Q (SUS)	-83	83	145	-145	-153	153	215	-215
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	17.0	19.0	17.0	19.0	17.0	19.0	17.0	19.0
Pm+Pl (SUS)	11.7	13.6	37.3	39.2	17.5	19.4	31.4	33.4
Pm+Pl+Q (Total)	96.1	80.4	167.3	162.4	170.4	149.3	237.5	226.9

Nozzle Stress Summation Comparison (N./mm²):



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Nozzle Calcs.: N4-Shell outlet

Nozl: 11

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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	19.00	130.00	Passed
Pm+Pl (SUS)	39.23	195.00	Passed
Pm+Pl+Q (TOTAL)	237.48	390.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N2-Tube Outlet

Nozl: 12

6:22pm

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Input, Nozzle Desc: N2-Tube Outlet

From: 50

Pressure for Reinforcement Calculations	P	528.822	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-240 304	
Parent Allowable Stress at Temperature	Sv	133.20	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	900.00	mm
Design Length of Section	L	1000.0314	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5944.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

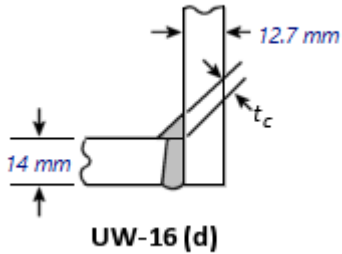
Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	133.20	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80S	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	424.0000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0750	mm
ASME Code Weld Type per UW-16		d	
Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 11.750 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 528.82 \cdot 450.0 / (133 \cdot 1.0 - 0.6 \cdot 528.82)$
 $= 1.7909 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 528.82 \cdot 149.22 / (133 \cdot 1.0 - 0.6 \cdot 528.82)$
 $= 0.5939 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8745 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	596.9000	mm
Parallel to Vessel Wall, opening length	d	298.4500	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		35.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 133.2 / 133.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 133.2)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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Nozl:

12

6:22pm

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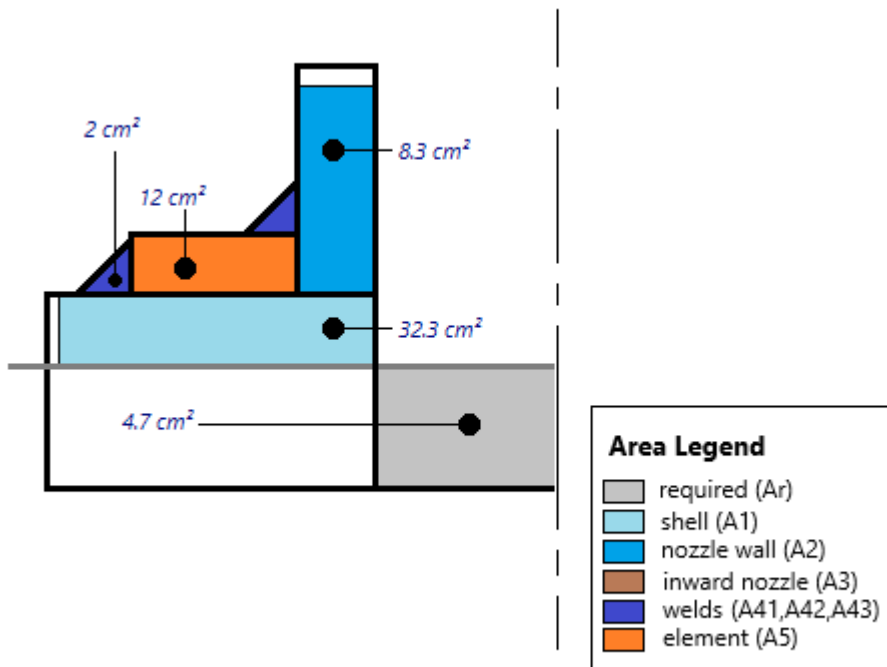


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5.345	4.721	NA
Area in Shell	A1	36.438	32.340	NA
Area in Nozzle Wall	A2	8.474	8.278	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.000	2.000	NA
Area in Element	A5	12.018	12.018	NA
TOTAL AREA AVAILABLE	Atot	58.930	54.636	NA

The External Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$= 0.5(d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F(1-fr1)) \text{ per UG-37(d)}$$

$$= 0.5(298.45 \cdot 3.1639 \cdot 1 + 2 \cdot 12.7 \cdot 3.1639 \cdot 1(1-1.0))$$

$$= 4.721 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 298.45(1.0 \cdot 14.0 - 1.0 \cdot 3.164) - 2 \cdot 12.7$$

$$(1.0 \cdot 14.0 - 1.0 \cdot 3.1639) \cdot (1 - 1.0)$$

$$= 32.340 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 35.0) \cdot (12.7 - 0.87) \cdot 1.0$$



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$$= 8.278 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 10.0^2 \cdot 1.0$
 $= 2.000 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot fr_4$
 $= (424.0 - 323.85) \cdot 12.0 \cdot 1.0$
 $= 12.018 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 0.8745 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 1.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 1.7909 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 1.7909 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 0.3496 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 1.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[8.331, \max(1.7909, 1.5)]$
 $= 1.7909 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(0.8745, 1.7909)$
 $= 1.7909 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 20.3	Allowable	: 133.2 N./mm ²	Passed
Expansion	: 0.0	Allowable	: 318.7 N./mm ²	Passed
Occasional	: 3.0	Allowable	: 177.2 N./mm ²	Passed
Shear	: 13.7	Allowable	: 93.2 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 11.113$, $t_r = 0.573$, $c = 0.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.052$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 0.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = -1.0$, Temp. Reduction = $0 \text{ }^\circ\text{C}$
 Attachment governing, Attachment stress = Shell stress per Code



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Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N2-Tube Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 12.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 12.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	6.0000 = 0.5*TminPad	7.0700 = 0.7 * Wp mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 3037 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 30.1425 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 244.1426 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	0.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	900.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 133.20	N./mm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	298.450	mm
Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-312 TP304	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 133.20	N./mm ²
Diameter of Reinforcing Pad	424.000	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	528.82	KPa.
Radial Load	P	-2294.51	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	ML	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No



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Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{463.0(14.0 - 0.0)}$
 $= 589.536 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 926.000 mm
 Nozzle Outside Diameter (d) = 323.850 mm
 Vessel Thickness used (T) = 26.000 mm
 Nozzle Thickness used (t) = 12.700 mm
 $T / t = 2.047$
 $d / t = 25.500$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.087$$

Nr/P = 0.079
 Mr/P = 0.078
 M0/P = 0.047
 N0/P = 0.151
 MrD/Mc = 0.186
 NrDL/Mc = 0.117
 M0d/Mc = 0.129
 N0DL/Mc = 0.171
 MrD/Ml = 0.114
 NrDL/Ml = 0.069
 M0D/Ml = 0.072
 N0DL/Ml = 0.267

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-15	-15	15	15
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	8	9	8	9
Outplane Stress Summary	-16	14	63	-4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-39	39	39	-39
Inplane Pressure Stress	4	4	4	4
Inplane Stress Summary	-20	26	65	-43

Vessel Stresses (N./mm²):



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LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			4	4	4	4
Shear Stress Summary			5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			24	28	67	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			5	5	5	5
Outplane Bending (P)			9	-9	9	-9
Outplane Membrane (Mc)			-9	-9	9	9
Outplane Bending (Mc)			-44	44	44	-44
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			4	4	4	4
Outplane Stress Summary			-35	34	73	-34

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			2	2	2	2
Inplane Bending (P)			15	-15	15	-15
Inplane Membrane (Mc)			-6	-6	6	6
Inplane Bending (Mc)			-64	64	64	-64
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			8	9	8	9
Inplane Stress Summary			-43	54	98	-61

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			4	4	4	4
Shear Stress Summary			3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress			44	54	99	62



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-15	-15	15	15
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	7	7	7
Hoop Stress Summary	-3	-2	27	28

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13
Axial Bending (ML)	-114	114	114	-114
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-90	74	165	-127

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	10	10	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	91	79	165	155

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	7	7	7
Hoop Stress Summary	2	2	22	22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	1	1	1	1
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-184	184	184	-184
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-160	144	235	-197
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8
Shear Stress Summary	6	6	10	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	163	144	235	221

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(457.0(14.0 - 0.0))}$$

$$= 587.809 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	914.000 mm
Nozzle Outside Diameter (d)	=	424.000 mm
Vessel Thickness used (T)	=	14.000 mm
Nozzle Thickness used (t)	=	62.775 mm
T / t	=	0.223
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.748$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.087
M0/P	=	0.027
N0/P	=	0.057
MrD/Mc	=	0.257
NrDL/Mc	=	0.090
M0d/Mc	=	0.077
N0DL/Mc	=	0.085
MrD/Ml	=	0.072
NrDL/Ml	=	0.051
M0D/Ml	=	0.021
N0DL/Ml	=	0.062

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside



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Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	16	17	16	17

Outplane Stress Summary	13	15	70	-4
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-65	65	65	-65
Inplane Pressure Stress	8	8	8	8

Inplane Stress Summary	1	11	146	-102
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	6	6	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	17	19	146	103

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-70	70	70	-70
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	8	8	8	8

Outplane Stress Summary	-49	53	116	-61
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	5	5	5	5



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Inplane Bending (P)	59	-59	59	-59
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-234	234	234	-234
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	16	17	16	17
Inplane Stress Summary	-165	183	330	-257

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		4	4	4	4
Shear Stress Summary		2	2	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		165	183	330	257

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	9	8	9	8	9	8	9
Circ. Pl (SUS)	-10	-10	20	20	-4	-4	8	8
Circ. Q (SUS)	-15	15	33	-33	-49	49	79	-79
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-1	-1	5	5	-4	-4	14	14
Long. Q (SUS)	-24	24	54	-54	-35	35	53	-53
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	8.0	9.0	8.0	9.0	8.0	9.0	8.0	9.0
Pm+Pl (SUS)	5.4	4.5	28.1	29.0	4.5	5.4	18.4	18.6
Pm+Pl+Q (Total)	24.4	28.7	65.2	45.2	45.8	54.5	96.0	62.9

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	133.20	Passed
Pm+Pl (SUS)	29.05	199.80	Passed
Pm+Pl+Q (TOTAL)	96.00	399.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	17	16	17	16	17	16	17
Circ. Pl (SUS)	-3	-3	15	15	-8	-8	18	18
Circ. Q (SUS)	-1	1	37	-37	-175	175	293	-293
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-2	-2	12	12	-6	-6	18	18
Long. Q (SUS)	-6	6	124	-124	-52	52	88	-88
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	16.0	17.0	16.0	17.0	16.0	17.0	16.0	17.0
Pm+Pl (SUS)	13.1	14.1	31.1	32.1	8.2	9.1	34.1	35.1
Pm+Pl+Q (Total)	15.6	18.7	144.1	104.1	167.1	184.1	327.1	258.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	133.20	Passed
Pm+Pl (SUS)	35.11	199.80	Passed
Pm+Pl+Q (TOTAL)	327.12	399.60	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	7	7	7	7	7	7	7
Circ. Pl (SUS)	-10	-10	20	20	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-12	-12	14	14	-12	-12	14	14
Long. Q (SUS)	-83	83	145	-145	-153	153	215	-215
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0
Pm+Pl (SUS)	9.6	9.6	27.4	27.4	12.6	12.6	21.8	21.8
Pm+Pl+Q (Total)	93.1	79.6	162.3	155.5	165.4	144.3	232.5	219.9

Nozzle Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	133.20	Passed
Pm+Pl (SUS)	27.39	199.80	Passed
Pm+Pl+Q (TOTAL)	232.47	399.60	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main shell

Shell Maximum Design Pressure	Psd,max	1300.00 KPa.
Shell Maximum Operating Pressure	Psox,max	1300.00 KPa.
Shell Minimum Operating Pressure	Psox,min	103.42 KPa.
Shell Thickness	ts	12.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	900.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	100.00 °C
Shell Material		SA-240 304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		S30400
Shell Allowable Stress at Temperature	Ss	137.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	520.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	520.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	12.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	900.000 mm
Channel Design Temperature	TEMPC	130.00 °C
Channel Flange Matl		SA-240 304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		S30400
Channel Allowable Stress at Temperature	Sc	133.20 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1010
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	4250.00 mm
Straight Tube Length (bet. inner tubsht faces) L		4170.00 mm
Design Temperature of the Tubes		130.00 °C
Tube Material		SA-213 TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		133.20 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	159.60 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	424.264 mm



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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	5397.3 cm ²
Perimeter of Tube Layout	Cp	2735.9 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	130.00 °C
Tubesheet Material		SA-965 F304

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		S30400
Tubesheet Allowable Stress at Temperature	S	106.20 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	40.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.0000 mm
Tubesheet Outside Diameter	A	924.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

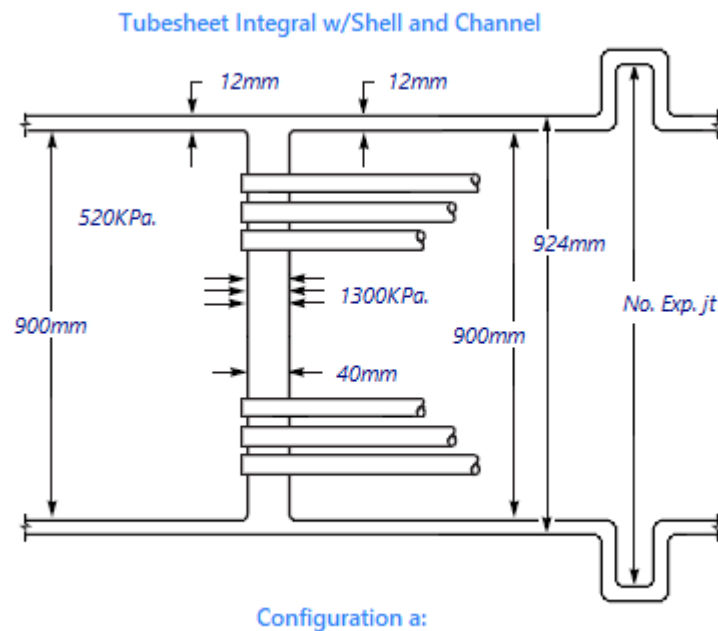
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	280.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Ps _{d,min}	-103.4190 KPa.
Channel Side Min. Design Pressure	Pt _{d,min}	-103.4190 KPa.
Mean Shell Metal Temp. along Shell len.	T _{sm}	0.00 °C
Mean Tube Metal Temp. along Tube length	T _{tm}	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Austenitic Group 3 Steels
Shell - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.18919E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Channel - TE-1 Austenitic Group 3 Steels
Channel - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	130.0 °C	0.18733E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels
Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	130.0 °C	0.18733E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	130.0 °C	0.18733E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TE-1 Austenitic Group 3 Steels
TubeSheet - TM-1 Material Group G

Elastic Mod. at Design Temperature	130.0 °C	0.18733E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Note:



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The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (623.42 \cdot (19.05/2 - 0)) / (133.2 \cdot 1.0 + 0.4 \cdot 623.42)$$

$$= 0.0445 + 0.0000 = 0.0445 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-1 at 130.00 °C
Elastic Modulus for Material 185038288.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	19.05	4170.00	11.55	50.0000	0.0082522	84.10
EMAWP = (4*B)/(3*(D/T)) = (4 *84.1037)/(3 *11.5455) = 9712.2012 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4362	19.05	4170.00	43.67	50.0000	0.0005767	45.98
EMAWP = (4*B)/(3*(D/T)) = (4 *45.9757)/(3 *43.6731) = 1403.5482 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4362 mm
Allowable Internal Pressure at Corroded thickness	24790.34 KPa.
Required Internal Design Pressure	623.42 KPa.
Allowable External Pressure at Corroded thickness	9712.20 KPa.
Required External Design Pressure	1403.42 KPa.
Required Thickness due to Shell Side pressure	0.4362 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/40.0$$

$$= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (p); dt - 2 \cdot tt)$$

$$= \max(19.05 - 2 \cdot 1.65 (0.18733E+09/1.8733E+09) \cdot (133/106) \cdot (0); 19.05 - 2 \cdot 1.65)$$

$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 424.264 + 19.05$$

$$= 867.578 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (25.4 - 19.05) / 25.4$$

$$= 0.250$$



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Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 867.5781/2$$

$$= 433.7890 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)}$$

$$= 25.4 / \sqrt{1 - 4 \cdot \min(0 \cdot 100.0 ; 4 \cdot 867.578 \cdot 25.4) / (\pi \cdot 867.578^2)}$$

$$= 25.4000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25.4 - 19.05) / 25.4$$

$$= 0.2500$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 450.0 / 433.789$$

$$= 1.0374$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 450.0 / 433.789$$

$$= 1.0374$$

Compute parameter [xt]:

$$= 1 - Nt / ((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 1010 / ((19.05 - 2 \cdot 1.65) / (2 \cdot 433.789))^2$$

$$= 0.6671$$

Determine parameter [xs]:

$$= 1 - Nt / (dt / (2 \cdot a_o))^2$$

$$= 1 - 1010 / (19.05 / (2 \cdot 433.789))^2$$

$$= 0.5130$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s (D_s + t_s) E_s / L$$

$$= \pi \cdot 12.0 (900.0 + 12.0) \cdot 1.8919E+09 / 4170.0$$

$$= 159067.3281 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 1.65 (19.05 - 1.65) \cdot 1.8733E+09 / 4170.0$$

$$= 413.1844 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 159067.328 / (1010 \cdot 413.184)$$

$$= 0.38117$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 159067.328 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.31^2))^{1/4}) / ((900.0 + 12.0) 12.0)$$



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= 0.0173 1/mm

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.017 \cdot 0.18919\text{E}+09 \cdot 12.0^3 / (6(1 - 0.31^2))$$

$$= 1045692480.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [ls]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(900.0 + 12.0) \cdot 0.10457\text{E}+1 / 40.0^3 \cdot (1 + 40.0 \cdot 0.017 + (40.0 \cdot 0.017)^2 / 2)$$

$$= 172970704.0000 \text{ KPa}$$

Shell coefficient [ds]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 900.0^2 / (4 \cdot 0.18919\text{E}+09 \cdot 12.0) \cdot (1 - 900.0 \cdot 0.31 / (2(900.0 + 12.0)))$$

$$= 0.75549\text{E}-01 \text{ mm/N./mm}^2$$

Channel coefficient [bc]:

$$= ((12(1 - \nu_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.31^2))^{1/4} / \sqrt{(900.0 + 12.0)12.0})$$

$$= 0.0173 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.017 \cdot 0.18733\text{E}+09 \cdot 12.0^3 / (6(1 - 0.31^2))$$

$$= 1035404160.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [lc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(900.0 + 12.0 / \cos(0)) \cdot 0.10354\text{E}+1 / 40.0^3 \cdot (1.0 + 40.0 \cdot 0.017 \cdot (1.0 + 1.0) / 2 + (40.0 \cdot 0.017)^2 / 2 \cdot 1.0)$$

$$= 171268896.0 \text{ KPa}$$

Channel coefficient [dc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 900.0^2 / (4 \cdot 0.18733\text{E}+09 \cdot 12.0) \cdot (1 - 900.0 \cdot 0.31 / (2(900.0 + 12.0)))$$

$$= 0.76300\text{E}-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 1.574803$; $\mu^* = 0.250000$; $E^* = 43259828 \text{ KPa}$
 $E^*/E = 0.230932$; $\nu^* = 0.378304$

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t)_{ao^2}) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.378^2)1010 \cdot 0.18733\text{E}+09 \cdot 1.65$$

$$(19.05 - 1.65)433.789^2) / (43259828 \cdot 4170.0 \cdot 40.0^3))^{1/4}$$

$$= 6.5322$$

Values from Table 4.18.3

$$Z_d = 0.005359 ; Z_v = 0.023593 ; Z_m = 0.220402$$

$$Z_a = 0.160579\text{E}+03 ; Z_w = 0.023593$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 924.0/867.5781$$



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= 1.0650

Compute coefficient [F]:

= (1 - v*) / E* * (λs + λc + E • ln(K))
 = (1 - 0.38) / 43259828 • (0.17297E+09 + 0.17127E+09 +
 0.18733E+09 • ln(1.07))
 = 5.1168

Compute parameter [φ]:

= (1 + v*) F
 = (1 + 0.3783) 5.1168
 = 7.0524

Compute parameter [Q1]:

= (ps - 1 - φ • Zv) / (1 + φ • Zm)
 = (1.0374 - 1 - 7.0524 • 0.0236) / (1 + 7.0524 • 0.2204)
 = -0.050509

Compute Parameter [Qz1]:

= (Zd + Q1 • Zw) / 2 • Xa⁴
 = (0.00536 + -0.05051 • 0.02359) / 2 • 6.53223⁴
 = 3.7938

Compute parameter [Qz2]:

= (Zv + Q1 • Zm) / 2 • Xa⁴
 = (0.02359 + -0.05051 • 0.2204) / 2 • 6.53223⁴
 = 11.3440

Compute parameter [U]:

= (Zw + (ps - 1) Zm) Xa⁴ / (1 + φ • Zm)
 = (0.0236 + (1.0374 - 1) 0.2204) 6.53223⁴ / (1 + 7.0524 • 0.2204)
 = 22.6879

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

= 0 (config a)

Compute Parameter [γ]:

= 0.000 mm (For Pressure only cases)

Calculate parameter [ws]:

= ps • ks • βs • δS (1 + h • βs)
 = 1.0374 • 0.10457E+1 • 0.0173 • 0.075549 (1 + 40.0 • 0.0173)
 = 2408.3311 mm²

Calculate parameter [ω*s]:

= ao² (ps² - 1) • (ps - 1) / 4 - ωs
 = 433.789² (1.037² - 1) • (1.037 - 1) / 4 - 2408.331
 = -2274.4778 mm²

Parameter [wc]:

= pc • kc [δC (CmTheta • βC + CqΔ • h • βC²) -
 ζc (CmTheta + h • CqTheta • h • βC / 2))
 = 1.037 • 0.10354E+1 [0.0763 (1.0 + 0.017 • 1.0 • 40.0) -
 0 (1.0/40.0 + 1.0 • 0.017/2))
 = 2408.3315 mm²

Parameter [ω*c]:

= ao² [(pc² + 1) • (pc - 1) / 4 - (ps - 1) / 2] - wc
 = 433.78903² [(1.03737² + 1) • (1.03737 - 1) / 4 - (1.03737 - 1) / 2] - 2408.33154
 = -2274.4778 mm²

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D



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Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 1300.0 \{ 0.513 + 2(1 - 0.513) 0.31 + [2/0.381 \cdot (900.0/867.578)^2] 0.31 - [(1.037^2 - 1)/(1.0 \cdot 0.381)] - [(1 - 1.0)/(2 \cdot 1.0 \cdot 0.381)] [(0^2 - (900.0)^2)/867.578^2] \}$$

 = 3075.3220 KPa.

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.667 + 2(1 - 0.667) 0.31 + 1/(1.0 \cdot 0.381)] 520.0$$

 = 1818.4567 KPa.

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 1010 \cdot 413.184 \cdot 0 / (3.142 \cdot 433.789^2)$$

 = 0.0000 KPa.

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 22.688 \cdot 0 / (2 \cdot \pi \cdot 433.789^2)$$

 = -0.0000 KPa.

Calculate the pressure [Prim]:

$$= -(U/a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_t)$$

$$= -(22.688/433.789^2) \cdot (-2274.478 \cdot 1300.0 - -2274.478 \cdot 520.0)$$

 = 213.9014 KPa.

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_c)$$

$$= (22.688/433.789^2) \cdot (2408.3311 \cdot 0 - 2408.3315 \cdot 0)$$

 = 0.0000 KPa.

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (p_s - 1)Qz2)) \cdot (P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.381 / (1 + 1.0 \cdot 0.381 (3.794 + (1.037 - 1)11.344)) \cdot (3075.322 - 1818.457 + 0 + 0 + 0 + 213.901)$$

 = 214.9864 KPa.

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega_s \cdot P_s - \omega_c \cdot P_t) - (\omega_s \cdot P_s - \omega_c \cdot P_c)) + W^* \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-2274.478 \cdot 1300.0 - -2274.478 \cdot 520.0) - (2408.331 \cdot 0 - 2408.332 \cdot 0)) \cdot 0. + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 7.05245 \cdot 0.2204)$$

 = -70.8253 Kg

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2)$$

$$= -0.051 + 2 \cdot -70.825 / (214.986 \cdot 433.789^2)$$

 = -0.08485

Fm Value from Table 4.18.3 = 0.042423

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe$$



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$$= (1.5 \cdot 0.0424 / 0.25) \cdot (2 \cdot 433.789 / (40.0 - 0))^2 \cdot 214.99$$

$$= 25.7444 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 106.2 = 159.3 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.0169 / 0.25) \cdot (2 \cdot 433.789 / (7.737 - 0))^2 \cdot 125.08$$

$$= 159.2924 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 7.7366 + 0 + 0$$

$$= 7.7366 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.25 \cdot 40.0 / 433.789) \cdot 84.96$$

$$= 3.9171 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 17.9017 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(7.7366; 3.81)$$

$$= 7.7366 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.253, F_{tmax} = 3.0563$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((1300.0 \cdot 0.513 - 520.0 \cdot 0.6671) - (214.986) \cdot -0.253) /$$

$$(0.6671 - 0.513)$$

$$= 2.4299 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((1300.0 \cdot 0.513 - 520.0 \cdot 0.6671) - (214.986) \cdot 3.056) /$$

$$(0.6671 - 0.513)$$

$$= -2.1872 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 2.43 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 133.1995 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.005 + -0.085 \cdot 0.024)6.532^4); 1.25)$$

$$= 1.7219 \text{ (Should be } \leq 2)$$



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Determine the factor [rt]:

$$= \left(\sqrt{\left(\frac{dt^2}{4} + (dt - 2 \cdot tt)^2 \right)} \right) / 4$$

$$= \left(\sqrt{(19.05^2 + (19.05 - 2 \cdot 1.65)^2)} \right) / 4$$

$$= 6.1794 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{\left(\frac{2 \cdot \pi^2 \cdot Et}{Syt} \right)}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.18733E+09 / 159)}$$

$$= 152.2161$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1.0 \cdot 280.0 / 6.179$$

$$= 45.3117$$

Buckling Allowable Stress [Stb]:

$$= (Sy / t / Fs) \cdot (1 - Ft / (2 \cdot Ct))$$

$$= (159 / 1.72) \cdot (1 - 45.312 / (2 \cdot 152.216))$$

$$= 78.894 \text{ N/mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 2.43 \cdot 0.902$$

$$= 22.35 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (19.05 - 1.65) 1.65$$

$$= 0.902 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 1.65 (19.05 - 1.65) 133.199$$

$$= 1225.059 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 1.65 (19.05 + 0.67 \cdot 1.65) 106.2$$

$$= 961.7053 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 133.199 / 106.2$$

$$= 1.254$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 1.65 (19.05 + 0.67 \cdot 1.65) 106.2$$

$$= 622.2800 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 1.65 \cdot (19.05 - 1.65) 1.254 \cdot 1.0} - 0.75 \cdot 19.05 \right)$$

$$= 2.580 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2901 mm
 Minimum Required Groove Weld Leg [agr]: 1.2901 mm



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Tube-Tubesheet Jt allowable, 1225.06 is >= tube strength 1225.06 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 137.0 \cdot 1.0 = 137.0 \text{ N./mm}^2$

Axial Membrane Stress in Shell $[\sigma_s, m]$:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 433.789^2 / ((900.0 + 12.0) 12.0) \cdot [214.99 + (1.037^2 - 1)(1300.0 - 520.0)] + 450.0^2 \cdot 520.0 / ((900.0 + 12.0) 12.0)$
 $= 14.3402 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 12.0 mm, the minimum Shell length adjacent to the tubesheet should be 187.061 mm

Shell Membrane Stress due to Joint Interaction $[\sigma_s, m]$:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 433.789^2 / ((900.0 + 12.0) 12.0) [214.99 + (1.037^2 - 1)(1300.0 - 520.0)] + 450.0^2 \cdot 520.0 / ((900.0 + 12.0) 12.0)$
 $= 14.3402 \text{ N./mm}^2$

Shell Bending Stress due to Joint Interaction $[\sigma_s, b]$:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E^*) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2/a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 0.10457 \text{E}+1 / 12.0^2 \{ 0.017 [0.075549 \cdot 1300.0 + 450.0^2 \cdot 0 / (.22703 \text{E}+1)] + 6(1 - 0.38^2) / (43259828) (433.79 / 40.0)^3 (1 + 40.0 \cdot 0.02 / 2) [215.0 (0.024 + 0.22 \cdot -0.051) + 2 / 433.79^2 \cdot 0.22 \cdot -70.825] \}$
 $= 83.6009 \text{ N./mm}^2$

Shell Stress Summation vs. Allowable
 $\therefore |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$
 $\therefore |14.34| + |83.6| \leq 205.5 \text{ N./mm}^2$
 $\therefore 97.94, \text{ must be } \leq 205.5 \text{ N./mm}^2$

Note:
 For a given Channel thickness of 12.0 mm, the minimum Channel length adjacent to the tubesheet should be 187.061 mm

Channel Membrane Stress due to Joint Interaction $[\sigma_c, m]$:
 $= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$
 $= 450.0^2 \cdot 520.0 / ((900.0 + 12.0) 12.0 \cdot \cos(0))$
 $= 9.622 \text{ N./mm}^2$

Channel Bending Stress due to Joint Interaction $[\sigma_c, b]$:
 $= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6(1 - \nu^2) / (E^*) \cdot (a_o / h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2/a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 0.10354 \text{E}+1 / 12.0^2 \cdot \{ (0.0763 \cdot 1.0 \cdot 0.017 - 0 \cdot 1.0) \cdot 520.0 - 6(1 - 0.378^2) / 43259828 \cdot (433.789 / 40.0)^3 \cdot (1.0 + 40.0 \cdot 0.017 / 2 \cdot 1.0) \cdot [215.0 (0.02359 + 0.2204 \cdot -0.051) + 2 / 433.789^2 \cdot 0.2204 \cdot -70.825] \}$
 $= 20.4304 \text{ N./mm}^2$

Channel Stress Summation vs. Allowable
 $\therefore |\sigma_c, m| + |\sigma_c, b| \leq 1.5 \cdot S_c$



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:: | 9.62 | + | 20.43 | ≤ 199.8 N./mm²
 :: 30.05 must be ≤ 199.8 N./mm²

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	25.7 <=	159.3 N./mm ²	Pass
Tubesheet Shear Stress	4.2 <=	85.0 N./mm ²	Pass
Maximum Tube Stress	2.4 <=	133.2 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-2.2 <=	-78.9 N./mm ²	Pass
Maximum Force on any one Tube	22.3 <=	1225.1 Kg	Pass
Axial Membrane Stress in Shell	14.3 <=	137.0 N./mm ²	Pass
Shell Stress (jt. inter.)	97.9 <=	205.5 N./mm ²	Pass
Channel Stress (jt. inter.)	30.1 <=	199.8 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	7.737	40.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.290	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.290	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 16 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Case Type	Pass/ Fail
D1uc	11.493	...	32	159	6 85 Ps+Pt-Th	D1 Ok
D2uc	19.178	...	59	159	11 85 Ps+Pt-Th	D2 Ok
D3uc	7.737	...	26	159	4 85 Ps+Pt-Th	D3 Ok
D4uc	3.810	...	1	159	... 85 Ps+Pt-Th	D4 Ok
O1uc	4.509	...	27	319	6 85 Ps+Pt+Th	O1 Ok
O2uc	8.962	...	53	319	10 85 Ps+Pt+Th	O2 Ok
O3uc	3.810	...	26	319	4 85 Ps+Pt+Th	O3 Ok
O4uc	3.810	...	...	319	... 85 Ps+Pt+Th	O4 Ok
D1c	13.489	...	34	159	7 85 Ps+Pt-Th-c	D1 Ok
D2c	21.172	...	63	159	11 85 Ps+Pt-Th-c	D2 Ok
D3c	9.734	...	28	159	4 85 Ps+Pt-Th-c	D3 Ok
D4c	5.810	...	1	159	... 85 Ps+Pt-Th-c	D4 Ok
O1c	6.507	...	30	319	6 85 Ps+Pt+Th-c	O1 Ok
O2c	10.958	...	57	319	10 85 Ps+Pt+Th-c	O2 Ok
O3c	5.810	...	28	319	4 85 Ps+Pt+Th-c	O3 Ok
O4c	5.810	...	...	319	... 85 Ps+Pt+Th-c	O4 Ok
Max:	21.1724	... mm	0.396	0.135	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance



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[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	3	136	...	...		...	...	...	...			Ok
D2uc	10	136	...	...		...	...	...	...			Ok
D3uc	14	136	...	...		...	...	...	...			Ok
D4uc	2	136	-2	-77		...	...	...	...			Ok
O1uc	4	410	...	...		...	...	...	...			Ok
O2uc	10	410	...	...		...	...	...	...			Ok
O3uc	14	410	...	...		...	...	...	...			Ok
O4uc	...	410	...	...		...	...	...	...			Ok
D1c	3	136	...	...		...	...	...	...			Ok
D2c	9	136	...	...		...	...	...	...			Ok
D3c	14	136	...	...		...	...	...	...			Ok
D4c	2	136	-2	-77		...	...	...	...			Ok
O1c	4	410	...	...		...	...	...	...			Ok
O2c	10	410	...	...		...	...	...	...			Ok
O3c	14	410	...	...		...	...	...	...			Ok
O4c	...	410	...	...		...	...	...	...			Ok
Max RATIO												
	0.105			0.021		...			...			

Tube, Shell and Channel Stress Summary:

---- Tube Stresses					Tube Loads		Shell Stress		Channel Stress		Stress Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	5	133	-3	-87	43	1225	34	205	64	199	Ok
D2uc	7	133	-7	-84	66	1225	120	205	43	199	Ok
D3uc	2	133	-2	-78	22	1225	98	205	30	199	Ok
D4uc	...	133	...	-100	3	1225	6	205	10	199	Ok
O1uc	4	266	-3	-88	38	2450	26	410	61	399	Ok
O2uc	6	266	-6	-84	58	2450	116	410	31	399	Ok
O3uc	2	266	-2	-78	22	2450	98	410	30	399	Ok
O4uc	...	266	...	...	...	2450	...	410	...	399	Ok
D1c	5	133	-3	-90	43	1225	34	205	64	199	Ok
D2c	7	133	-7	-86	67	1225	119	205	43	199	Ok
D3c	2	133	-2	-80	22	1225	98	205	30	199	Ok
D4c	...	133	...	-103	3	1225	6	205	10	199	Ok
O1c	4	266	-3	-90	38	2450	26	410	61	399	Ok
O2c	6	266	-6	-86	59	2450	116	410	31	399	Ok
O3c	2	266	-2	-80	22	2450	98	410	30	399	Ok
O4c	...	266	...	...	...	2450	...	410	...	399	Ok
Max RATIO											
	0.054			0.084		0.054		0.582		0.319	

Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	21.172	40.000	Ok
Tube thickness :	0.436	1.650	Ok



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Tube-tubesheet fillet weld leg :	1.290	1.650	Ok
Tube-tubesheet groove weld leg :	1.290	1.650	Ok

Min Shell length of thk, (12.000) adj. to tubesheet: 187.061 mm

Min Channel length of thk, (12.000) adj. to tubesheet: 187.061 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.043 mm
 Due to Pressure + Thermal Shell Compresses by : -0.043 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	3014.54	0.995	3813.25	0.998
Tubesheet Shear Stress	7523.51	1.000	10743.02	1.000
Tube Tensile Stress	16577.57	1.000	27006.76	1.000
Tube Compressive Stress	5602.46	0.328	17320.54	1.000
Tube-Tubesheet Joint load	16577.56	1.000	27006.73	1.000
Shell Stress (Axial, Junction)	3014.54	0.995	3813.25	0.998
Tubesheet-Channel Junction	3014.54	0.995	3813.25	0.998
Tube Pressure Stress	24790.33	1.000	9712.19	1.000
Minimum MAWP	3014.54		3813.25	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	3930.24	1.000	5066.10	1.000
Tubesheet Shear Stress	10020.53	1.000	14329.17	1.000
Tube Tensile Stress	17319.58	1.000	28322.89	1.000
Tube Compressive Stress	20804.25	1.000	21744.96	1.000
Tube-Tubesheet Joint load	17319.58	1.000	28322.91	1.000
Shell Stress (Axial, Junction)	3930.24	0.476	4641.90	1.000
Tubesheet-Channel Junction	3507.50	1.000	4641.90	0.270
Tube Pressure Stress	25683.31	1.000	11364.51	1.000
Minimum MAPnc	3507.50		4641.90	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0100	0.0100	...	...	...	...
2	-0.0427	-0.0427	...	...	...	...
3	-0.0346	-0.0347	...	...	...	...
4	0.0020	0.0020	...	...	...	...
5	0.0067	0.0067	...	...	...	...
6	-0.0413	-0.0414	...	...	...	...
7	-0.0346	-0.0347	...	...	...	...
8	...	...	...	...	...	...



PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: